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Bereich für Mathematik und Naturwissenschaften Fachrichtung Physik, Institut für Strahlenphysik

Probing of ultra-intense Laser-Solid Interaction Dynamics with Resonant Small-Angle X-ray Scattering

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Abstract

This thesis provides a contribution to the long-term goal pursued at Helmholtz-Zentrum Dresden-Rossendorf (HZDR) to establish small-angle X-ray scattering (SAXS) with X-ray free-electron lasers (XFELs) as a standard technique to probe transient laser-plasma processes inside solids. The main part of this thesis is a pump-probe experiment conducted at the Matter in Extreme Conditions endstation at the Linac Coherent Light Source. This experiment allowed to employ resonant SAXS for the first time at relativistic pump laser intensity, which reflects the experimental conditions present in ultrahigh intensity laser-driven proton acceleration. By actively tuning the XFEL probe energy to specific K_α transition energies in nanostructured copper grating targets, the potential of resonant SAXS as an in-situ probe for specific ionization states and therefore temperature was demonstrated. Changes in a measured asymmetry in the scattering patterns were correlated with opacity changes in a laser-heated copper plasma by applying a semi-analytical model for the asymmetry. In combination with numerical simulations, this allowed to deduce a lower limit temperature estimate of 300 electronvolts for the hot copper plasma present within the first 300 femtoseconds upon laser pump.
These experiments represent a stepping stone to use SAXS in the future to probe laser-induced electron density structures, e.g. caused by plasma instabilities. The latter represent a main driver for the efforts to establish XFEL probing techniques for laser-solid interaction: In 2011, net-like modulated proton beams were measured in laser-proton acceleration experiments with the Draco-150 TW laser system at HZDR in combination with few micrometer thick Titanium foil targets[1]. Numerical simulations identified the Rayleigh-Taylor instability or Weibel-like instability as possible candidates serving as the root cause for their generation. Yet, to this day, the uncertainties regarding the specific mechanism and instability type causing such transverse proton beam modulations are a topic of intense discussion.
As a complementary part of this thesis, different laser-target configurations have been investigated in experiments with the Draco PW laser system with respect to the generation of net-like modulated proton beam profiles. By a controlled variation of the temporal laser intensity contrast via prepulses applied onto the front side of 2 µm thick Titanium targets, net-like modulations as measured in 2011 could not be reproduced at the Draco PW laser system. However, a frequent generation of net-like proton beam modulations was found in a regime of high laser contrast in combination with few hundred nanometer thick Formvar (plastic) foil targets. A particularly prominent visibility of net-like proton beam modulations was found in a thickness range between around 100 and 300 nanometers. For a fixed target thickness of 240 nm, the net-like beam modulations remained unaffected by a variation of the laser pulse energy on target between around 4 and 18 J. While the experiments can give answers to the underlying mechanisms for these proton beam modulations only to a limited extent, experimental settings have been identified in which modulated proton beam structures – and potentially laser-induced electron density structures inside solid density plasma – are generated in a nearly stable manner, which serves as a groundwork for systematic SAXS studies at XFEL facilities in the future.

Zusammenfassung

Diese Dissertation liefert einen Beitrag zu dem langfristigen Ziel des Helmholtz-Zentrums Dresden-Rossendorf (HZDR), Small-angle X-ray Scattering (SAXS) mit Freie-Elektronen Lasern im Röntgenbereich (XFELs) als Standardmethode zur Untersuchung von flüchtigen Plasmaprozessen in Festkörpern zu etablieren. Der zentrale Teil dieser Arbeit behandelt ein Anrege-Abfrage Experiment, welches am MEC Labor am LCLS durchgeführt wurde. Hierbei wurde SAXS zum ersten Mal im Bereich der relativistischen Pumplaserintensität durchgeführt, was die typischen Bedingungen von Experimenten der Laserprotonenbeschleunigung widerspiegelt. Als experimentell neues Konzept in diesem Kontext wurde resonantes SAXS durchgeführt. Hierbei wurde die Photonenenergie des XFELs aktiv auf resonante K_α-Übergangsenergien im lasergeheizten Kupfer gestimmt. Damit wurde das Potential von resonantem SAXS als in-situ Sonde zur Messung spezifischer Ionisationszustände in Plasmen und somit der Plasmatemperatur demonstriert. Hierzu wurde ein semi-analytisches Modell für die Asymmetrie in den resonanten Streusignalen verwendet, welches an die spezifischen nanostruktierten Gittertargetgeometrien angepasst wurde. Mithilfe dieses Modells konnte die gemessene Asymmetrie mit Veränderungen der Plasmaopazität verknüpft werden. In Kombination mit numerischen Simulationen erlaubte dies, eine Untergrenze der Plasmatemperatur von 300 eV abzuleiten, die bereits während der ersten 300 fs der Laser-Plasma-Wechselwirkung erreicht wurde.
Solche Experimente stellen einen Ausgangspunkt dar, um SAXS in der Zukunft zur Messung von laser-induzierten Plasmastrukturen zu verwenden, zum Beispiel in Form von Plasmainstabilitäten. Letztere sind ein zentraler Treiber für die Bemühungen des HZDRs, XFEL Probing-Techniken für flüchtige Laser-Plasmaprozesse zu etablieren: Im Jahre 2011 wurden in Experimenten mit dem Draco-150 TW Laser netzartig modulierte Protonenstrahlprofile in Kombination mit wenigen um dicken Titanfolien gemessen[1]. Numerische Simulationen identifizierten eine Rayleigh-Taylor- oder Weibel-artige Instabilität als mögliche Ursache für deren Erzeugung. Dennoch konnte bislang kein eindeutiger Mechanismus experimentell nachgewiesen werden, sodass derartige Strahlmodulationen auch aktuell intensiv diskutiert werden. Als komplementäres Thema dieser Arbeit wurden verschiedene Laser-Target-Konfigurationen am Draco-PW Laser hinsichtlich des Auftretens transversaler Strahlmodulationen getestet. Die 2011 gemessenen, netzartigen Strahlprofile konnten durch kontrollierte Variation des Laserkontrasts mittels Vorpulsapplikation auf 2 µm dicke Titan-Folien nicht reproduziert werden. Jedoch konnte eine häufige Erzeugung von netzartigen Modulationen bei hohem Laserkontrast in Kombination mit wenigen hundert Nanometer dicken Formvartargets (Plastik) beobachtet werden. Die netzartigen Modulationen waren besonders prominent sichtbar für Targetdicken zwischen etwa 100 und 300 nm. Bei einem Scan der Laserpulsenergie zwischen 4 und 18 J bei fester Targetdicke von 240 nm wurde kein Einfluss auf die Erzeugung dieser Strahlmodulationen festgestellt. Somit wurde eine experimentelle Konfiguration gefunden, in der Protonenstrahlmodulationen – und potentiell die damit verbundenen Plasmainstabilitäten – häufig auftreten, was eine Grundlage für zukünftige, systematische Fallstudien mit SAXS an XFEL-Anlagen darstellt.

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1 Introduction

The interaction of powerful lasers with solids enables the generation of matter in extreme density and temperature conditions in laboratories, which otherwise only exists in the universe, e.g., inside planetary cores, stars or astrophysical jets. This unique capability initiated a variety of aspiring research fields such as laboratory astrophysics, laser-fusion research and laser-plasma-based particle acceleration. Laser-proton acceleration (LPA) is of particular interest for envisioned applications in translational radiobiology research[2, 3, 4, 5], radiotherapy[6, 7] or for the proton fast ignition scheme[8] in inertial confinement fusion (ICF)[9, 10]. This thesis has been prepared at the institute for radiation physics at Helmholtz-Zentrum Dresden-Rossendorf (HZDR), which operates the ultrahigh intensity (UHI) ultrashort pulse laser system Draco for research in laser-proton acceleration. One of the central missions at HZDR is to provide an infrastructure for translational research in radiobiology with laser-accelerated protons.

Presently, the most robust and widely used LPA mechanism is Target Normal Sheath Acceleration (TNSA)[11]. Its basic principle relies on the absorption of an ultrashort HI laser pulse at the front side of few micrometer thin solid density foil targets and subsequent acceleration of electrons to relativistic energies. These fast electrons propagate through the target towards the rear side, where they set up a quasistatic sheath field of extremely high electric field strength in the order of TV/m, in which protons and other ions originating from a hydrocarbon surface contaminant layer present in the experimental vacuum conditions can be accelerated to kinetic energies of several tens of MeV per nucleon.

Translational radiobiology research with laser-accelerated protons requires a high control over the acceleration process to ensure the stable generation of desired beam parameters, e.g., with respect to the maximal kinetic energy, spectral shape or transverse beam uniformity. However, plasma instabilities such as the Rayleigh-Taylor instability (RTI) or Weibel-like instabilities may lead to a break up of uniform electron transport through the plasma bulk by a decay into electron filaments, which may result in transversely modulated electromagnetic field structures at the target rear and transversely modulated proton beams. The mentioned instability types have been identified as potential causing agents of net-like modulated proton beam profiles measured by Metzkes-Ng et al.[1] in 2011 at the Draco 150 TW laser system in experiments with few micrometer thick Titanium foil targets. However, until today, an unambiguous proof of a specific instability has not been possible due to a lack of experimental access to the transient solid density plasma dynamics and limited predictive capability of existing numerical simulation codes. Plasma instabilities also play a critical role in other fields

of laser-solid interaction, such as in inertial confinement fusion with lasers, which exploits the ability of kilojoule lasers to compress and heat fusion pellets to ignite fusion. In this case, the RTI diminishes laser-energy absorption and hydrodynamic instabilities inhibit the uniform compression needed to fulfill the Lawson-criterion[12]. These examples show the strong demand in the scientific community to study solid density plasma upon interaction with HI and high energy (HE) lasers in experiments directly.

X-ray free-electron lasers (XFELs) provide unique capabilities to probe solid density plasmas and to resolve the transient, nonlinear plasma processes due to their short pulses down to few tens of femtoseconds and high photon energies enabling to penetrate through solids, which is not possible with established optical probing techniques without use of higher harmonics. Accordingly, there are intense, worldwide efforts, e.g., at European XFEL, LCLS or SACLA, to combine XFEL facilities with state-of-the-art HI and HE laser systems and to extend X-ray probing techniques for pump-probe experiments of laser-solid interaction.
The HZDR has been strongly contributing to push ahead the Helmholtz International Beamline for Extreme Fields (HIBEF), an international user consortium that operates an UHI and HE laser at the European XFEL, dedicated to research in the fields of LPA, Warm Dense Matter and High Energy Density. One central HIBEF goal is to establish small-angle X-ray scattering (SAXS) as a probing technique for laser-solid interaction.
With SAXS, the complex plasma dynamics become accessible via the temporal evolution of the nanoscale Fourier components of the scattering length density $\rho(\vec{r}, t) = n_e + n_i f_i$, where n_e and n_i are the free electron and ion density and $f_i = f_0 + f' + if''$ with f_0 being the Thomson term and f' and f'' being the dispersion correction terms. SAXS offers a resolution down to the few nanometer scale, which is ideally suited for solid-density probing[13, 14, 15] and fills a gap between the resolutions of sub-nanometers for wide-angle X-ray Scattering (WAXS) and hundreds of nanometers for phase-contrast imaging (PCI)[16].
SAXS promises to measure laser-induced structures in the electron density or ionization state distribution in plasmas, e.g. due to instability-induced electron beam filamentation. This potentially allows to discriminate in the future between different instability types and to characterize their growth rate in order to identify the specific causing agents for transverse proton beam modulations. Achieving this goal poses two requirements: On the one hand, it requires the establishment of SAXS in the complex context of LPA experiments and the demonstration of its potential on the example of science cases of reduced complexity. On the other hand, it requires to test experimental laser-target configurations, in which laser-generated structures in the electron density are generated in a stable manner in order to allow for systematic studies in beam times granted at XFEL facilities. This thesis is a contribution to both of these aspects.

Content of this thesis

The main topic of this thesis deals with an XFEL pump-probe experiment of UHI laser-solid interaction performed in 2018 at the Matter in Extreme Conditions (MEC) endstation at the Linac Coherent Light Source (LCLS) at Stanford Linear Accelerator Center (SLAC). The experiments were realized in the frame of a big collaborative effort between more than 37 scientists and 14 research facilities. This thesis is to a large extent based on the contribution to the preparation and conduction of the experiment, as well as the analysis of the experimental SAXS data. The results led to a publication in Physical Review Research[17]. The experiment is a follow-up of a first proof-of-principle experiment at MEC, in which SAXS has been demonstrated by a mea-

surement of the plasma front surface expansion of nanostructured Silicon grating targets with nanometer resolution upon laser irradiation at intensities of around 10^{17} W/cm^2[14], which is close below the relativistic intensity regime. In this regime, the SAXS signal is dominated by Thomson scattering on free electrons and the signal is connected to the modulus square of the Fourier Transform of the projected electron density along the XFEL probe direction. Prestructured grating targets probed in a transmission geometry have been proven to be a suitable experimental platform for systematic XFEL studies, as they provide a strong, reliable reference signal for every pump-probe experiment and are sensitive to the nanoscopic, ultrafast dynamics of the plasma electron density. In the previous experiment, an effective plasma scale length attributed to the grating expansion could be deduced from the decay of the scattering signal at high scattering vectors[14], which is linked to the smoothing of the grating edges.

The recent experiment, content of this thesis, was performed in the regime of relativistic pump laser intensity of around 10^{19} W/cm^2, which allowed for the first time to perform SAXS in conditions reflecting those typically present in LPA experiments with UHI short pulse lasers. As a conceptually new approach in this experimental context, resonant SAXS has been demonstrated and shown to extend the present XFEL capabilities to probe solids. For this, the XFEL photon energy was tuned to on- and off-resonant energies of specific atomic bound-bound transition channels in a laser-heated copper plasma, which made the SAXS signal sensitive to changes in the plasma opacity due to X-ray absorption. This enabled to exploit the ionic contribution to the scattering, which is primarily given by the imaginary part of ρ. With the scattering length being complex-valued, this allows for non-point symmetric or *asymmetric* signals. Based on theoretical groundwork by T. Kluge, in a collaborative effort a semi-analytical model has been developed, which links the asymmetry in the scattering patterns to plasma expansion and opacity changes upon the optical laser pump. This model has been adapted to the specific target configurations used in the experiments and applied on the experimental data to obtain additional information on the expansion and opacity changes.

Two specific science cases have been investigated with dedicated targetry. In *case 1* silicon-copper grating targets have been used to study the heating and expansion of a buried grating structure following moderate pump laser irradiation of around 10^{17}W/cm^2. The Si-Cu targets were obtained by sputtering of a Cu-layer onto the Si grating targets used in the precedent MEC experiment. This resulted in a buried grating structure and a rear side grating in form of a smooth replication of the buried structure. Comparison of the measured asymmetry with the model enabled to prove an expansion of the *buried layer* grating structure of around 10 nm starting at 6 ps after laser pump. This insight is non-trivial, as in the experimental transmission probe geometry, the SAXS signal is linked to the *2D areal density* due to projection along the XFEL probe direction.

In *case 2* only the sub-micrometer thin copper grating targets were pumped at relativistic pump laser intensity of around 10^{19}W/cm^2 in order to study the ultrafast expansion and ionization dynamics following laser irradiation. In this case, an increased asymmetry was measured in the scattering signals obtained from the copper targets at the on-resonant XFEL probe energy of 8165 eV within the first 300 femtoseconds after laser pump. This increase in asymmetry has been associated with opacity changes, and more specifically with the existence of nitrogen-like[1] ionization states in the heated copper plasma. Since the opacity

[1] This nomenclature denominates ionization states by the ground state atoms with the according number of electrons bound to the inner atomic shells. Thus, N-like refers to ionization states with 7 electrons bound to the K and L shells.

depends on the existence of specific ionization states in the plasma, which in thermal equilibrium is linked to the plasma temperature, with according modeling this method resembles an ultrafast in-situ temperature measurement. With this regard, it is similar for example to X-ray Thomson Scattering (XRTS)[18] but with the potential of high spatial resolution similar to the element-specific imaging technique in [19]. However, PIC simulations revealed that in the studied case the plasma passes through highly non-equilibrium states. During the first few hundred femtoseconds upon the UHI laser-solid interaction, the average kinetic electron energy reaches the keV range. Yet, due to a comparatively slow ionization, the average ionization in the PIC simulations does not correspond to a non-local thermodynamic equilibrium (NLTE) temperature in the keV range. However, we can identify an *NLTE equivalent temperature* of around 300 eV, which matches with the average ionization in our simulations. Novel methods, such as the one presented here, are a valuable contribution to extend XFEL probing capabilities to fully exploit the large-scale XFEL experiments.

A complementary part of this thesis addresses experiments at the Draco PW laser system, in which different laser-target conditions were tested with respect to the generation of net-like modulated proton beams, as they were associated with the above mentioned instabilities. These experiments aimed to systematically study the influence of different laser-target parameters on the generation of transverse beam modulations. The experiments are driven by indications from the literature. Yet, they have an heuristic and explorative character and can provide answers on the exact underlying mechanisms in the generation of proton beam modulations only to a limited extent.

In a first approach, we aimed to reproduce the laser contrast conditions present in the experiments from Metzkes-Ng et al. at the Draco PW laser system by controlled prepulse application onto the front side of 2 µm Titanium foils. While in these experiments, the net-like modulated beam profiles could not be observed, experiments with few hundred nm thick Formvar (FV) foil targets at conditions of high laser pulse contrast have shown to feature prominent, net-like structures at a high occurrence rate. In a scan of different target thicknesses modulated proton beam profiles with structures of arbitrary type were observed to be generated up to the maximal tested thickness of 900 nm. Net-like structures were most prominently visible in a thickness range between around 100 and 300 nm. This thickness range is above the onset of target transparency for plasma mirror cleaned laser contrast conditions ($d \gtrsim 60$ nm). For a fixed target thickness of 240 nm, a scan of the laser pulse energy on target in the range of 4 to 18 J ($20 \leq a_0 \leq 45$) did not reveal a dependence on the generation of these beam profile structures. Moreover, FV targets coated with a thin, ~ 30 nm Aluminium (Al) layer located either at the target front or rear side were employed to test a potential influence on the source location of the modulated proton beam. It was observed that an Al layer at the rear side leads to a strong reduction in the acceleration performance, while Al present at the front side yields similar performance and also net-like modulated proton beam structures compared with the pure FV foils.

A central important observation is the robustness in the generation of modulated structures for fixed laser settings. We received the impression of similarities between beam modulations at fixed laser-target conditions. With *similarity* we here refer to common modulation wavenumbers being located in same regions within the proton beam spot. Yet, we *did not* observe identical proton beam profiles from shot to shot in the sense that identical features were generated at the exact same positions. The latter is rather to be expected in scenarios, in which spatial information from the laser is imprinted into the proton beam. From this we deduce that the measured proton beam modulations can be understood in a radiographical picture, in which sheath-accelerated protons probe transversely modulated electromagnetic

field structures self-generated in the plasma at the target rear. Yet, the experiments cannot prove the exact underlying mechanism or exclude either seeding of a RTI at the target front side or Weibel-like instabilities or a combination of both.
The remaining uncertainties are a strong endorsement of the HIBEF program – in particular to measure and characterize laser-induced electron and ion density structures in solid density plasma. Additionally, they emphasize the need for systematic simulations of the proton beam propagation during the acceleration process in transiently evolving modulated electromagnetic fields in close vicinity to the source location. This would be of great relevance for an understanding of how those field structures translate into different types of proton beam modulations.

The outline of this thesis is the following: Chapter 2 gives an introduction on TNSA as a starting point for relevant effects to be considered in the generation of modulated proton beam profiles. Additionally, a brief introduction to the relevant plasma instabilities being discussed with respect to the net-like modulated proton beams measured by Metzkes-Ng et al. and the link to the HIBEF program will be given. Building on this, evidences of modulated proton beams in the literature will be presented and additional plasma processes and acceleration mechanisms will be discussed. In Chapter 3, the experiments at the Draco laser system will be presented and put into the context of the literature. Chapter 4 describes the theory of resonant scattering for XFEL probing on hot laser-generated plasmas and presents the derivation and adaptation of the semi-analytical model for the asymmetry in the scattering signals for the targets used in the experiments at MEC. In Chapter 5, the main results of the pump-probe resonant SAXS experiment at MEC will be presented.

2 Principle Mechanisms and Challenges in Laser-Proton Acceleration

2.1 Laser-proton acceleration and its applications

Laser-proton acceleration (LPA) exploits extremely high electric fields in the order of TV/m, which can be generated in plasmas as a result of relativistic laser-matter interaction. Thereby, this novel acceleration concept greatly exceeds the achievable field strengths of conventional radiofrequency cavity based accelerators, which are limited to tens of MV/m. The regime of relativistic laser-matter interaction refers to laser intensities above 10^{18} W/cm^2, for which a laser can excite electrons to relativistic oscillation velocities via the Lorentz force. At presently available intensities of UHI ultrashort pulse lasers, protons cannot be accelerated directly. Instead, one makes use of strong, quasistationary charge separation fields that are set-up between directly accelerated electrons pushed forward by the laser and heavier, plasma ions. Potential applications of LPA can be found, e.g., in the proton fast ignition scheme[8] in ICF, proton radiography[20] or translational radiobiology research [2, 3, 4, 5]. Their realization relies on a high control over the acceleration process to meet the respective demands regarding beam parameters such as the spectrum, maximal kinetic energy and transverse beam uniformity. One of the key requirements for the realization of envisioned applications is to increase the maximal kinetic energy of the accelerated protons, which up to now is limited to around 100 MeV[21]. Improving the achievable kinetic proton energy is highly relevant to reach clinically relevant proton energies of up to 250 MeV to reach deep seated tumors in hadron therapy or to enable proton radiography schemes on heavy element objects[20]. Accordingly, the scientific community seeks to fully exploit theoretical scalings for the maximal proton energy of the most established acceleration mechanism TNSA[11] and to develop and realize advanced acceleration schemes.

In TNSA, the accelerating field is located at the target rear side as a result of electrons being accelerated from the target front towards the rear side, where they form an electron sheath. The starting point of TNSA is an intense, ultrashort laser pulse, which is tightly focused on a thin ($\lesssim$ µm) solid density foil target. In this case, the laser pulse interacts with an overdense

target and can only penetrate up to the critical density surface defined by

$$n_c = \frac{m_e \varepsilon_0}{e^2} \cdot \omega_L^2, \tag{2.1}$$

with e, m_e, ε_0 and ω_L being the electron charge, electron mass, vacuum permittivity and laser circular frequency, respectively. When the electron density, n_e, exceeds n_c, the plasma frequency

$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \varepsilon_0}} \tag{2.2}$$

becomes larger than ω_L so that the target turns opaque for the laser. At the critical density surface, the electromagnetic wave of the laser pulse interacts with electrons via the Lorentz-force and forces them to perform a quiver motion. In the relativistic intensity regime, electrons experience a longitudinal force via the B-field directed into the target, and can escape the restoring $\vec{v} \times \vec{B}$-force. As a result, electrons are inserted into the target in bunches with a frequency of $2\omega_L$[22]. The laser-accelerated energetic electrons reach relativistic kinetic energies (up to few MeV) and propagate towards the target rear side, where they set-up a Gaussian-shaped electron sheath. In the resulting, quasistationary charge separation field between the electron sheath and the heavier plasma ions, contaminant atoms (C, H, O) present in a surface layer at experimental vacuum conditions are field-ionized and can be accelerated to tens of MeV/nucleon. Protons are most efficiently accelerated compared to other ions due to their favorable charge to mass ratio. Additionally protons are accelerated earliest and shield the electric field for heavier ions, when they follow the electron expansion into vacuum.
The ion acceleration field is characterized by a high-field gradient, i.e., it is maximal at the target rear surface and drops within a few µm. The TNSA mechanism generates exponential proton spectra with a maximal kinetic energy[1] (cut-off energy) and a beam directed in target-normal direction. The beam divergence is energy-dependent with typical maximal half-angles of around 20° and decreasing divergence for higher proton energy. In the idealized TNSA model, the acceleration process generates smooth and highly laminar beam emission as protons are accelerated along the gradient of the local electron density within the Gaussian-shaped Debye-sheath.
In order to improve the proton cut-off energies, it is of interest to increase the average energy of the laser generated fast electrons, T_e, and their density, $n_{e,h}$, and to minimize the scale length, L_S, of the rear sheath, as the TNSA sheath field, E_S, scales in the isothermal Mora model[23] with

$$E_S \sim \frac{T_e}{L_s} \sim \sqrt{n_{e,h} k_B T_e} \text{ [11].} \tag{2.3}$$

The hot electron temperature (T_e) commonly refers to the ensemble-averaged kinetic energy of the laser-accelerated electrons. The average kinetic energy of a single electron can be identified as its cycle-averaged quiver motion in the electromagnetic wave considering the electron proper time as introduced by Kluge et al.[22]. For $a_0 \gg 1$, which is the case for intensities achieved with the Draco laser system, the average kinetic energy of the laser-accelerated electrons then scales with

$$T_{e,hot} = \frac{\pi a_0}{ln(16) + ln(a_0)} - 1 + O(a_0^{-3}) \text{ [24].} \tag{2.4}$$

[1] In the following the term *proton energy* will refer to the cut-off energy in the proton spectrum.

Here, a_0 is the dimensionless laser amplitude defined by

$$a_0 = 0.85 \left(\frac{I\lambda_{\mu m}^2}{10^{18}\,\mathrm{W/cm^2}} \right)^{1/2} \tag{2.5}$$

with I and λ being the laser intensity and wavelength, respectively. Accordingly, higher laser intensities promise to increase the hot electron temperature and to achieve higher proton cut-off energies. Yet, the TNSA-scaling of cut-off energies with laser intensity is not ideal and faces limitations as described later.
Other methods to increase the hot electron temperature include optimized target concepts, e.g., by use of near-critical foam targets[25] to enable volumetric electron heating. Volumetric electron heating can also arise as a result of relativistic induced transparency[26] (RIT) allowing for increased laser penetration into the target. At highly relativistic laser intensities, the laser can penetrate through the plasma, if the electron density, n_e, gets lower than the relativistic critical density $n_c' = \gamma n_c$. When electrons reach relativistic kinetic energies due to the thermal expansion and their oscillation due to the ponderomotive force of the laser, the electron mass is corrected by the cycle-averaged[2] Lorentz factor $\gamma = \sqrt{1 + a_0^2}$. Another approach is to use Reduced Mass Targets (RMTs) with the goal to confine the lateral target size to minimize the dilution of the expanding electron sheath and to increase the accelerating field strength[27]. Another well-known key element for efficient proton acceleration is to generate high density gradients at the target rear. This require minimal expansion of the target prior to the main pulse arrival and thus a high contrast in the temporal laser curve (laser contrast). The laser contrast refers to the ratio of laser light preceding the temporal peak of the laser pulse (main pulse). Typically, this is realized by use of a plasma mirror setup[28], an intensity- dependent temporal switch, in order to suppress the Amplified Spontaneous Emission (ASE) level and to clean the laser contrast curve from prepulses. High laser contrast also allows to reduce the target thickness, which is beneficial to increase the hot electron density in the rear sheath, as the lateral sheath extension due to the geometric effect of the divergent electron beam propagation through the target may be reduced.

To this day, the scaling of the proton cut off energy is an intensively discussed topic due to the fact that theoretical models tend to overestimate the experimental measurements. Scaling models proposed in the literature suggest power law scalings, $E_p \sim E_L^{\zeta}$, with ζ ranging from 0.5 for long pulse laser systems ($\tau \gg 500\,\mathrm{fs}$) to 1 for short pulse laser systems such as Draco[29]. Keppler et al.[30] explain the overestimation of theoretical scaling models at high laser intensities by a spatial mismatch between the location of highest field strength close to the rear surface and the expanding proton source at the target rear side. Besides, Keppler et al. point out the importance to include the preplasma scale length in theoretical scaling models and emphasize the role of the temporal intensity contrast in determining the preplasma conditions. Additionally, plasma instabilities are discussed to be limiting for the TNSA scalings, as they may lead to a breakup of a uniform electron transport through the target, which is not accounted for in existing scaling models.

To circumvent limitations inherent to TNSA, there is a plethora of other acceleration mechanisms proposed in literature, which might become accessible with the advance of available laser intensities. These include schemes exploiting co-moving fields with the accelerated protons, for example in the RTF-RPA scheme[31]. Other mechanisms are for example coulomb

[2] A plane laser field is assumed here.

explosion[32], radiation pressure acceleration (RPA)[33] or Magneto-Vortex-Acceleration (MVA) [34, 35, 36]. For more information on these acceleration mechanisms the reader is referred to the reviews from Daido[37] or Macchi[38].

2.2 Plasma instabilities and their link to the HIBEF program at HZDR

One of the central missions at HZDR is to provide an infrastructure for translational research in radiobiology with laser-accelerated protons. The institute of radiation physics at HZDR operates the UHI ultrashort pulse laser system Draco (30 J, 30 fs) to perform first irradiation studies with laser-accelerated protons and currently devolopes a self-constructed long pulse laser system Penelope[39] (150 J, 150 fs) to extend the experimental platform as a user facility for radiobiology research. The overarching goal is to study the (tumor) cell response on ultrahigh peak dose rates, in particular to study the so-called Flash radiotherapy effect[40]. Laser-driven proton sources are particularly interesting in this context, as the achievable peak dose rates exceed those of conventional accelerators, such as cyclotrons, by many orders of magnitude.
Recently, a first successful irradiation pilot study on a mouse tumor model has been performed at the Draco laser facility[5]. Realization of such experiments relies on a high control over the acceleration process, e.g., with regard to the spectrum, cut-off energy and dose uniformity. The latter is of particular relevance in the context of radiobiological irradiation studies to ensure the transverse dose uniformity and to avoid particle losses due to passive scattering foils, when aiming for ultrahigh peak dose rate experiments. Besides, transverse beam uniformity is also crucial in the proton FI scheme of ICF[8], to generate uniform compression of the fusion pellet[41] to reach fusion conditions or for proton radiography[20]. Yet, during the last two decades of laser plasma acceleration research, transverse proton beam modulations[3] have been consistently observed at different laser laboratories. Since modulated proton beams are closely related to plasma instabilities, their understanding is crucial to control the acceleration process as a whole, which makes them an intensively studied research subject until today.

At Draco, net-like proton beam modulations have been first observed in 2011 in measurements performed by Metzkes-Ng et al.[1] (see Fig. 2.2). It had already been well-known at this time, that the laser-target interaction is strongly affected by the temporal laser contrast curve as it defines the preplasma conditions, i.e., the plasma density gradient and preionization at the main pulse arrival time. Thus, it was of particular interest that the occurrence of net-like modulations coincided with changes in the Draco laser system, which resulted in a modification of the laser pulse contrast (see Fig 2.2a, details in Sec. 2.3.2). By means of 2D PIC simulations, the Rayleigh-Taylor or Weibel-like instability were found to be suitable candidates in two distinct preplasma scenarios. Yet, up-to-date, a clear identification of a certain instability type as a causing agent of the proton beam modulation has not been possible due to the following challenges. On the one computational side, numerical simulations can provide answers only to limited extend as the exact target condition at the arrival time of the main pulse is typically unknown, yet strongly determines the plasma dynamics. With this regard, numerical simulations are challenging as both the picosecond and femtosecond time scale

[3] Throughout this thesis, the term *proton beam modulations* refers to transverse structures in proton beam profiles.

need to be covered, which for necessary time and space resolution exceeds the computational resources. Additionally, the matter passes through the regimes of solid state, Warm Dense Matter and plasma physics, for which a simulation code covering all relevant physics in the necessary detail does not exist. On the experimental side, identification of a specific instability type requires to access the solid density plasma dynamics at the laser-solid interaction site directly, which is not possible with standard optical probing techniques, which are limited to the critical density surface.
One way forward to experimentally access the solid density plasma dynamics at the relevant timescale of femtoseconds represent the ultrashort X-ray pulses at XFEL facilities. In particular, small-angle X-ray scattering (SAXS) has been proposed as a promising technique to resolve the transient plasma evolution in solid density plasmas in experiment. Thus, the HZDR has been strongly contributing to push ahead the Helmholtz International Beamline for Extreme Fields (HIBEF). HIBEF is an international user consortium that operates an UHI and high energy laser at the European XFEL, dedicated to research in laboratory astrophysics and WDM or HED research. With regard to LPA and plasma instabilities, novel X-ray probing capabilities promise to measure and characterize the growth of instabilities and to answer open questions in the interpretation of experimental measurements such as the measured proton beam modulations. Additionally, such measurements would enable to benchmark PIC simulation codes in order to improve their predictive capabilities. Finally, these measurements would also help to identify the role of plasma instabilities and their contribution to experimental deviations from theoretical scaling models for the proton cut-off energy.

In the next sections, a brief introduction to the Rayleigh-Taylor instability and Weibel instability will be presented as these two instability types have been suggested as likely candidates causing the proton beam modulation measured by Metzkes-Ng et al.. Additionally, the 2-surface plasmon decay of plasma oscillations[42] as a potential seed for RTI growth will be presented. Then, a review on a plethora of other evidences of modulated proton beam profiles and associated instabilities and mechanisms in the literature will be given.

2.2.1 Rayleigh-Taylor instability

The RTI is a hydrodynamic instability that occurs at the interface of two fluids of different densities, when the denser fluid is above the less dense fluid in the gravitational field[43]. The same dynamics takes place, when a lighter fluid is pushed into a heavier fluid[44]. There are various examples for the occurrence of the RTI, ranging from every day situations, for example, when milk is poured into a cup of coffee, to the Crab Nebula. The RTI leads to bubbles of the lighter fluid to rise into the heavier fluid and spikes of the heavier fluid to enter the lighter one[45]. Transferred to laser-plasma interaction, a RTI can occur, when a lighter photon fluid is pushed via the radiation pressure against the dense plasma interface at the target front side. The generation of the RTI is a highly relevant topic in laser plasma physics. In inertial confinement fusion, limiting the ablative RTI[46] instability growth rate is crucial to achieve fusion conditions in the core of a fusion pellet. In LPA, controlling RTI generation is relevant for the performance and stability of the acceleration process.
The starting point of the RTI growth are small perturbations of the interface of the two fluids. The growth begins with a linear stage, for which the fluid dynamics can be described by linear equations leading to an exponential growth of the perturbation amplitudes. Subsequently, the unstable system often enters a stage of non-linear growth, which requires non-linear equations. Theoretical models for the RTI evolution typically consider a sinusoidal perturba-

tion of the interface with a spatial wave number k. When in the linear stage the amplitude of the perturbation is small compared to its wave length, the amplitude increases exponentially with time ($\sim e^{\gamma t}$) and the growth rate reads

$$\gamma = \sqrt{ka}, \tag{2.6}$$

with a being the accelerating force.
In the context of LPA, it is a question of intense discussion, by which type of initial perturbation a RTI is triggered and how the dominant wavenumber in the RTI wavenumber spectrum is predetermined. This question is also relevant in the interpretation of modulated proton beam profiles as one needs to check the modulation scales in the proton beam and the RTI wavenumber spectrum for consistency.
According to eqn. 2.6, the RTI favors high instability wavenumbers, thus enhancing the smallest perturbations present at the target front side. However, in the context of RPA in combination with thin foil targets, Sgattoni et al.[47] demonstrated theoretically that surface rippling at the target front side generates a laser polarization dependent modulation of the radiation pressure and enhances plasmonic modes close to the laser wavelength, which then define the dominant mode in the RTI wave number spectrum. As other potential seeds the surface roughness of the target or intensity variations in the UHI laser profile are being discussed. For initially flat, micrometer thick targets, surface modulation by the Two-surface-plasmon decay of plasma oscillations has been proposed by Kluge et al.[42] (see Sec. 2.2.3).

2.2.2 Weibel(-like) instabilities

Systems of two counter-streaming electron flows in plasmas are known to be unstable to any magnetic fields perpendicular to the flow axis. In laser proton acceleration we encounter this situation, when a laser-accelerated electron current generated at the target front side propagates towards the rear side and is compensated by a cold bulk electron return current to achieve charge balance. As a result plasma instabilities lead to a decay of uniform electron flow into electron filament currents surrounded by magnetic fields, whose amplitudes grow in time. There is a variety of instability types in the literature, whose definition depends on the orientation of the instability wave vector $\vec{k}$. If the instability wave vector is perfectly aligned with the current flow this refers to a two-stream instability (TSI)[48]. If $\vec{k}$ is oriented in transverse direction, this refers to a filamentation instability (FI), and a mixed state of the latter two is called two-stream/filamentation instability (TSF)[49].
Important theoretical foundation for the description of the unstable system of two counter-propagating electron flows was introduced by Weibel[50]. The theory of the Weibel instability (WI) in its original scenario refers to a plasma instability, which occurs when a plasma possesses an anisotropy in the velocity space, i.e., when two temperatures in *different directions* can be assigned to the electron population. Yet, in literature the term *Weibel-like instability* is often used as a comprehensive term for any of the aforementioned instability types, as in realistic situations superposition of these instabilities occur.
In the LPA context Weibel-like instabilities are associated with filamented magnetic fields located in preplasma at the target rear with amplitudes of more than 10 kT and modulation wavelengths on the um scale[51, 52, 53].

2.2.3 Two-surface plasmon decay

By means of PIC simulations, Kluge et al. identified two distinct mechanisms for a break up of a uniform transport of the laser-accelerated electrons by electron filamentation at the target front side[42]. As a result of this, the electron density modulation imprints on the ion density at the target rear for TNSA with micrometer thick foils without occurrence of relativistic transparency. Realization of one of the two mechanisms depends sensitively on the preplasma conditions at the time, when the main laser pulse arrives at the target front side.
In case of an exponential preplasma being present at the target front side the $2\omega_L$ electron bunches are observed to be prone to a Weibel-like instability and enter the target directed almost normal to the foil surface as filaments surrounded by strong magnetic fields. In this case the ion density gets structured *from the beginning* of the acceleration process in contrary to the mechanism described by Quinn et al.[54], where a Weibel instability grows at the rear on the picosecond time scale *during* the acceleration process.

The second case comprises a high density gradient at the front side, i.e., without the presence of a preplasma. Here, a transverse structure of the foil surface is generated in form of density ripples, which may trigger the growth of a RTI and results in a filamented electron injection. In the literature it is intensively discussed which type of initial density fluctuation may serve as the seed for growth of the RTI. One discussed mechanism comprises the parametric decay of an initially longitudinal surface oscillation[47, 55, 56] in the laser field with a driver frequency of $\omega_D = 2\omega_L$ into a 2D electron density surface oscillation via interference of an upwards and downwards moving surface wave with $\omega_S = \omega_D/2$ resulting in a transverse standing surface wave. For vanishing electron temperature, the dispersion relation then reads

$$k_S = \omega_S\sqrt{\omega_p^2 - \omega_S^2}/\sqrt{\omega_p^2 - 2\omega_S^2}\ [57]. \tag{2.7}$$

For the case $\omega_p \gg \omega_s$ found in the PIC simulations by Kluge et al., this would correspond to $k_s \sim \omega_L$. Yet, the value of k_s extracted from the PIC simulations was found to be significantly greater. In contrast, Kluge et al. found a good numerical agreement, when the *plasma oscillations themselves* were considered as the driving source resulting in electron surface oscillations by parametric decay. In this scenario and for non-vanishing plasma temperature the theoretical dispersion relation is then approximately

$$k_s \approx \frac{\omega_p}{2}\ [57]. \tag{2.8}$$

The proposed mechanism is supported by a good agreement between the theoretical dispersion relation and peaks at $\frac{\omega_p}{2}$ and ω_p being present in the wave number spectrum extracted from the front surface density modulation in the PIC simulations.

2.3 Transverse proton beam modulations in laser-proton acceleration

Modulated proton beam profiles have been consistently measured at different laser laboratories during the last two decades and a whole plethora of effects and interpretations exists for their origin. Whereas in recent years, ultrashort pulse lasers strongly shift in the focus of attention for laser particle acceleration, various evidences of proton beam modulation go back to a decade of long pulse high energy laser experiments. Moreover, little unifying mate-

rial exists and there is a lack of comparisons between measured data. Publications often refer to single-shot evidences in combination with lower dimensional simulations (< 3D). The latter are often tweaked to reflect assumed experimental preplasma conditions and to support the interpretation, while the exact target conditions realized in the experiments are usually not completely known. This circumstance leads on the one hand to a lack of reproducibility of measurements and on the other hand to various causing agents being discussed for proton beam modulations.
This section aims to provide the theoretical context of the net-like modulated beam profiles measured by Metzkes-Ng. et al. accompanied by other examples of beam modulations and respective effects discussed in the literature, yet without claim of completeness. A few examples of experimental evidences of transverse proton beam modulations are shown in Fig. 2.1 and the experimental settings are summarized in Table 2.1.

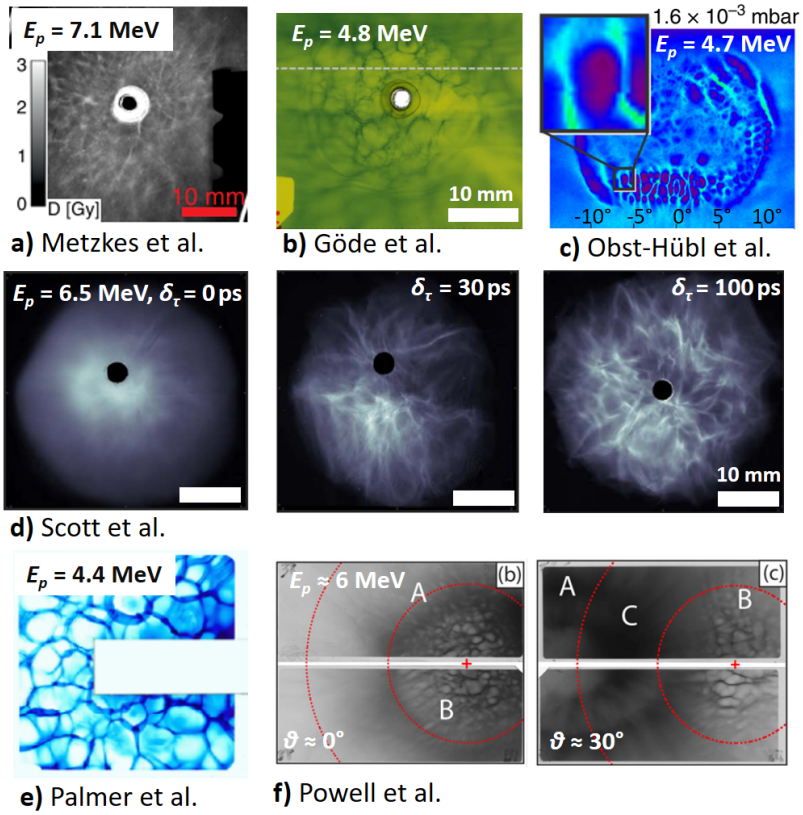

Figure 2.1: Excerpt from the literature of RCF measurements of modulated proton beam profiles. **a)** Metzkes-Ng et al.[1], 2 µm Ti targets, Draco laser (30 fs, 1.5 J), cp. Fig. 2.2. **b)** Göde et al.[52], 5 µm H_2-Jet, Draco laser (30 fs, 3 J). **c)** Obst-Hübl et al.[58], 10 µm tungsten wire, Draco laser (30 fs, 3 J). High-frequency spatial information (here the acronym HZDR) from the laser near-field is imprinted into the proton beam. **d)** Scott et al.[51], 45 µm graphite targets, PHELIX laser (500 fs, 72 J) with prepulses at different delays δ_τ (prepulse to main pulse energy ratio of 1:10). **e)** Palmer et al.[59], 5 nm DLC (diamond-like carbon) targets, Vulcan PW laser (660 fs, ∼ 300 J). **f)** Powell et al.[60], 10 nm Al targets, Vulcan PW laser (1 ps, 200 J), different laser incidence angle θ. A bubble-like structure (feature B) is observed centered along the laser axis (red cross) unaffected by a change of the laser incidence angle.

Reference, Year	Laser system	pulse dur., energy	pol., inc. angle	target material, thickness	Linked mechanism
Fuchs et al. [61], 2003	LULI 100 TW	350 fs, 25 J	0°	SiO_2 and CH insulators or CH+Au targets, various thickness (see Ref.)	target conductivity, laser imprinting on las.-acc. electrons
McKenna et al. [62], 2011	Vulcan	1 ps, 200 J	p	allotropes of Carbon, 500 um	resistive filamentation
Palmer et al. [59], 2012	Vulcan	660 fs, 300 J	p, 0°	diamond-like foils, 5 to 50 nm	RPA + RTI
Metzkes-Ng et al. [1], 2014	Draco 150 TW	30 fs, up to 3 J	p, 0°	Ti, few um	Weibel-like instab. or param. instab. + RTI
Powell et al. [60], 2015	Vulcan	1 ps, (200±15)J	p, 0...30°	Al, $\lesssim$ 40 nm	RPA + RTI
Scott et al. [51], 2017	PHELIX	500 fs, 72 J	s, 34°	gold, graphite, 25 and 45 um	Weibel instability
Göde et al. [52], 2017	Draco 150 TW	30 fs, 3J	p	cylindrical H_2 jet, 5 to 10um	Weibel-like instability
Obst-Hübl et al. [58], 2018	Draco 150 TW	30 fs, 3 J	p	cylindrical H_2 jet, 5 um, W-wires, 10 um	All-optical imprinting

Table 2.1: Overview of the experimental settings for the evidences of proton beam modulations from the literature discussed in this section (without claim of completeness).

The major experimental challenge for an unambiguous identification of a specific plasma process serving as a causing agent for non-uniform proton emission, lies in a lack of experimental access to the solid density plasma evolution directly at the laser-target interaction site. To understand which aspects are relevant to consider in the generation of modulated proton beam profiles it is useful to recall the basic steps of the most established acceleration mechanism TNSA. The role of nonlinear effects such as plasma instabilities or the realization of (or interference with) advanced acceleration mechanisms can then conveniently be integrated in the discussion.

The TNSA scheme consists of three fundamental steps:

1. Laser absorption by electrons at the critical density surface at the target front side and acceleration of these electrons to relativistic kinetic energies.
2. Transport of the laser-accelerated electrons through the plasma bulk towards the target rear, electron thermalization and set-up of the Debye sheath.
3. Field-ionization of contaminant atoms and expansion of the quasineutral electron-proton plasma into vacuum.

All of these three phases need to be considered, as non-linear plasma effects may occur at the target front surface, target bulk or rear side during the TNSA dynamics.
In the idealized model of TNSA, protons are accelerated normal to the local isodensity contour of the Gaussian-shaped Debye-sheath and the beam emission is highly laminar. This means, the emission is characterized by a high correlation between the source position and emission angle and between the acceleration time and final energy of the particle. This property gave rise to make use of laser-accelerated proton for application in proton radiography, e.g., to

probe electromagnetic fields in plasmas[4] or to characterize materials, e.g., objects related to ICF[20].
In the context of the acceleration process itself, strong transverse electric or magnetic field structures may lead to proton deflection and caustic formation[65]. Another mechanism for proton beam modulations may result from a modulated rear surface topology. In 2004, Cowan et al.[66] demonstrated experimentally the mapping of artificially introduced, periodic or non-periodic rear surface topologies into the sheath accelerated proton beam. The interpretation contains an initial acceleration directed normal to the rear surface contour, which generates transverse momenta and translates the topology structure into the angular emission distribution (virtual cathode phase), followed by a magnification effect during the laminar sheath acceleration phase.

In realistic LPA conditions, i.e., when flat foils are used, proton beam modulations may be a result of non-uniform electron transport. When an UHI intensity laser interacts with a solid density target, electrons absorb a fraction of the laser pulse energy at the critical density surface at the target front side and are accelerated towards the target rear side at relativistic kinetic energies. During the electron transport through the target, Mega-Ampere currents are generated exceeding the Alfvén limit[67]. In first order, this electron current is compensated by a return current of cold plasma bulk electrons. The system of the two opposing current flows tends to be unstable and has been linked to a variety of plasma instabilities such as the two-stream instability[68], filamentation instability or Weibel instability (WI)[50, 69, 70] often interchangeably referred to as a Weibel-like instability. As a result of Weibel-like instabilities uniform electron transport may break-up by decay into electron filaments surrounded by strong magnetic fields, which may perturb the sheath acceleration field.
While the important role of non-uniform electron transport is rather uncontroversial, the spatial location of the break-up of uniform electron transport is a matter of debate in literature. Different mechanisms attributed to the target front side, bulk or rear plasma scale length have been proposed and will be discussed in the following. Various experimental parameters of the laser-target interaction have been discussed to be relevant for triggering certain instability types. This concerns on the one hand target-specific properties such as the material, density, resistivity or surface quality and on the other hand laser parameters such as incidence angle, laser polarization, intensity, pulse length and contrast.

2.3.1 Instability seeding in the target bulk

Modulated proton beam profiles were observed by Fuchs et al.[61] (in 2003) and McKenna et al.[62] (in 2011) in experiments with targets of few tens of micrometer thickness and laser pulse durations of 350 fs (Fuchs) and 1 ps (McKenna). In both references a dependence of the electron transport uniformity and smoothness of proton emission on the target conductivity is claimed. Fuchs et al. observed smooth electron transport and proton emission for gold foils of high conductivity, whereas CH insulator targets resulted in non-uniform electron transport and modulated proton beam profiles. McKenna et al. claimed a dependence of the electron transport uniformity on the sample's conductivity by using different allotropes of carbon[62]. Homogeneous electron transport was reported for diamond targets supposed to get metallic-like during heating as opposed to non-uniform electron transport in case of vitreous and

[4] Of special interest is the promised high spatial resolution in point-projection imaging schemes, as Borghesi et al.[63] found experimental evidence that the laminar type of beam emission allows the definition of a virtual, point-like source of a few micrometer size located in front of the target. This is much smaller than the *real* source size of around 100 µm at the target rear reported in the literature[64].

pyrolytic carbon targets. The electron transport uniformity is attributed to the sample's lattice structure and conductivity during its initial Warm Dense Matter state while the return currents heat the sample.
A difference in the electron transport through insulator (CH) and metal foil (Al) targets has also been observed by Manclossi et al.[71] in experiments with the ultrashort pulse laser system at Laboratoire d'Optique Appliquée ($\tau_L = 40$ fs, $E_L = 0.7$ J). For CH targets of more than 30 um thickness a filamentation of the laser-accelerated electron beam has been reported. The electron beam filamentation has been deduced from a spatially non-uniform Cherenkov emission and has been attributed to an instability of the ionization front[72] inside the target. The theoretical prediction of the growth rate of the ionization front instability is in the order of $\Gamma_c \sim 10^{-13}$ s [72]. This growth rate corresponds to a filamentation distance of around 100 um, which makes this instability relevant to be considered for targets with thicknesses of that order.

2.3.2 Instability seeding at the target front side

While the experiments from Fuchs et al. and McKenna et al. were performed with long pulse high energy lasers and comparably thick targets of a few ten micrometers, the evidence of net-like modulated proton beam profiles by Metzkes-Ng et al. (see Fig. 2.2) was made in experiments with the Draco 150 TW ultrashort pulse laser at HZDR in combination with few micrometer thick Ti targets. The experiments were performed with a peak intensity of around

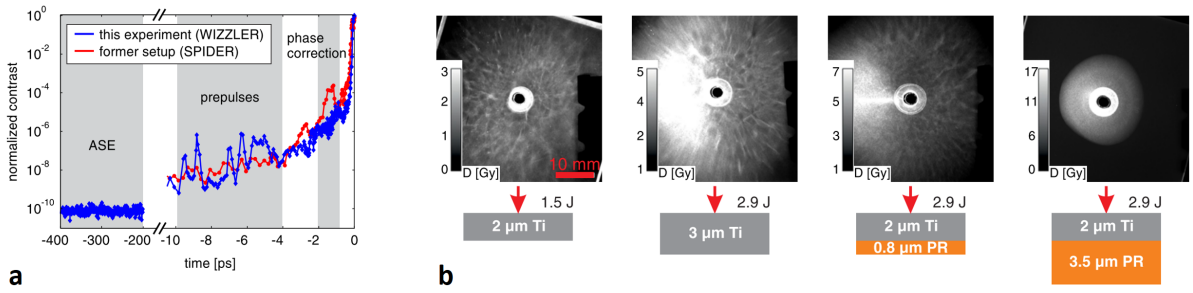

Figure 2.2: Temporal intensity contrast of the Draco 150 TW laser (**a**) and proton beam profiles (**b**) for different target thicknesses and configurations from the experiments from Metzkes-Ng et al.[1]. Net-like structures can be observed for different target thicknesses until a total thickness of 3 µm. Increased target thicknesses required higher laser energy thresholds for the onset of net-like patterns. The measurements coincided with the implementation of a Wizzler that modified the laser pulse contrast.

10^{20} W/cm^2 and normal incidence. The degree of modulation has been found to depend on the laser energy with a thickness-dependent optimum. Thicker targets required higher laser pulse energy to reach strongest modulation. Tests with a photoresist (PR) layer at the target rear side did not suggest an influence of the target material or rear surface roughness but rather of the total target thickness.
It has been well-known, that the laser-target interaction is strongly affected by the temporal laser contrast curve as it defines the preplasma conditions, i.e., the plasma density gradient and preionization at the main pulse arrival. Thus, it was of particular interest that the occurrence of net-like modulations coincided with changes in the Draco laser system in 2011, which resulted in a modification of the laser pulse contrast (see Fig 2.2a). Replacement of a SPIDER

system[73] by a Wizzler[74, 75] used for spectral phase correction resulted in increased laser contrast between 4 and 0 ps prior to the main pulse arrival by removal of ps-prepulses, but additional ps prepulses between 10 and 4 ps prior to the main pulse.
The Rayleigh-Taylor or Weibel-like instability were found to be suitable candidates in two distinct preplasma scenarios to be seeded *at the target front side*, by means of 2D PIC simulations (see Fig. 2.3). The first case comprises full ionization of Titanium and a sharp density gradient at the target front side, where surface rippling may be seeded by the two-surface plasmon decay of plasma oscillations forming a standing wave pattern[42]. This surface structure is proposed to be the seed of a RTI at the target front side. As a result of this, spatially bunched electron jets are generated at the front side and propagate towards the target rear side with an increased modulation strength of the electron density at higher a_0. The second case deals with different preplasma gradients at the target front and suppressed ionization and surface rippling. At optimized preplasma lengths at the target front side, electron filaments surrounded by strong magnetic fields may be generated by a Weibel-like instability. The optimal preplasma scale length at the front side is assumed to be given by an interplay between increased electron acceleration in the underdense preplasma (which enhances the WI) and laser depletion (reducing the growth of the WI). In both cases, the generated electron filaments can reach the target rear, which leads to a transverse modulation of the electron density at the target rear that is translated into the accelerated ion beam.
In summary, these PIC simulations suggest instability seeding at the target front side, which affects the proton beam uniformity via electron filaments reaching the target rear sheath field. The preplasma conditions used in these simulations were chosen to showcase the two distinct plasma instabilities scenarios. Yet, in experimental conditions a superposition of both effects might occur.

2.3.3 Instability seeding at the target rear side

In contrast, experimental evidence of seeding of a Weibel instability in preplasma scale lengths *at the rear side* has been claimed by Scott et al.[51] and Göde et al.[52]. The experiment from Scott et al. was performed at the PHELIX laser system with s-polarized pulses of 500 fs pulse length, oblique laser incidence, peak intensity of $4 \cdot 10^{19}$W/cm^2 and gold and graphite foils of 25 and 45 µm thickness. The rear side scale length was varied by application of additional prepulses at the target front side with a fixed intensity ratio of 1/10 compared to the peak intensity and delays between 30 and 100 ps. The main finding is that in increased rear scale lengths the lifetime and length of filamented magnetic field structures can be increased leading to enhanced proton deflection.
Similarly, Göde et al. attributed modulated proton profiles obtained from a solid density hydrogen jet target to the onset of Weibel-like instabilities and filamented electromagnetic fields in the rear side preplasma scale length. The experiments were performed at the Draco 150 TW system using a cryogenic hydrogen monospecies target of cylindrical shape and diameters between 5 and 10 µm. PIC simulations revealed filamented electric and magnetic field amplitudes of up to around 0.5 MV/µm and 10 kT that are maximized at the rear side critical density surface. In accordance with Scott et al., the simulations revealed longer magnetic filaments in increased rear scale lengths. Projection of protons on a virtual detector yielded smooth profiles for protons originating from the front and caustic structures, as measured, for protons from the rear.

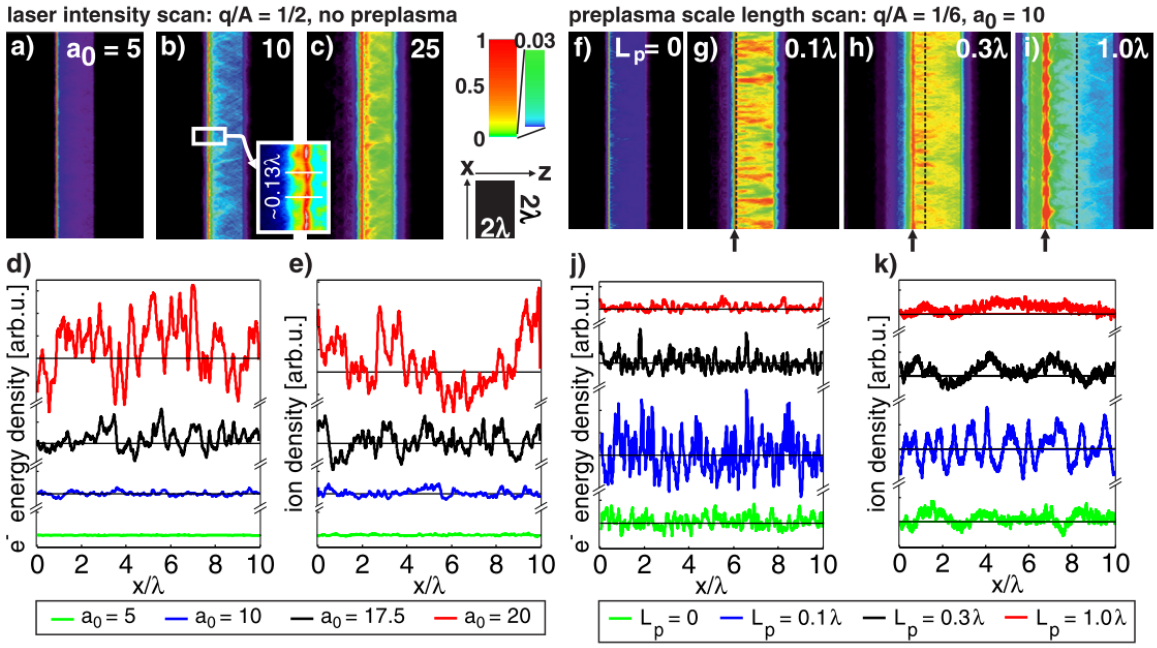

Figure 2.3: Results of PIC simulations (adopted from [1]) showcasing distinct preplasma conditions for the onset of a Rayleigh-Taylor instability (**left**) or Weibel-like instability (**right**) as potential candidates responsible for the observed net-like proton beam modulations by Metzkes-Ng et al.. The upper row shows the electron energy density for different laser intensities for the RTI scenario (**a-c**) and front preplasma scale lengths for the Weibel-like instability case (**f-i**). The lower row shows transverse lineouts of the electron density at the target rear (**d, j**) and of the ion density (**e,k**).

2.3.4 Particle-In-Cell simulation study on filamented magnetic field generation at the target rear side

Simulations performed by Branco[53] support the important role of preplasma formation at the target rear side in the generation of magnetic field structures at the target rear as claimed by Scott et al. and Göde et al.. In the multiparametric study with 2D3V PIC simulations, different front and rear preplasma conditions were investigated for the case of the hydrogen jet target used by Göde et al. by variation of the front and rear plasma scale length and laser intensity.

The magnetic field structure at the rear side was found to be composed by an amibipolar dipole field that extends over several micrometers and a filamented magnetic field structure with wavelengths on the sub-micrometer scale. Both field structure types compete in their degree of expression in dependence on the rear side preplasma scale length, with the filamented magnetic field structure dominating the ambipolar field at increased rear side plasma scale lengths (see Fig. 2.4). This behavior can be explained by looking at the motion of the fast, laser-accelerated electrons at the rear side and by the respective electron phase space plots in Fig. 5.15 in [53]. When the rear scale length is too small, e.g. at a value of 0.1 µm, the fast electrons cannot escape the target as they are forced by the ambipolar field to gyrate back into the target. Accordingly, the generation of magnetic filaments is suppressed. In an increased rear preplasma scale length of 1 µm, the electrons can escape the rear side since

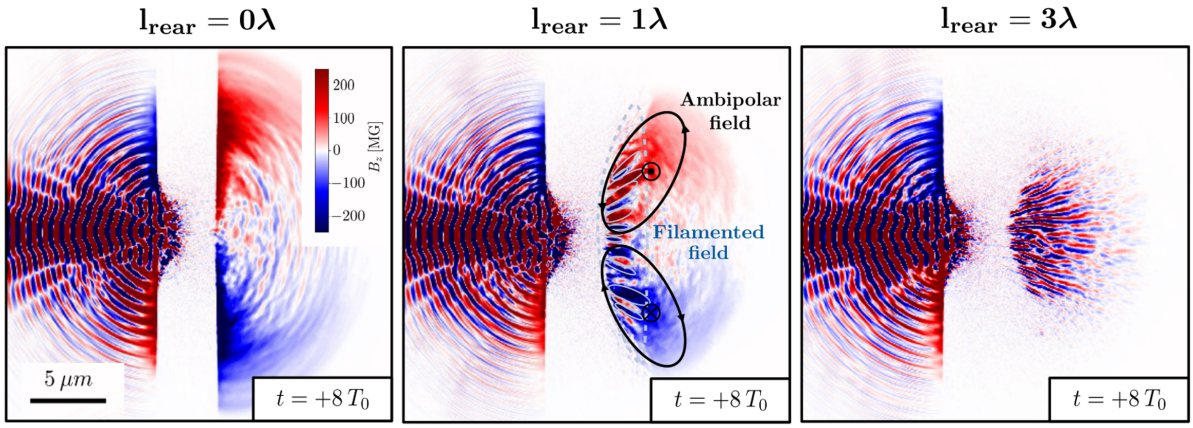

Figure 2.4: Magnetic field distribution from the PIC simulations from Branco[53] for different rear preplasma scale lengths (without front preplasma) for a laser amplitude of $a_0 = 30$ around 20 fs after the laser peak E-field interacted with the target front side. An increase of the rear preplasma scale length leads to a suppression of the ambipolar magnetic field and a stronger expression of the filamented magnetic field.

the rear side preplasma screens the electric sheath field. In this case, the escaping electrons can generate a filamentary magnetic field at at the rear side.
For the case of the simulated hydrogen jet target, the filamented magnetic field structures were found to dominate for small front scale lengths below approx. 0.5 µm and at rear scale lengths of up to around 3 µm. For these preplasma conditions, the filamented B-field structure dominates in a wide range of laser intensity for a_0 between 10 and 30. Higher laser intensities resulted in increased maximal field strengths but also in increased formation of the ambipolar field compared to the filamented contribution. The robustness of the filamented magnetic field generation is an important observation, since proton beam modulations have also been observed in this intensity range in experiments at the Draco PW system in combination with Formvar foils in a scan of different laser pump energies (see Sec. 3.2.3).

These findings demonstrate that the presence of the rear side preplasma scale length is a requirement for strong filamentary magnetic fields at the target rear as claimed by Scott et al. and Göde et al.. However, the PIC simulations from Branco give several strong indications that the *primary* instability is seeded *at the target front side* via a 2-plasmon decay of plasma oscillations at the target front surface coupled to a subsequent RT-like instability generation as proposed by Kluge et al.[42]. First, in the PIC simulations from Branco we can observe a rippling of the front surface and a subsequent generation of electron filaments propagating towards the target rear side (see Fig. 4.17 in [53]), which *precedes* the generation of filamented magnetic field structures at the target rear. Second, the wavelength of the filamented magnetic field in the PIC simulations was found to be in the order of 1 µm, which is significantly smaller than the theoretical prediction for a Weibel-like instability of several µm (see Fig. 5.8 in [53]). Neither does the wavelength of the filamented magnetic field depend on the relativistic Lorentz factor γ_r of the return current (or a_0) as predicted by the Weibel theory. On the contrary, the projection of the filamented magnetic field structure onto the target front surface matches the plasma frequency, the front surface rippling period and the filamented electron distribution at the target front side. Third, it is observed that the magnetic field induced by the fast electron current originating from the target front side is consistent with the saturating

magnetic field amplitude of the filamented magnetic field at the target rear side (see Fig. 5.13 in [53]). Fourth, the induced B-field is also in agreement with the observed trends for the filamented magnetic field in dependence on the laser intensity and the front preplasma scale length. Fifth, the theoretical growth rates of the Weibel and Two-stream instability overestimate the observed growth rate in the PIC simulation, while the theoretical growth rate of a RTI is in agreement with the latter. In conclusion, the PIC simulations from Branco support the 2-plasmon decay of plasma oscillations in combination with a RTI seeded at the target front side.

Summarizing, the experiments from Scott and Göde give indications for seeding and growth of WI-induced electromagnetic fields *in rear side* plasma scale lengths. Yet, as proposed by Metzkes-Ng et al. and supported by the simulations from Branco, also a RT-like instability may be seeded at the *target front surface* via the two-surface plasmon decay of plasma oscillations. Also, as claimed by Metzkes Ng et al., a WI generation in a *front side* preplasma remains a possible scenario.

2.3.5 Proton source location and acceleration regimes beyond TNSA

Highly relativistic laser intensities in combination with the use of ultrathin, overdense foils allow to enter advanced acceleration regimes linked to the generation of modulated proton beams such as RPA, the regime of relativistic transparency or hybrid acceleration regimes in combination with TNSA[76, 77]. The possible realization of these acceleration mechanisms needs to be considered in the discussion of modulated proton beams, since the source protons being transversely modulated may originate purely from the target front by RPA or from the rear by TNSA, or the proton beam may consist of contributions of both mechanisms.

Radiation Pressure Acceleration and Rayleigh-Taylor instability

When an UHI laser interacts with an overdense target, the ponderomotive force of the laser leads to a pile-up and denting of the electron density at the front surface due to the radiation pressure of the laser, which is known as hole boring. When the radiation pressure is sufficiently strong, protons from the front side can be accelerated in the resulting charge separation field between the compressed electron layer and quasi-stationary, heavier plasma ions. This mechanism is known as RPA. One distinguishes between the regimes of hole boring-RPA (HB-RPA)[78], when comparably thick targets of several laser skin depths are involved, and Light Sail RPA (LS-RPA)[33], when the target is thin enough that all target ions are accelerated. Theoretical work by Esirkepov et al.[33] showed that RPA is dominant for circular polarized light at laser intensities exceeding 10^{23} W/cm^2. The use of circular laser polarization in combination with normal laser incidence is favorable for RPA as it provides a constant pressure via the non-oscillating ponderomotive force and as the generation of hot electrons and thus TNSA is suppressed due to absent $\vec{j} \times \vec{B}$ electron heating. Yet, recent work by Qiao et al.[76] proposed a hybrid acceleration regime of RPA-LS and TNSA for linearly polarized light that can be realized at 2 orders of magnitude less intensity. The hybrid scheme is based on a double-peak electric field structure by RPA and TNSA and can be realized if coupling conditions between the laser intensity, pulse length and target thickness are fulfilled so that LS-RPA accelerated ions reach the TNSA sheath field and overtake the purely sheath accelerated ions from the rear. Recently, Higginson et al.[77] reported on a RIT-induced enhancement of RPA-TNSA hybrid acceleration.

RPA is known to be unstable to the RTI at the laser-solid interface[79]. Evidence for net-like modulated proton beams due to RPA and RTI seeded at the target front side was claimed by Palmer et al.[59] and Powell et al.[60] for experiments at the Vulcan laser. In both cases, RPA accelerated protons originating from the target front side are claimed to constitute the net-like modulated proton beam. The experiment from Palmer et al. was performed with linearly polarized pulses of 660 fs, laser intensity of around $5 \cdot 10^{20}$ W/cm^2 with plasma mirror cleaned laser contrast and diamond-like foil targets with thicknesses between 5 and 50 nm. Net-like beam modulations were found to be persistent in stacks of RCF and to be dependent on the target thickness. Strongest modulations were found for the minimal thickness of 5 nm and the degree of expression was found to decrease with increasing target thickness. Additionally, an observed anticorrelation between the intensity of the transmitted light and proton beam profile was interpreted as a selective channeling of the laser pulse through the RTI unstable surface resulting in a modulated ion density structure.
Powell et al. reported on a bubble-like proton beam modulation due to RPA and RTI in experiments with similar laser parameters and 40 nm Al targets. Remarkably, rotation of the target (p-polarization for $\theta = 30°$) revealed that the bubble-like modulated proton beam is directed at the laser propagation axis and could be separated from the TNSA accelerated proton beam directed to the target normal direction. Furthermore, Powell et al. reported on relativistic transparency during the rising edge of the laser contrast curve.

Transmitted light

Modulated proton beams have also been attributed to light transmitted through the target either by relativistic transparency or by laser light passing around targets of limited lateral extent. Gonzales-Izquierdo et al.[80] introduced that a partially transparent plasma may represent a relativistic plasma aperture during the rising edge of the laser contrast curve, which leads to diffraction of the transmitted light through 10 nm thin Al foil targets. The spatial structure of the accelerated electron beam is a result of the near-field diffraction pattern of the transmitted light and depends on the laser polarization and laser focal spot size. In a further simulation study[81] it was shown that for RPA accelerated protons from the target front side, streak-like proton beam structures can be generated for linearly or elliptically polarized light and net-like modulations (due to RPA and RTI) for circular polarized light. In the proposed mechanism, the spatial profile of the electron beam is mapped into the proton beam profile via modulation of the electric field[81].

For targets of limited width ($\lesssim 10$ µm) proton beam structuring may occur via the so-called all-optical imprinting mechanism discovered by Obst-Hübl et al.[58]. In this case, a fraction of the laser pulse passing around the target generates electric field structures in a low-density plasma behind the target via ionization of the residual rest gas in the vacuum chamber. Thereby spatial information from the laser near-field is imprinted into the proton beam profile. The modulated electric field structures extend to the low density plasma volume, where the laser is intense enough to ionize the residual gas. The mechanism enhances imprinting of higher frequencies of the laser near-field as the opaque target acts as a high frequency filter in the laser far field at the focal position.

2.4 Conclusions

More than 10 years after the NJP publication from Metzkes et al., a clear identification of one specific causing agent for the net-like proton beam modulation has not been possible. Other evidences of proton beam modulation related to TNSA by Scott and Göde et al. identify Weibel-like instability induced magnetic filaments in the rear side preplasma to be responsible for proton deflection. Yet, also the proposed mechanisms of RTI or Weibel-like instability seeding at the target front surface proposed by Metzkes-Ng et al. remain possible scenarios. Additionally, all mechanism may superpose in the experiment, which further complicates pinning down a single instability type. Moreover, with the advance of achievable laser intensities, hybrid acceleration schemes shift in focus and raise the additional question of the source location, i.e., whether modulated protons originate from the target front side by RPA, or from the rear side by TNSA, or if a hybrid acceleration scheme is realized. For ultrathin targets also relativistic transparency effects need to be considered in addition.
On the experimental side, the remaining questions require systematic studies to investigate the effect of different laser target conditions, in particular the influence of different pre-plasma conditions. Such experimental studies are important to achieve reproducibility of experimental evidences between different laser laboratories. Moreover, the uncertainties underline the need for XFEL probing techniques to access solid density plasma evolution in experiment to study and characterizes plasma instabilities.
On the simulation side, studies are required to understand the exact mechanism how transverse electromagnetic structures close to the source location at the rear side translate into proton beam structures on a detector at few centimeters distance. This is especially necessary as usually a virtual point source and the intercept theorem are applied to compare spatial wave numbers of electromagnetic field structures at the target or source location with those of proton beam modulations on the detector. The latter assumptions may turn out to be too strong simplifications, as both the spatial distribution and evolution of the magnetic field and of the accelerated proton source overlap and are highly transient.

3 Study of Transverse Proton Beam Modulations in Different Experimental Laser-Target Configurations at Draco

In this part of experimental work, different laser-target configurations have been tested at the Draco laser facility with the goal to find experimental settings, in which modulated proton beam profiles are generated in a stable manner. The experiments aimed to further investigate inconsistencies in the interpretation of measured proton beam modulations in recent years. Certainly, one of the origins for discrepancies in the literature arises from a lack of knowledge of the exact laser-target condition in particular during the interaction of the laser main pulse with the target. This topic has been intensely studied recently in parallel thesis projects in the ion acceleration division at HZDR[82, 83, 84].
As discussed in the previous section, proton beam modulations have been consistently linked to plasma instabilities and the generation of laser-induced filamentary structures in the electron plasma. Thus, experiments as carried out here serve on the one hand to investigate limits for scaling laws of LPA models and applications. On the other hand, they serve as testbed examples for future X-ray pump-probe experiments within the framework of HIBEF to study plasma instabilities at the EUXFEL. Accordingly, the experiments are closely linked to the SAXS program at HZDR, which aims to study plasma instability generation in foil targets and their consequences on ion acceleration. Thus, this work can be seen as a preparatory process for future SAXS proposals. Yet, the preliminary character of the presented studies needs to be emphasized as they raise further issues regarding the generation of proton beam modulations and provide answers only to some extent.

The following experimental studies have been performed (see Table 3.1): First, Ti foil tar-

gets have been tested under variation of the preplasma conditions realized by controlled application of prepulses onto the target front surface. These experiments are motivated by the experimental work of Metzkes-Ng et al.[1] and aimed to reproduce similar laser contrast conditions at the Draco PW laser system (Draco PW) as present in the experiments at the Draco 150 TW laser system, when at Draco net-like proton beam profiles were observed for the first time (cp. Sec. 2.3.2).
Secondly, proton beam profiles obtained from few hundred nm thick FV targets were investigated in detail. The impulse for this study arose from the empirical observation at Draco PW that the proton beams obtained with these targets featured modulated structures which reminded us of net-like profiles in the literature. Additionally, their origin was of interest, as no evidence of imprinting was found, no transmitted light has been observed and the modulations were directed in target normal direction. In this study, different laser-contrast settings have been tested and a target thickness and laser energy scan have been carried out. Thirdly, FV targets covered with a thin Al coating located either at the target front (towards the laser) or rear side have been tested to study the effect of changes in the atomic number of the front or rear side target material.
In the following, all measured proton beam profiles will be shown as raw RCF data and ad-

Study	Exp. settings	Target thickness
Controlled variation of the temporal laser contrast and preplasma conditions with Ti foils (see Sec. 3.1)	PM cleaned laser contrast & prepulses applied to the target front side, 12.1 J laser energy on target	2 um
Empirical study of various laser-target settings with FV foil targets (see Sec. 3.2.1)	inherent and PM cleaned contrast, variable pump laser energies, spectral phase settings and target thicknesses	30…400 nm
FV — target thickness scan (see Sec. 3.2.2)	Optimized laser contrast conditions (PM, spectral phase optimized), max. pump laser energy	30…900 nm
FV — Laser energy scan (see Sec. 3.2.3)	Optimized laser contrast conditions, fixed target thickness	240 nm
Al-coated FV targets (see Sec. 3.2.4)	Optimized laser contrast conditions, max. pump laser energy, Al coatings at the front or rear	FV: 120 nm & 300 nm, Al: ∼ 30 nm

Table 3.1: Overview of the performed experiments at the Draco PW laser system with regard to the generation of proton beam profile modulations. In all experiments the laser incidence angle was 45°.

ditionally in an image-processed data representation (see Fig. 3.1). In order to carve out the modulated structures of interest, the dose converted RCF data is shown in the representation D_{HP}/D_{LP}, where D_{HP} is a high pass filtered dose distribution, which is normalized by a low pass filtered dose distribution D_{LP} in order to obtain a relative measure of the degree of

redistribution within the proton beam.

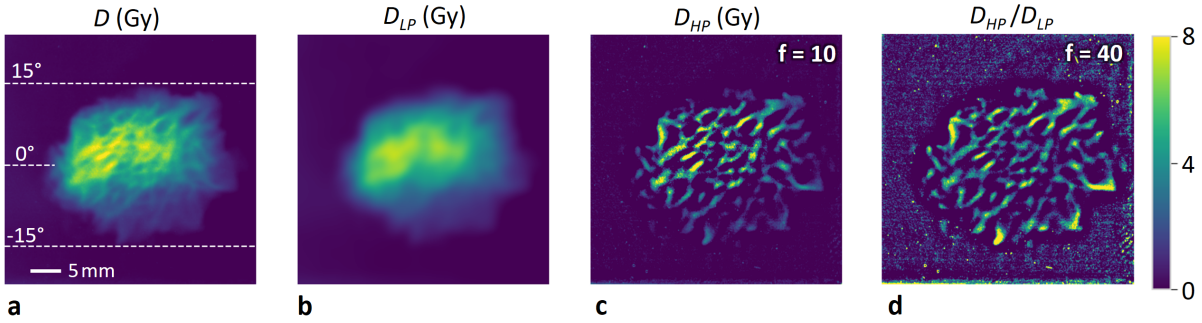

Figure 3.1: Data processing applied to highlight modulated structures in the proton beam profiles. **a)** Exemplary 2D Dose distribution, D, obtained from a 120 nm FV target measured on RCF (46 MeV layer). **b)** A low pass filtered dose distribution, D_{LP}, is obtained by applying a Gaussian Filter with a Kernel size of $\sigma = 10$ px ($\sim$ 1 mm). **c)** The high pass filtered dose distribution, D_{HP}, is obtained by subtraction of D_{LP} from D in order to carve out the modulated structures of interest. **d)** D_{HP} is normalized by D_{LP} to normalize the modulated structures by the dose in different angular ranges in the beam spot. The scaling factor f is applied on the data to adapt to the color bar.

3.1 Variation of preplasma conditions by controlled prepulse application in combination with Titanium foils

As described in detail in the previous section, Metzkes-Ng et al. observed net-like modulated proton beam profiles in experiments with few micrometer thick Titanium foils at the Draco 150 TW laser system. This evidence coincided with the measurement of additional prepulses on the few ps time scale before the main pulse arrival in combination with increased temporal intensity contrast during the last picosecond (see Fig. 2.2 a) after changes in the laser system had been made. The temporal laser contrast is composed of the ASE pedestal and prepulses. The ASE pedestal defines the base contrast level and is hard to control experimentally. Prepulses constitute a problem as they are known to be ambiguous in Sequoia contrast measurements, as prepulses may be generated by nonlinear coupling with post pulses. As this connection was not clear at that time, the existence of the prepulses measured by Metzkes-Ng et al. is not unambiguous. Therefore, the goal of the following experiments was to generate similar laser-target conditions with the Draco PW system as present in the experiments by Metzkes-Ng et al. and thus to reproduce the net-like proton beam modulations in an empirical approach.

The Draco PW laser system delivers p-polarized pulses of up to around 30 J in 30 fs before compression. After the plasma mirror, 18 J pulse energy are available on target in a focal spot characterized by a Gaussian distribution of around 2.6 um FWHM. This yields a peak intensity of around $4 \cdot 10^{21}$ W/cm^2 or $a_0 \sim 45$. The experiments at this intensity level are of additional

interest, as theory and numerical simulations predict a higher likelihood of instability generation with the increase in laser intensity[1, 47].
In order to vary the temporal laser contrast and thus the preplasma conditions in a controlled manner, we used a plasma mirror (PM) cleaned laser contrast in combination with additional prepulses applied to the front side of 2 µm Ti-targets. The prepulses were generated by insertion of a 1"-pick-off mirror (prepulse mirror) in front of the last guiding mirror (DD) before the off-axis-parabola (OAP) (see Fig. 3.2 a). The setup allowed to scan delays between the prepulse and main pulse between 0 and a few ten picoseconds. The maximal prepulse intensity level is around 4 orders lower than the main pulse intensity (see Fig. 3.2 b) and could be further attenuated by up to around 3 orders of magnitude by blocking a part of the prepulse mirror with an attenuation plate (see Fig. 3.2 c).
The inline prepulse setup used here leads to an increase of the prepulse focal spot size compared to the main pulse focus, as the prepulse illuminated area at the OAP is much smaller. The unattenuated prepulse has a FWHM spot size of around 20 µm. The prepulse focal spot size increases, when an increased region of the prepulse mirror is blocked by the attenuation plate, e.g. up to a FWHM of 100 µm at the attenuation plate position $x = 20$ mm.
The experiments from Metzkes-Ng et al. were carried out under 0° incidence angle. In the study at the PW system presented here, an incidence angle of 45° was chosen, as handling of back reflections in combination with PW experiments are at the technical limit. Additionally, oblique incidence results in an separation of the laser propagation and the target normal axes (particle acceleration axis in case of TNSA). Thus, this scheme is analogue to the oblique setting in the experiments by Powell et al.[60].

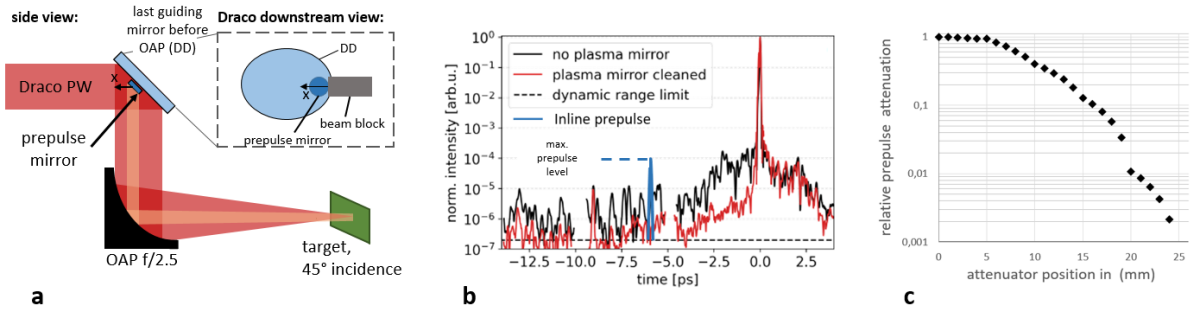

Figure 3.2: **a)** Side view of the inline prepulse setup. Inset shows the attenuation plate used for reduction of the prepulse intensity. **b)** Laser contrast curve on the ps time scale measured with single-shot time extended self-referenced spectral interferometry technique[85] (adopted from[82]). The inline prepulse and its maximal intensity level are indicated in blue color. **c)** Calibration curve of the relative prepulse intensity as a function of the position of the attenuator plate.

Firstly, the influence of different prepulse delays and attenuation levels on the acceleration performance, i.e., the cut-off energy, has been investigated by proton spectrum mea-

surements with a Thomson parabola spectrometer (TPS). In general, a clear impact of the additional prepulses can be noticed (see Fig. 3.3). For a delay of 30 ps one can observe decreased cut-off energies at full prepulse intensity (no prepulse attenuation) compared to the case of unperturbed laser contrast. With increasing prepulse attenuation, the acceleration performance successively increases until the average cut-off energy without application of prepulses is restored. For delays of 10 and 5 ps, there are several shots which exceeded the average acceleration performance without prepulse application and its statistical shot-to-shot fluctuation from that experiment day (1σ margin). This suggests the existence of an optimal front preplasma scale length allowing for increased laser absorption and hot electron generation and thus higher achievable cut-off energies. We expect that the optimal front preplasma is given by the simultaneous interplay of improved laser absorption and a reduction of the acceleration field gradient at the target rear in analogy to the mechanism described in [86]. We assume that this optimal front preplasma scale length is generated at delay-dependent prepulse intensity levels. This assumption is supported by the observable trend that a reduction of the prepulse delay requires a higher prepulse intensity, i.e. less prepulse attenuation, to reach this condition: While for 10 ps delay increased acceleration performance is reached at strong prepulse attenuation by 1 to 2 orders of magnitude (absolute prepulse intensities between around $4 \cdot 10^{15}$ and $4 \cdot 10^{16}$ W/cm^2), for 5 ps delay it is reached at less attenuation of only around 1 order of magnitude ($4 \cdot 10^{16}$ W/cm^2). At 2.5 ps delay, no improved acceleration performance can be observed. Presumably, in this case the prepulse arrives too early before the main pulse, so that one would need to apply a higher prepulse intensity than realizable here to generate the optimal preplasma scale length.

Next, the effect of the applied prepulses on the proton beam profiles has been investigated. For this part, RCF stacks measurements were performed (see Fig. 3.4) for prepulse delays of 10, 5 and 2.5 ps at the indicated prepulse attenuation levels in Fig. 3.3. These prepulse settings were chosen heuristically, motivated by the increased cut-off energies measured during the respective prepulse intensity scans.
From the raw RCF data (see Fig. 3.4 a), it can be generally ascertained that streak-like structures of different degree of expression are present in all RCF stack measurements. These structures are embedded in a rather smooth beam profile structure. The high pass filtered data (see Fig. 3.4 b) allows to analyze the high frequency structures in the beam profiles more clearly. In case of no prepulse application one observes well-visible streak structures with an apparent favorable direction parallel to the horizontal plane (polarization plane of the laser). While these structures are strongly pronounced for the measurement shown in the top row, they are less pronounced in the measurement shown in the second. This emphasizes the shot-to-shot fluctuation in the experiments and the individual character of each measurement. Yet, qualitatively the horizontal streaks are reproduced in the two measurements shown here. The measurements at 5 and 2.5 ps prepulse delay also reveal structures, yet of less degree of order compared to the measurements without prepulse application. At 10 ps

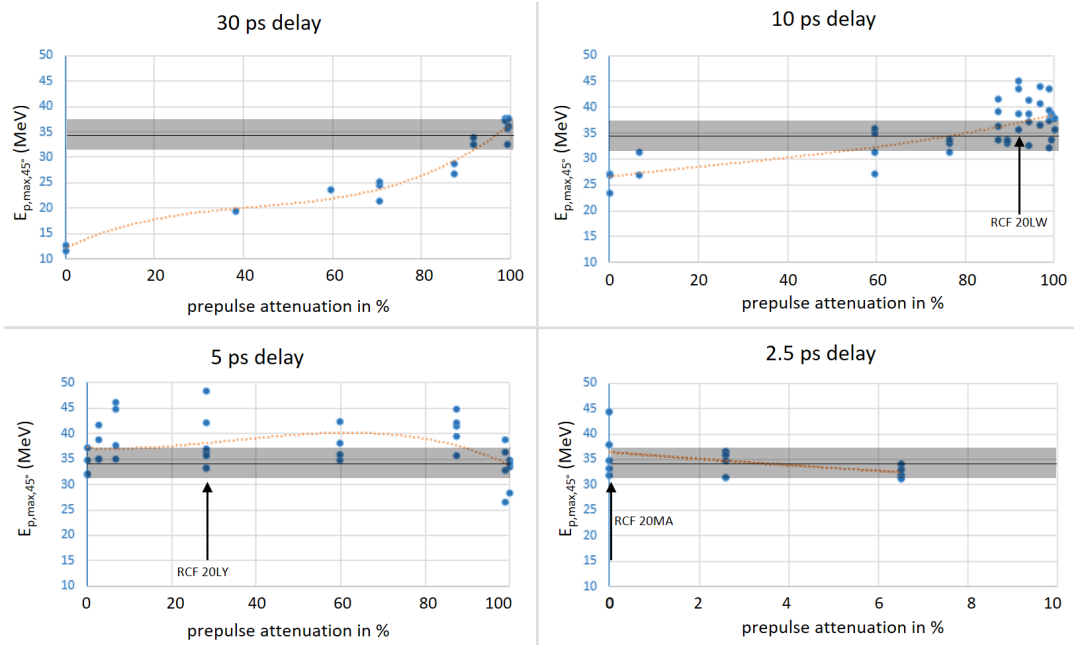

Figure 3.3: Proton cut-off energies (blue data points) in target normal direction for different prepulse delays and prepulse attenuation levels for a laser energy on target of 12.1 J. Orange line: Polynomial fit as a guide to the eye. Black solid line and gray shaded area: Average cut-off energies and standard deviation of shots without prepulse application.

delay the beam profile turns out to be the smoothest in this data set. As a general trend it seems that modulated structures in the proton beam profiles are most prominent for highest laser contrast, i.e., when no prepulses are applied.

This subjective impression is supported by a quantitative analysis of the modulation strength (see Fig. 3.19 described in Sec. 3.3). This analysis indicates a trend of decreasing modulation strength with increasing prepulse delay for the prepulse intensity levels chosen in these measurements. This suggests that a higher laser contrast is favorable for the generation of pronounced beam modulations. From this one might infer that also in the measurements by Metzkes-Ng et al. the net-like profiles coincided with an increased laser contrast at the Draco 150 TW system. With this regard, it could be that the ps prepulses present in the laser contrast measurement at that time were either measurement artifacts or otherwise real but were not a decisive factor in the generation of the transverse proton beam modulations. However, one needs to consider that the recent experiments were performed with the PW laser system with a plasma mirror, whereas the former experiments were performed at Draco 150 TW without a plasma mirror, which complicates the comparability. Generally, one may assume that due to the lower pump laser energy in the TW experiments, the absolute prepulse intensity was lower than in the recent PW experiments. In conclusion, modulated beam profiles were observed in all measurements, while the net-like modulations measured by Metzkes-Ng et al. could not be reproduced at the Draco PW laser system with the chosen prepulse settings.

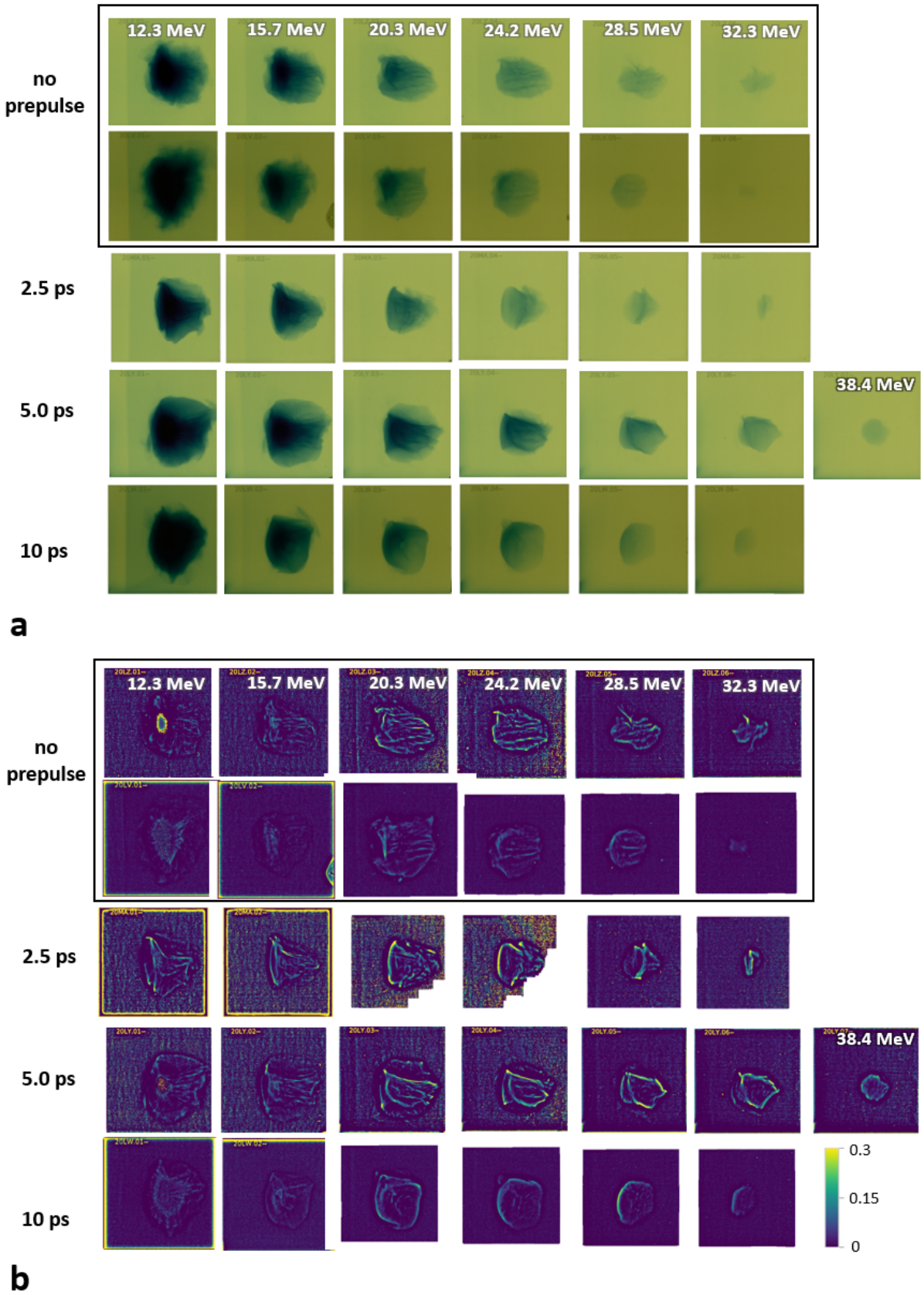

Figure 3.4: Proton beam profiles measured with RCF stacks for plasma mirror cleaned laser contrast (no prepulse) and additional prepulses at different delays relative to the main pulse arrival time. The prepulse attenuation level is indicated in Fig. 3.3. **a)** raw data, **b)** data in the representation D_{HP}/D_{LP} (cp. Fig. 3.1)

3.2 Formvar foil targets tested in different laser-target configurations

Formvar (FV) targets of few hundred nm thickness in combination with plasma mirror enhanced laser contrast have shown a high acceleration performance with respect to achievable cut-off energies and shot-to-shot stability in experiments at Draco. Thus, this target platform was chosen for a first in-vivo irradiation study with laser-accelerated protons performed at Draco[5]. In the course of the preparation process, different target thicknesses and laser contrast conditions have been tested at the Draco PW laser system. Highest acceleration performance has been observed for plasma mirror cleaned laser contrast and by manipulation of the laser pulse shape during the last picoseconds before the main pulse arrival via changes of the spectral phase[82]. During this optimization process, proton beam profiles were routinely measured with a Lanex screen (proton beam profiler), which provides an online diagnostic for the proton beam profiles for that part of the proton spectrum which is transmitted through absorber plates placed in front of the screen. With this detector the beam profile becomes accessible by imaging of the 2D intensity distribution of the emitted light from the scintillation screen (see Fig. 3.5 a). The proton beam profiler featured either a central pin hole or a horizontal slit, which allowed simultaneous measurements of proton spectra with a TPS.
In the generated data set, different types of beam profile features were observed. These include, e.g., rather *unordered* beam structures but also *streak-like* and in particular *net-like* structures (see Fig. 3.5 b). The net-like structures aroused interest as they reminded us of measurements of similar kind in the literature, which were linked to the generation of plasma instabilities as described in section 2.3. The objective was to identify experimental conditions that allow for a stable generation of these net-like structures and to test the dependence on different laser-target parameters. Certainly, this study requires a clear definition of what to classify as a net-like proton beam profile. In the following, the term net-like profile shall refer to beam profiles that contain several (i.e. more than 2) cell-like structures.
The study starts with an heuristic approach to investigate the influence of different laser target configurations on the generation of net-like proton beam modulations using an existent data set of measurements with the proton beam profiler. Based on these heuristic observations, further experiments have been performed at optimized laser contrast settings that yielded the highest protons energies, i.e., with plasma mirror cleaned contrast and spectral phase optimized laser pulse shape. These experiments include a scan of different FV thicknesses, a laser energy scan, and experiments with Al-coated FV targets. For all experimental parts, proton beam profiles have been measured with stacks of RCF and a (segmented) proton beam profiler. Since the RCF data represents dose-calibrated measurements of high spatial resolution, this data has been used to quantify the modulation strength in the proton beam profiles complementary to the qualitative and phenomenological discussion.

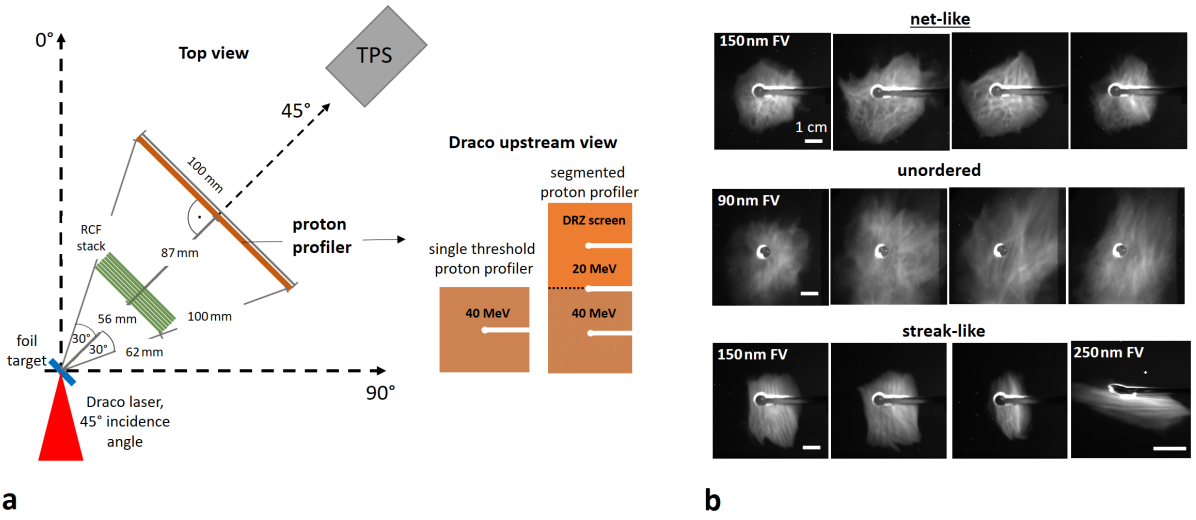

Figure 3.5: **a)** Experimental setup for online proton beam profile measurements with a Lanex scintillator screen (proton beam profiler). A single threshold proton beam profiler featured absorber plates stopping protons below 40 MeV. A segmented proton beam profiler featured an additional area with a threshold of 20 MeV. Either a central pinhole or slits in the proton beam profilers allowed for simultaneous proton spectra measurements with a Thomson parabola spectrometer. **b)** Examples of different beam profile types (all cases with PM cleaned laser contrast). Different profile types were heuristically classified by the terms *net-like*, *unordered* and *streak-like*. The statistical analysis was focused on the generation of net-like structures. Several examples obtained from same shot days, i.e., comparable laser-target settings, are shown here to emphasize the impression of similarity within the generated proton beam modulations.

3.2.1 Empirical analysis of different laser-target configurations

In a first, heuristic approach, an already present data set of around 500 laser shots on FV targets has been reviewed visually with regard to the occurrence of net-like proton beam profiles. The data set includes FV thicknesses between 30 and 400 nm, shots with inherent or plasma mirror cleaned laser contrast and different spectral phase settings. For this data set, the proton beam profiler featured absorber plates stopping protons with kinetic energies below 40 MeV. The data has been further reduced to 300 shots with $E_p > 50$ MeV to filter only strong signals on the detector allowing for best possible identification of the caustic structures of interest. This filter implicitly reduces the data set to shots with maximal laser energy on target of around 18 J. Furthermore, only shots at optimized longitudinal focus position were considered.

Among the 204 shots remaining after filtering, 106 shots were classified to feature net-like structures. The first important observation is that all of those net-like beam profiles were obtained in the plasma mirror setup, which suggests a great relevance of high laser contrast for their generation. While inherent laser contrast favors the generation of rather unconfined or diffuse beam profiles (Fig. 3.6 a,b), PM cleaned laser contrast results in more confined beam

spots and seems to be a requirement for the generation of net-like structures. Indeed, for one experiment day on which around 140 shots have been performed with inherent laser contrast, net-like structure could only be observed for those two consecutive shots for which the plasma mirror was used (see Fig. 3.6 c, d).

The experimental observation raises the question whether the generation of net-like struc-

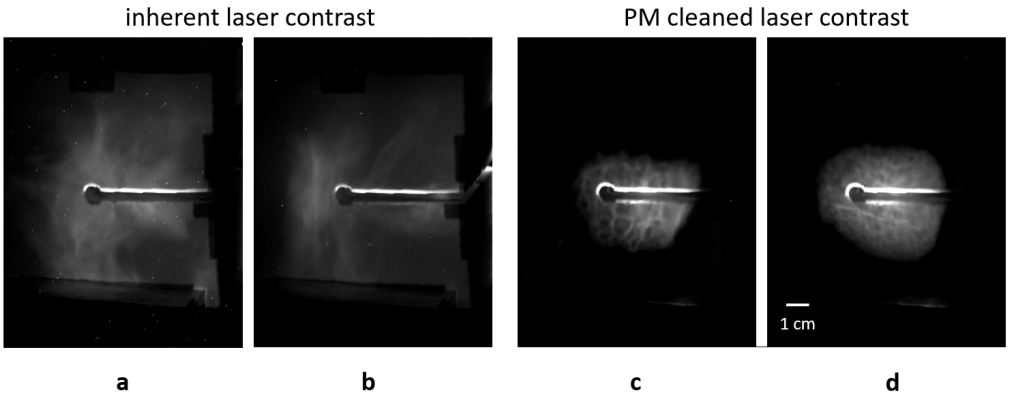

Figure 3.6: Single shot evidence for the relevance of a PM cleaned laser contrast for the generation of net-like beam profiles. **a, b)** Inherent laser contrast resulted in rather unordered or diffuse beam profiles. The examples were chosen to show a case of a FV thickness above (**a**, 375 nm) and below (**b**, 110 nm) the onset of transparency. **c, d)** Net-like profiles were generated promptly after implementation of the plasma mirror in two consecutive shots at same spectral phase settings for a FV thickness of 130 nm. Note: For these shots an additional OD1 filter was used in front of the Lanex imaging camera.

tures goes along with light being transmitted through the target, i.e., whether the findings coincide with a regime of transparency of the target. This question is relevant with regard to the imprinting mechanisms suggested by Gonzales-Izquierdo et al.[80, 81] or Obst-Hübl et al.[58] (see Sec. 2.3.5) for which the transmitted light leads to proton beam modulations via diffraction of the transmitted light through a partially transparent plasma aperture or via ionization of the residual rest gas behind the target, respectively. With regard to target transparency, the laser contrast plays a crucial role as it defines the thickness for the onset of light being transmitted through the target (see Fig. 3.7). Analysis of the transmitted light suggests an onset of the transparency regime for a FV thickness below $\sim$ 230 nm for inherent and $\sim$ 60 nm for PM cleaned laser contrast. Net-like beam profiles, however, were measured routinely for target thicknesses above the onset of transparency.

To investigate a potential thickness dependence in the generation of net-like beam profiles, their statistical occurrence rate, i.e., their rate of visibility on the proton beam profiler, was analyzed as a function of the target thickness for the available data set (see Fig. 3.8). One can observe that the regime of target transparency ($\lesssim$ 60 nm) is less favorable for the occurrence of net-like beam profiles than thicknesses above this regime at around 200 nm. Yet, there is no apparent limitation for the occurrence of net-like beam profiles to a specific thickness range. It is important to note that the data set contains various different laser spectral phase

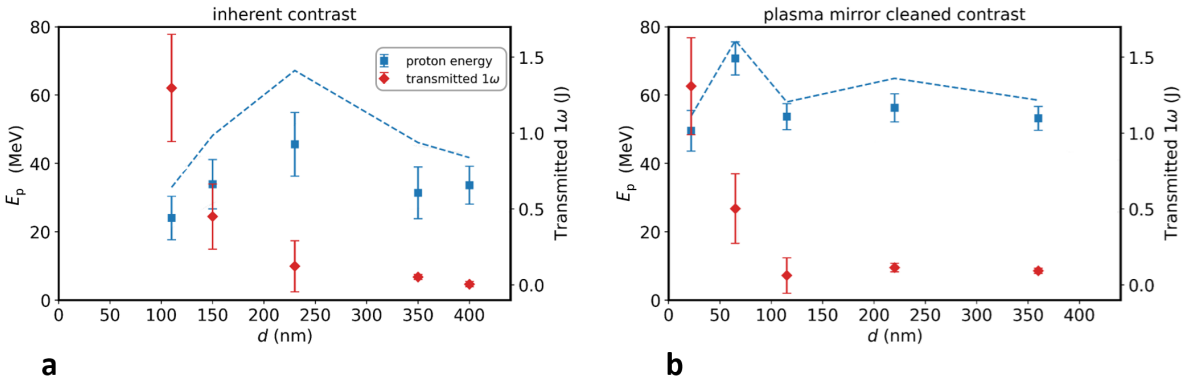

Figure 3.7: Transmitted light (red data points) measured in laser forward direction for inherent (a) and plasma mirror cleaned contrast (b) [82, 87]. The onset of light transmission can be observed below around 230 nm for inherent laser contrast and around 60 nm for PM cleaned contrast.

settings, which complicates the comparability of data obtained from different thicknesses. Yet, for the applied data filtering, the considered shots show comparable average cut-off energies around 55 and 60 MeV for different thickness bins.

A separated analysis for the subsets of data that feature or do not feature net-like structures[1] (see red and blue data points in Fig. 3.8), does not reveal differences in the average cut-off energies . However, one needs to emphasize that the absorber threshold of the proton beam profiler gives only information on the beam profile of the spectral part above 40 MeV. Also, the chosen filters applied for this statistical analysis represent a strong bias.

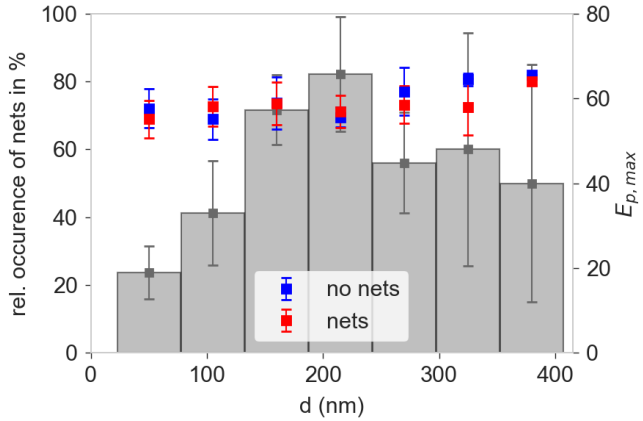

Figure 3.8: Histogram of the occurrence rate of net-like beam profiles for different FV thicknesses (including 1σ-uncertainty). Red and blue data points show the average proton cut-off energies for the subsets that do or do not feature net-structures.

[1] Note: Only shots with E_p > 50 MeV have been considered in this analysis.

3.2.2 FV targets – Thickness scan

A scan of different FV target thicknesses in the range between 30 and 900 nm has been performed to study the thickness dependence on beam profile modulations at optimized laser parameters. This study aims to provide information on whether the generation of net-like profiles correlates with a specific acceleration scheme beyond TNSA. For example, in case of proposed RPA-TNSA hybrid acceleration schemes, it is expected that spatio-temporal coupling conditions between foil thickness, laser intensity, and pulse duration need to be fulfilled, so that RPA front-side accelerated protons reach the TNSA rear sheath field[76].

RCF stack measurements have been performed in order to provide information on the proton beam profiles in the complete spectral range. Fig. 3.9 shows a representative subset of proton beam profile measurements for the FV thickness scan.

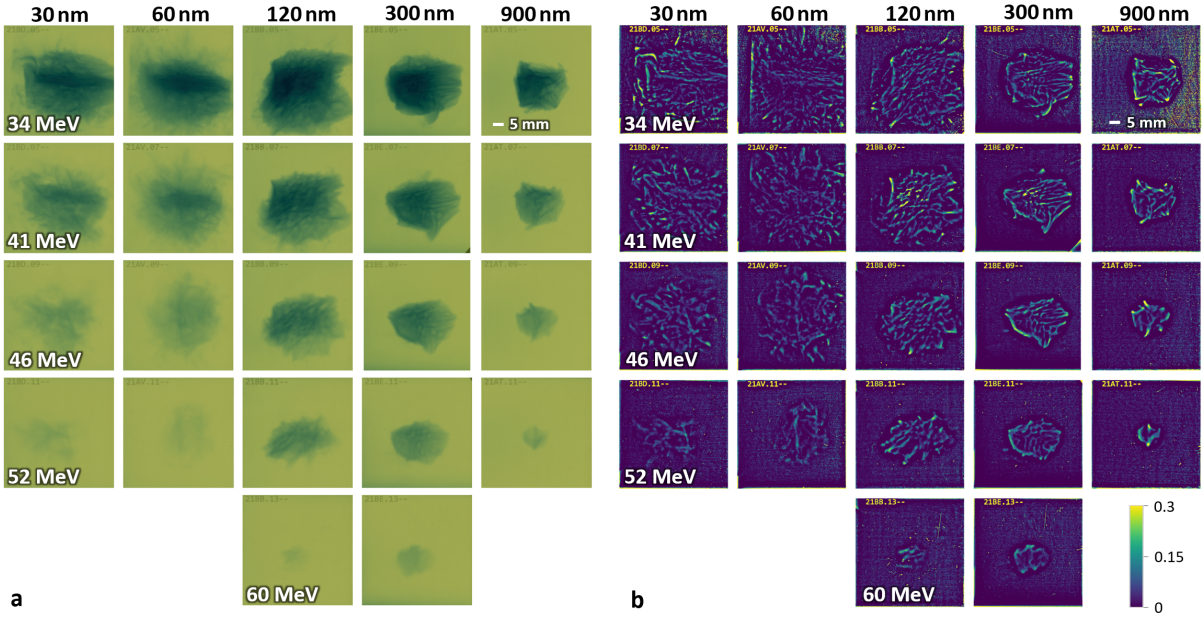

Figure 3.9: FV – thickness scan. Proton beam profiles were measured at PM-cleaned laser contrast and TOD/GVD-optimized pulse shape. **a)** RCF raw data. **b)** Data shown in the representation D_{HP}/D_{LP} (cp. Fig. 3.1). Energy labels refer to the Bragg-peak energies corresponding to the RCF layers. Data from 30 and 60 nm FV targets is linked to the regime of RIT, since the onset of light transmission through the target has been observed for $d \lesssim 60$ nm (see Fig. 3.7).

Observations

Several general observations can be made:

1. Modulated beam structures can be observed in the whole experimental thickness range.
2. The modulated structures extend over the whole beam spot area and are present on all irradiated layers of the RCF stacks.
3. The beam spot size decreases with increasing depth of the RCF layers in the stack, i.e.,

for higher proton energies (see Fig. 3.10). In combination with observation 2, this implies that modulated structures exist in the whole proton beam spectrum. Additionally, the beam divergence decreases with increasing foil thickness.

4. The measurements for target thicknesses of 120 and 300 nm feature closed and not completely closed cells, i.e., net-like structures, and streak structures. They are prominently visible at the 41, 46, and 52 MeV layers.
5. The beam profiles for 30 and 60 nm target thickness are less confined compared to the measurements at higher target thickness. In other words, the beam spot is more diffuse or less ordered. These measurements stand out from the others as they are clearly in the transparency regime.

Preliminary conclusions

To draw a few preliminary conclusions, no thickness dependence for the generation of arbitrarily modulated structures in the proton beam profiles can be observed. Additionally, no angular shift between the smooth and structured parts of the proton beam profiles is found, in contrary to the experiments by Powell et al.[60] at oblique laser incidence (cp. Fig. 2.1). While no thickness dependence on the mere generation of arbitrary structures is found, there seems to be a favorable thickness range for a pronounced visibility of net-like structures, i.e. closed cell structures. In this data set this is the case for target thicknesses of 120 and 300 nm. This observation is consistent with the statistical analysis of the occurrence rate of net-like beam profiles based on the Lanex data set shown in Fig. 3.8.
Furthermore, the behavior of beam divergence in the data set is consistent with TNSA as it features target normal emission and decreasing divergence angles with increasing depth within the RCF stack, i.e., with increasing proton energies (see Fig. 3.10). Finally, the single shot character of these measurements needs to be emphasized. One can not expect to reproduce these measurements in the sense of being able to generate identical beam profiles. Yet, from the proton beam profiler data, one finds an indication of similar beam profile types being generated for shots at same laser-target settings (cp. Fig. 3.5 b).

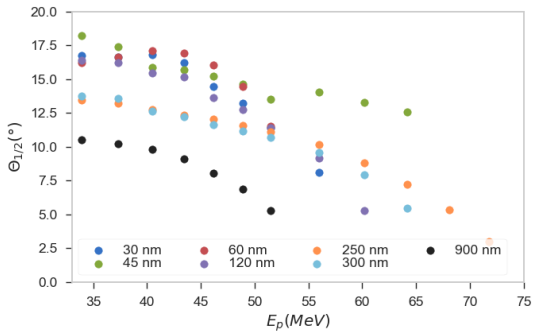

Figure 3.10: Divergence angle $\Theta/2$ (half angular aperture) for the FV thickness scan RCF data (see Fig.3.9) obtained by circular approximation of the beam spot contour. The decrease of the divergence angle with proton energy typical for TNSA can be confirmed.

3.2.3 FV targets – Laser energy scan

This experimental study aimed to investigate a potential laser intensity dependence on the generation of net-like beam profile modulations by a variation of the pump laser energy at otherwise same experimental settings as in the thickness scan. Here, one needs to consider that a variation of the pump laser energy may also alter the temporal laser contrast due to the trigger of the plasma mirror, which may thus have an additional impact on the intensity on target. The laser energy on target, E_L, for different pump laser energies, $E_{L,p}$, was derived from a reflectivity curve measurement of the plasma mirror[2] under the assumption of a constant compressor transmission of 70 %. The effective transmission of the compressor and plasma mirror is between 56 % (at maximal pump laser energy of $E_{L,p} = 31.9$ J) and 49 % at $E_{L,p} = 7.9$ J.

FV targets of 240 nm thickness are tested in the laser energy scan as in the FV thickness scan net-like beam profile structures were prominently visible for targets around this thickness. Proton beam profile measurements were performed with the segmented proton beam profiler (see Fig. 3.11).

While it had to be expected that the cut-off energies increase with higher laser pulse energies, the influence on beam profile modulations had been unclear. On the one hand, it is known from numerical simulations by Kluge et al. that seeding of a RTI instability is increasingly favorable for higher laser intensities[1]. With this regard, the scan of pump laser energies also aimed to create a link to the former Draco 150 TW experiments by Metzkes-Ng et al.. Simulations performed by Branco on the other hand (see Sec. 2.3) predict a generation of filamented magnetic field structures in rear preplasma scale lengths in a wide range of laser intensities.

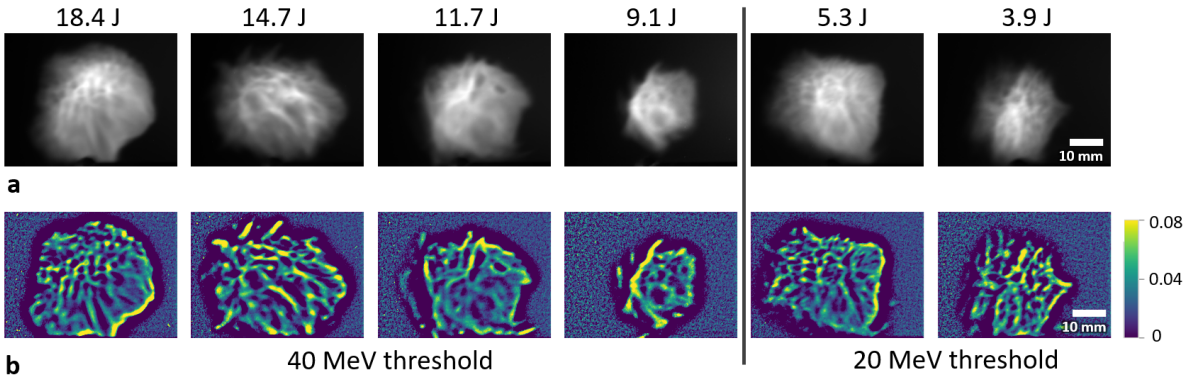

Figure 3.11: FV foil targets of 240 nm thickness – laser energy scan. Proton beam profiles measured with the segmented proton beam profiler at different laser pulse energies on target, E_L, at PM cleaned laser contrast and TOD/GVD-optimized pulse shape. Absorber threshold is 40 MeV for $E_L \geq 9.1$ J, otherwise 20 MeV. **a)** raw data, **b)** data filtered in analogy to Fig. 3.9 b, yet the data was not dose converted.)

[2] The measurement has been performed by H-P. Schlenvoigt. E_L in dependence on $E_{L,p}$ was fitted by $E_L = 7.9 \cdot 10^{-5} E_{L,p}^3 + 4.6 \cdot 10^{-4} E_{L,p}^2 + 0.5 E_{L,p}$.

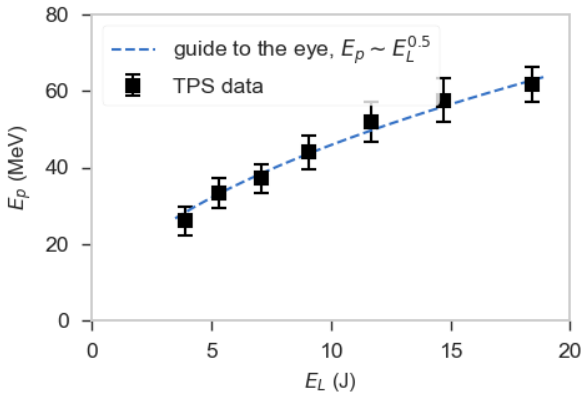

Figure 3.12: Proton cut-off energies obtained from 240 nm FV measured in the laser energy scan. The proton cut-off energy correspond well to a square root scaling with the laser energy on target.

Observations

Most importantly, for the present target thickness of 240 nm, net-like beam modulations are visible in the whole range of laser energies on target from 18.4 down to 3.9 J ($45 \geq a_0 \geq 20$). The cut-off energies show the expected increase for higher laser energy (see Fig. 3.12). This behavior is consistent with theoretical power scaling laws for TNSA with $E_p \sim E_L^{\zeta}$ with ζ between 0.5 and 1 discussed for ultrashort pulse laser systems such as Draco[29, 30, 88].
The beam divergence decreases with reduced pump laser energy. This behavior can be explained in the scope of the TNSA model: By a reduction of the laser energy, the cut-off energy is reduced and thus, the measurement point at the fixed absorber threshold is shifted further towards the higher energetic part of the proton spectra, which is known to feature lower beam divergence. The jump in beam divergence for the measurement at 10.7 J is explained by the switch from the 40 MeV to the 20 MeV absorber threshold of the segmented proton beam profiler.

Preliminary conclusion

These experiments do not reveal a dependence of the laser energy on the generation of the net-like proton beam modulations for the chosen target thickness and experimental range of laser energies. This observation is quite remarkable as it demonstrates the robustness of the underlying beam modulation mechanism in this regime. It is worth mentioning that the multiparametric simulation study from Branco identified a robust generation of filamented magnetic fields in target rear preplasma, which could cause such proton beam modulations, in a similar range of laser intensities (see Sec. 2.3.4).

3.2.4 Al-coated FV targets

With the goal to study the role of the source location in the generation of beam profile modulations, FV targets coated with a 30 nm-thick Aluminium layer either at the target front side (directed towards the laser) or rear side have been tested. The thin Al layer of higher atomic number and density ($Z = 22, n_e = 445\, n_c$) allowed to introduce an asymmetry in the target configuration and is expected to alter the plasma dynamics at the front or rear side, which may affect the proton beam emission and acceleration process. One would expect an Al coating at the target front side to affect laser absorption and hole boring, and an Al coating at the rear to affect rear sheath acceleration. For a pure FV target, we assume that bulk protons can continuously fill up the volume close to the rear surface and thereby replace expanding protons during the intrapulse time scale. The volume close to the rear surface is of particular relevance, as the highest accelerating field strength is expected there. If an Al layer is located at the rear surface, bulk proton acceleration would be inhibited and mainly protons from the pre-expanded contaminant layer would constitute the proton source.

Observations

By looking at the cut-off energies for the different target configurations, it becomes immediately evident that the acceleration performance is comparable for pure FV targets and front side coated targets, while Al coating at the rear side leads to a reduction of the cut-off energies of up to 30 % (see Fig. 3.13). This trend is consistently observable within the experimental thickness range between 120 nm and 1.5 µm. In Fig. 3.14 and 3.15 RCF and Lanex measurements are put together for 120 and 300 nm thicknesses, respectively. These target thicknesses were of interest, as they fell within the range of target thicknesses for which net-like structures were prominently observed in the FV thickness scan.

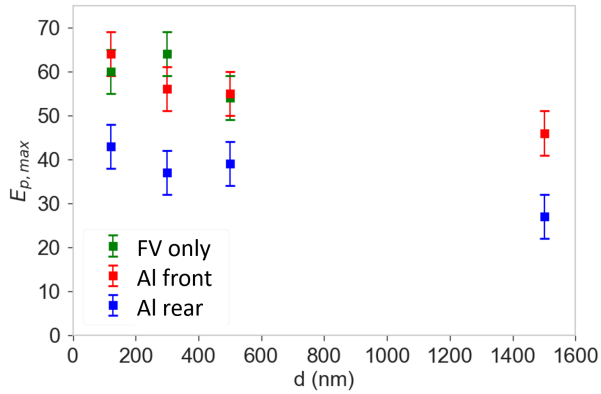

Figure 3.13: Cut-off energies for pure FV and Al-coated FV targets at the front or rear. While Al coating at the front leads to similar cut-off energy as uncoated FV, Al coating at the rear leads to strongly reduced cut-off energy.

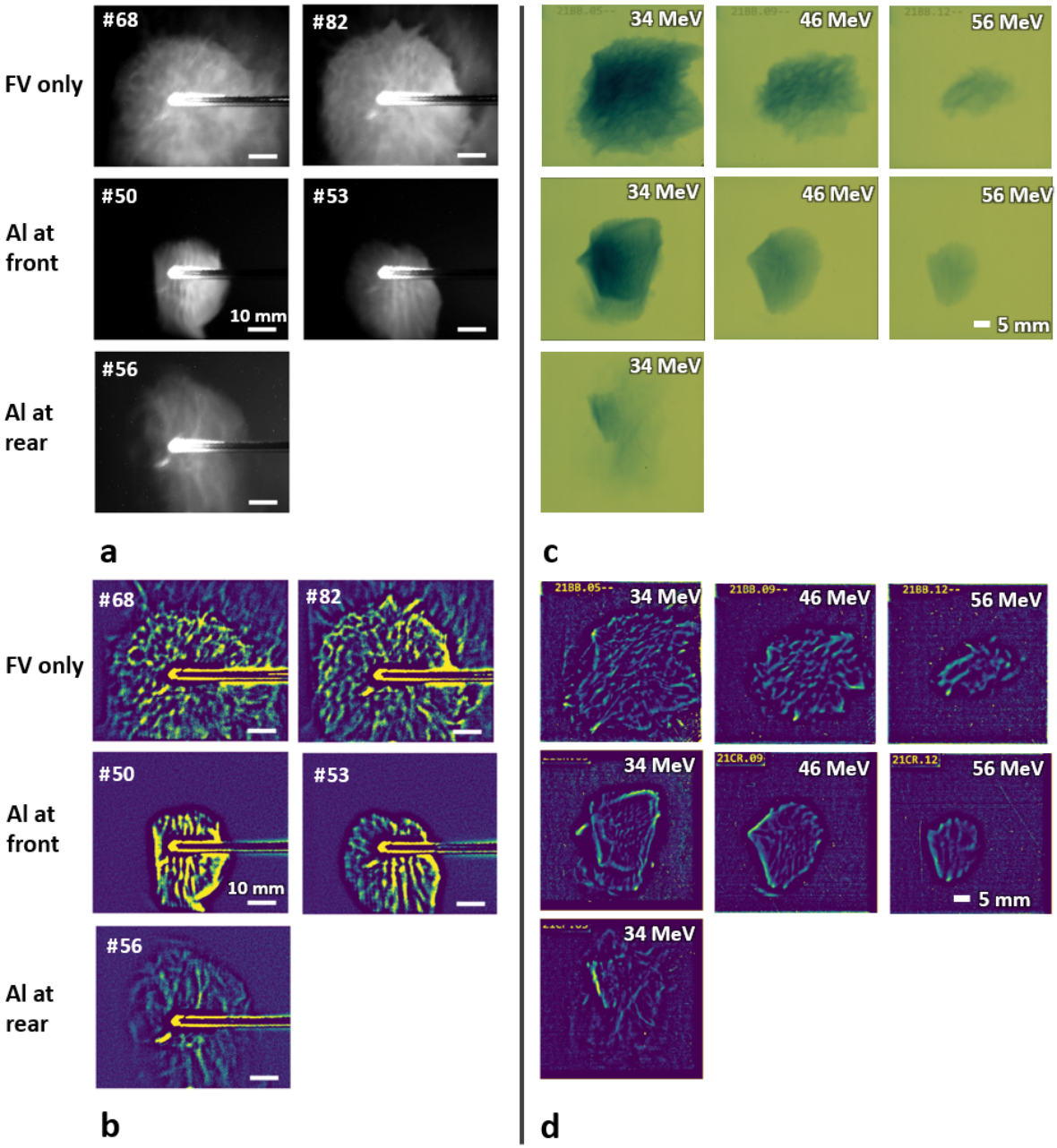

Figure 3.14: Proton beam profile comparison for pure and Al-coated FV targets of 120 nm thickness. a, b) Lanex data c, d) RCF data. a, c) raw data. b, d) filtered data (cp. Fig. 3.1). Note: The horizontal feature in the center of the proton beam profiler is a slit, which allowed simultaneous measurements of proton spectra with a TPS.

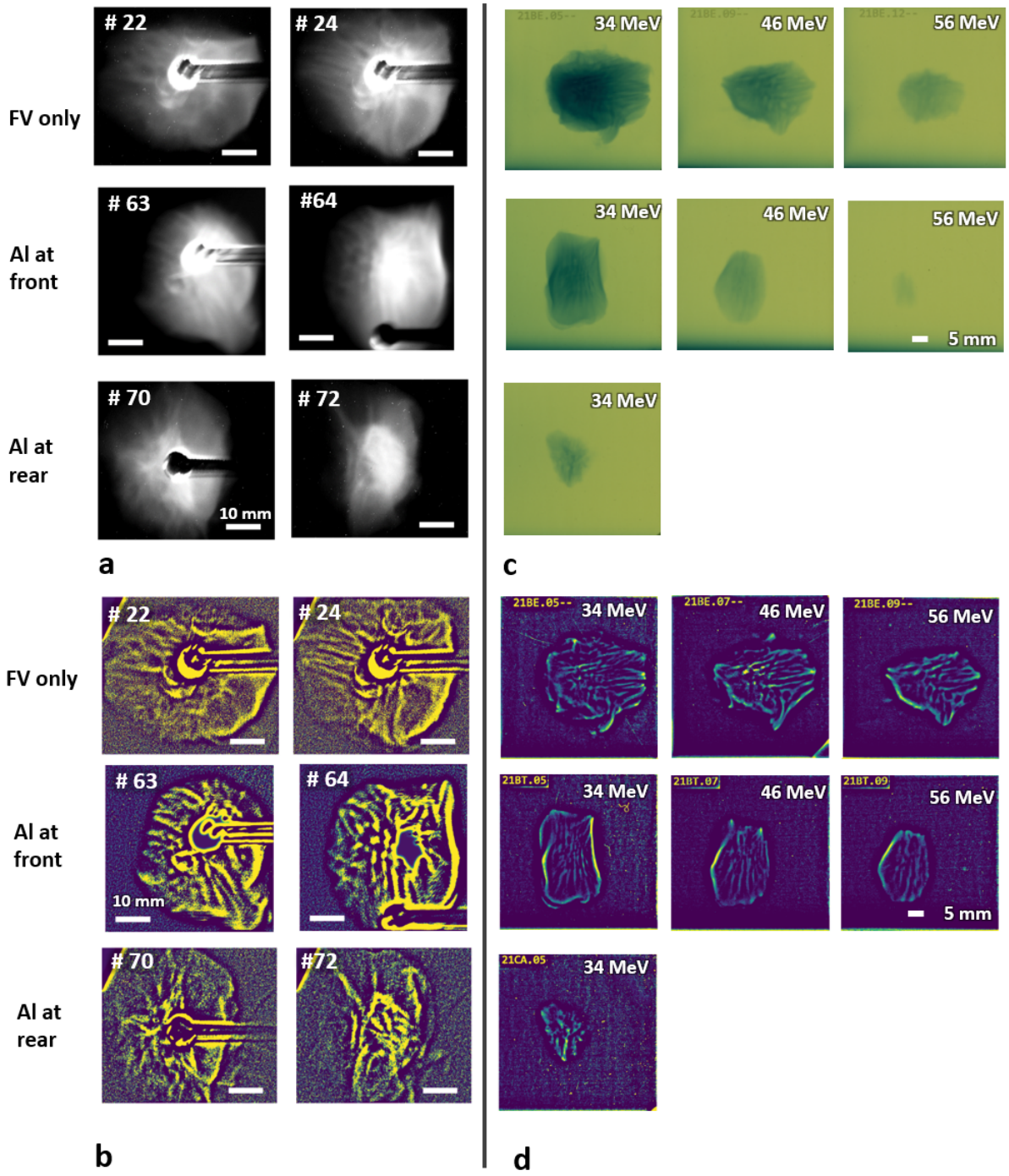

Figure 3.15: Beam profile comparison for pure and Al-coated FV targets of 300 nm thickness. a, b) Lanex data c, d) RCF data a, c) raw data. b, d) filtered data (cp. Fig. 3.1). Note: The horizontal feature in the center of the proton beam profiler is a slit, which allowed simultaneous measurements of proton spectra with a TPS.

Based on the (limited) data set, there are indications for several trends: First, beam profiles obtained from FV targets with an Al layer at the front side are smaller than those obtained from pure FV targets, while the cut-off energies are similar. Second, similarities can be ascertained between beam profiles generated in same target configurations with regard to the beam spot shape and prominently visible modulated structures. For example, by looking at Fig. 3.14, the beam profiles from pure FV targets would rather fall within the class of net-like profiles. For the beam profiles from FV with Al at the front on the other hand, vertical streak structures are prominent. Yet, it is important to note that such vertical streak structures have also been observed for pure FV targets (cp. Fig. 3.5 b). Considering the RCF measurements, it can be observed that while pure FV targets generated rather circular beam spots, FV targets with Al layer at the front generated rather elliptical profiles with the major axis directed in the vertical direction.
Third, FV targets with Al coating at the rear generated rather diffuse or unordered beam profiles. Additionally, they showed strongly reduced signals on the detector, which goes along with the reduced cut-off energies.

Preliminary conclusions

The introduction of the asymmetric target configuration has shown a significant impact on the acceleration performance. While an Al layer present at the front side of FV targets resulted in comparable cut-off energies compared to pure FV targets, an Al layer at the rear led to a break down of the acceleration process. The latter effect is consistent with the initial hypothesis of inhibited bulk proton acceleration, which means highest potential fields are not being explored. Another influence of the Al layer could be related to an increased rear side scale length. It is known that Al ionizes earlier than FV. One may assume that during the early stage of laser-target interaction, the laser penetrates through the underdense FV layer and is absorbed first at the critical density surface at the earlier ionized Al layer at the rear side. This could lead to an increased rear plasma scale length and additional reduction of the field gradients. At same laser settings the Al layer present at the front side induced qualitative changes in the beam profiles compared to the case of pure FV targets.

Analysis of spatial frequencies in the modulated beam profiles

Triggered by the subjective impression of differences in the spatial frequencies in the beam profile modulations from pure compared to front side coated FV targets, a quantitative analysis was performed for the cases of 120 and 300 nm target thickness. T. Kluge et al. suggested a parametric decay of plasma oscillations to be the seed for an electron density fluctuation at the front surface and Rayleigh Taylor instability growth[42]. In the numerical simulations it has further been observed that subsequently generated electron filaments reach the target rear side and modulate the ion density at the target rear side at the beginning of the acceleration process. In this theoretical frame work, the impact of the Al layer at the front side is given by the higher material density (Al: 445 n_c, FV: 230 n_c) and thus by increasing the plasma frequency at the target front side. Accordingly, when full ionization in both cases is assumed, the dominant spatial scale of the front surface modulation is relatively higher for the Al-coated targets by a factor of $\frac{\omega_{p,Al}}{\omega_{p,FV}} = \sqrt{\frac{n_{e,Al}}{n_{e,FV}}} \approx 1.4$. Thus, this analysis aims to study whether the Al layer at the front side correlates with higher spatial frequencies being present in the modulated proton beam structures.

In the following, quantitative analysis (see Fig. 3.16), the beam profiles at the 46-MeV RCF layer have been evaluated as for these layers, prominent beam modulations could be observed. First, the power spectral density (PSD) of contrast-optimized beam profiles has been calculated. It becomes apparent that the PSDs show an anisotropy with highest spatial frequencies being directed towards a certain angular range (cp. red, solid lines in Fig. 3.16 c). This correlates with caustic structures being mostly non-radially symmetric and stretched towards the orthogonal direction, in some parts of the profiles even forming directed streaks. Also, caustics of different sizes are present in the beam profiles so that a dominant spatial frequency in form of a Fourier correlation ring in the PSDs could not be observed.

As an alternative approach, the angular range of highest frequencies has been assessed by evaluation of the average peak distances in comb-like distributed profile lines on the beam profiles. The peak finding algorithm is based on the peak prominence parameter that was chosen such that the visible structures in Fig. 3.16 a are detected. The comb–like peak-finding analysis along the angular directions of highest spatial wave numbers (enclosed by the red lines in Fig. 3.16 b) confirms the subjective impression of smaller average structure sizes, μ, being present in case of the Al-coated targets for both analyzed thicknesses. The respective ratios are given by $\frac{\mu_{\mathrm{Al,300nm}}}{\mu_{\mathrm{FV,300nm}}} = 1.18 \pm 0.08$ and $\frac{\mu_{\mathrm{Al,120nm}}}{\mu_{\mathrm{FV,120nm}}} = 1.39 \pm 0.09$.

At the same time, we observe significant differences in the beam spot dimensions, d, measured along the angular directions analyzed in the peak-finding analysis. We find that the beam spots obtained from pure FV targets are bigger. Note, the beam spots only seem of similar size in Fig. 3.16 due to the different image sections displayed. For this, the respective ratios are $\frac{d_{\mathrm{FV,300nm}}}{d_{\mathrm{Al,300nm}}} = \frac{(20 \pm 4)\,\mathrm{mm}}{(16 \pm 2)\,\mathrm{mm}} = 1.25 \pm 0.29$ and $\frac{d_{\mathrm{FV,120nm}}}{d_{\mathrm{Al,120nm}}} = \frac{(28 \pm 2)\,\mathrm{mm}}{(20 \pm 2)\,\mathrm{mm}} = 1.40 \pm 0.17$. In consequence, we find that the absolute number of modulation periods or the average structure size normalized by the beam spot area are equal for both target configurations within the

error margins of the analysis.
Fig. 3.17 illustrates two examples, in which the interplay of source size, beam divergence and ion modulation wavenumber at the target rear result in the same amount of observable modulation periods in the beam spot, or in other words same caustic sizes normalized by the beam spot size. Thus, from the observation of higher spatial wavenumbers obtained from the Al-coated targets, one cannot make a direct link to the ion modulation wavenumber at the source location. In consequence, there is a need to measure the source size in future experiments. One approach in doing so could be a repetition of the experiments, i.e., with targets of different Z material at the front side, while adding a known, periodic rear surface topology in analogy to the experiments by Cowan et al.[66]. Given sheath acceleration of laminar type, e.g., a sinusoidal rear surface structure would translate into an additional beating pattern in the proton beam profile and would thus allow to deduce the source size from the amount of observable periods in the beating pattern.

In conclusion, the visual impression of higher spatial frequencies present in the beam profiles from Al-coated targets at the front side could be confirmed. However, the validity of this result is limited: While here only two measurements are considered, the shot-to-shot fluctuations in the measurements requires more statistics to validate the theoretical hypothesis from above. Second, the analysis itself turned out to be extremely challenging, since modulated structures of different degree of expression were found in the beam profiles. Accordingly, results from simple peak-finding algorithms vary with the chosen value of peak prominence, which defines which peaks are counted. Third, the considerations only apply, if the assumption of a virtual point source is valid and the intercept theorem is applicable for the proton beam propagation from the source location to the detector plane. However, when transiently evolving magnetic filament structures cause proton beam deflection, the situation is by far more complex. In this case, the initial ion density distribution at the source location *and* the spatial distribution and evolution of the magnetic filament structure need to be considered. Also, the magnetic rigidity of protons of different momenta needs to be taken into account as it defines the proton deflection: Particles of different momenta contribute to the dose distribution on a considered RCF layer. When TNSA-like acceleration is assumed, the central part of the beam spot constitutes protons of higher momenta (higher magnetic rigidity), while the exterior part constitutes protons of lower momenta (lower magnetic rigidity). Thus, for a periodic magnetic field structure of constant amplitudes along the lateral target dimension, one would expect caustics of increasing sizes towards the exterior parts of the beam spot. In consequence, one would need to evaluate the size of caustics in dependence of the radial distance to a symmetry center of the beam spot. However, in the measurements analyzed here, a clear symmetry center in the beam profiles was not observed.

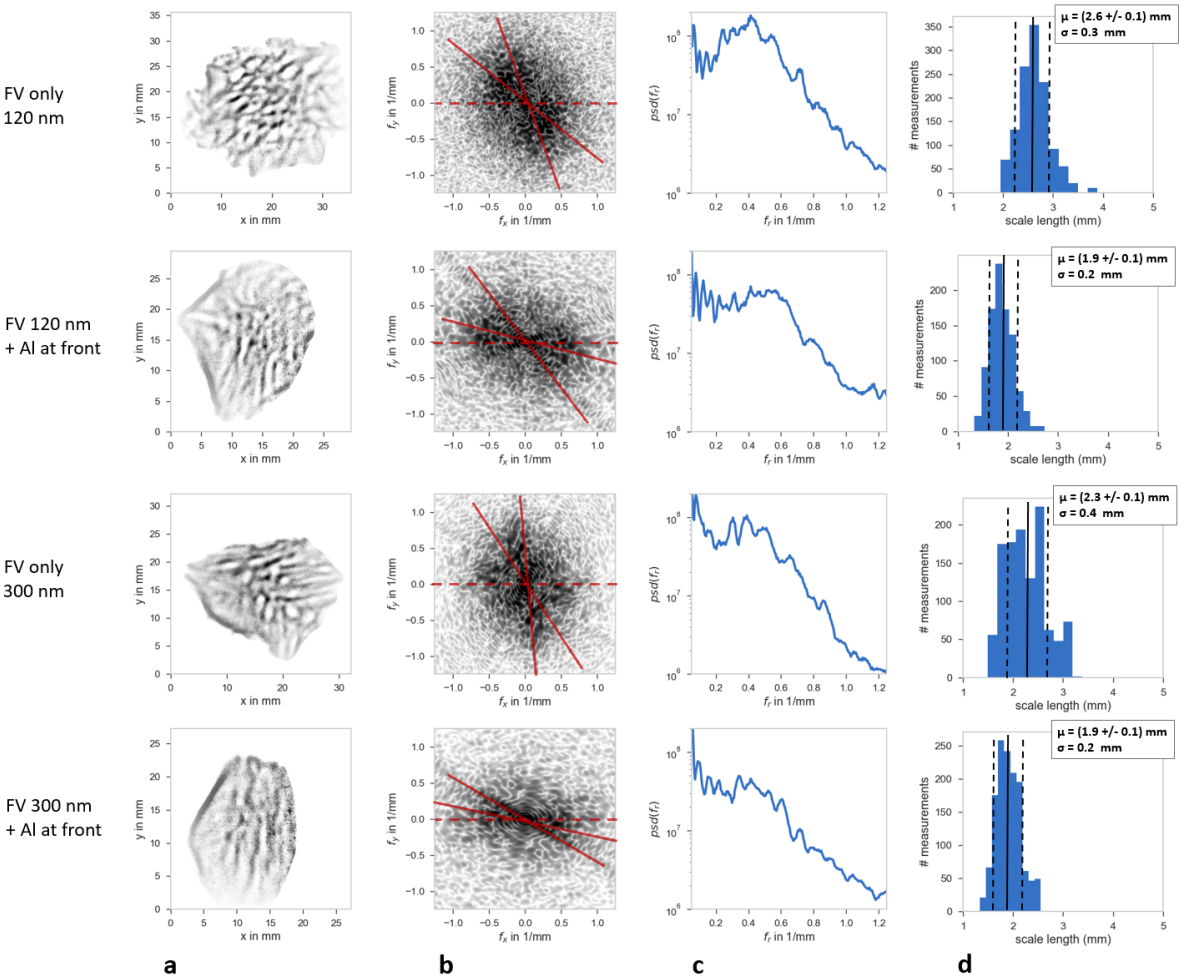

Figure 3.16: Analysis of spatial frequencies of modulated beam structures for pure and Al-coated FV targets of 120 and 300 nm thickness. **a)** contrast optimized proton beam profiles (including Hanning filter) **b)** 2D PSD **c)** angular-averaged PSD **d)** histogram of the measured scale lengths (avg. distance between neighboring peaks in 1D-profile lines). In total, 1500 profile lines were comb-like distributed along the angular directions of highest spatial frequencies observed in the PSDs (enclosed angular range by the red, solid lines in **c**). Vertical solid and dashed lines show the mean (μ) and standard deviation (σ) of the measurement distribution.

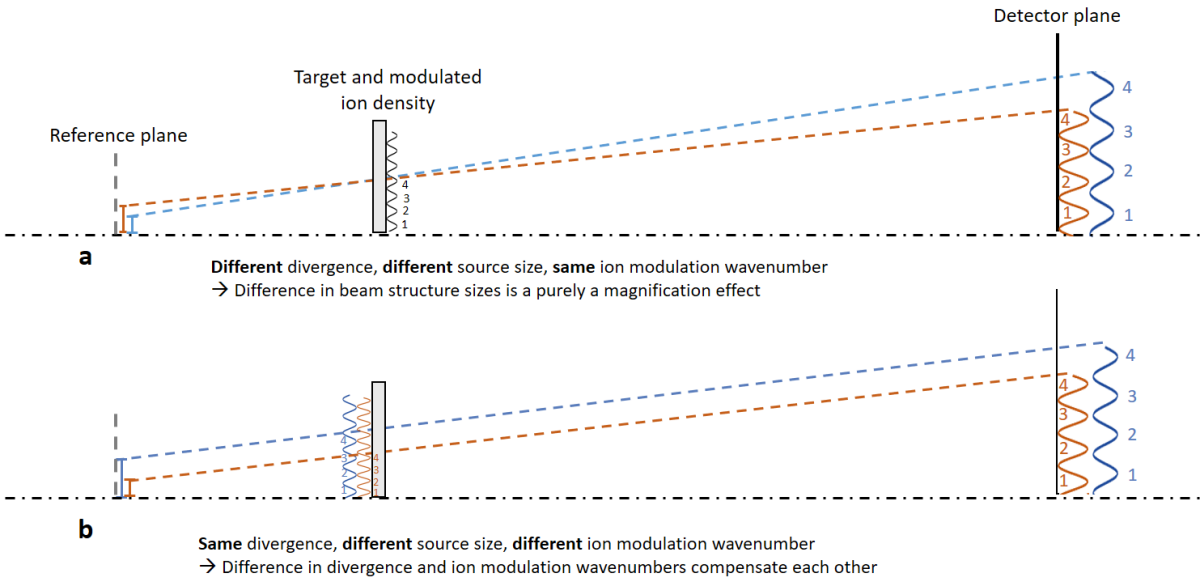

Figure 3.17: Schemes (not to scale) showcasing how the interplay of the source size, beam divergence and ion modulation wavenumber at the target rear side defines the structure size in the detector plane. In both cases, the number of modulations normalized by the beam spot dimension is the same. **a)** Scenario for a Weibel-like instability induced mechanism: In this case, a Weibel-like instability is seeded at the critical density surface at the target rear side so that the ion density modulation wavenumber is independent of the front surface material. **b)** Scenario for a mechanism including a two-plasmon decay of plasma oscillation in combination with a RTI. In this case, the wavenumber of electron filaments and thus the ion density modulation wavenumber is coupled to the plasma frequency given by the front side material.

3.3 Quantitative analysis of the modulation strength for different experimental settings

In order to complement the qualitative discussion and the subjective impression of differences in the prominence of modulations, the modulation strength in the beam profiles has been evaluated quantitatively with a method similar to the one introduced by MacLellan et al.[89] (see Fig. 3.18 & 3.19). While the data sampling was copied from MacLellan et al., the RCF data has been preprocessed differently by calculating $|\frac{D_{HP}}{D_{LP}}|$. This data representation has been chosen, since D_{HP} carves out the modulated structures in the beam profiles and normalization by D_{LP} sets these structures in relation to the overall number of protons emitted in this angular range.
Then, the variation coefficient $C_p(r,\theta) = \overline{|\frac{D_{HP}}{D_{LP}}|}$ was chosen as a quantitative measure of the redistribution in the emitted proton beam. It has been sampled in quadratic regions of interest (ROIs) of 50 pixel width ($\sim$ 4.3 mm) as a function of the radius r and angle θ relative to the geometric beam profile center. The sampling size was empirically chosen such that the ROIs cover the size of the caustic structures of interest. It was found empirically that the trends in the quantitative analysis shown in Fig. 3.19 are preserved for a wide range of different sampling sizes as long as it is not chosen smaller than the size of the caustic structures. By averaging over all evaluated angles, $C_p(r)$ has been calculated. As $C_p(r)$ was found to be quasi-constant, the mean variation coefficient, $\overline{C_p(r)}$ for a RCF layer was obtained by averaging over all radii. Finally, the mean variation coefficient has been evaluated for all RCF layers within the stack, respectively as a function of the corresponding Bragg-peak energy, E_p[3].
The analysis was performed for the conducted experiments presented in this thesis as well as for the measurements at the Draco 150 TW laser system from Metzkes-Ng et al. and Sommer et al.[90]. The data from Metzkes-Ng et al. was analyzed in order to compare the variation coefficient in the beam modulations obtained in the former Draco 150 TW experiments with those from the recent PW experiments. The experiments performed by Sommer et al. aimed to investigate the influence of ablative prepulses applied to the target front side of Ti foil targets on proton beam modulations. The ablative prepulses aimed to clear the target front side from source protons, which could otherwise –potentially– be accelerated from the front towards the rear and there be post-accelerated. These experiments were performed at target normal incidence.

Fig. 3.19 turns out to support the precedent, subjective discussion of trends regarding the modulation strength in the different experimental configurations.

[3] The spectral part below 30 MeV has been excluded from the analysis due to oversaturation by the high flux of the low-energy part of the proton spectrum.

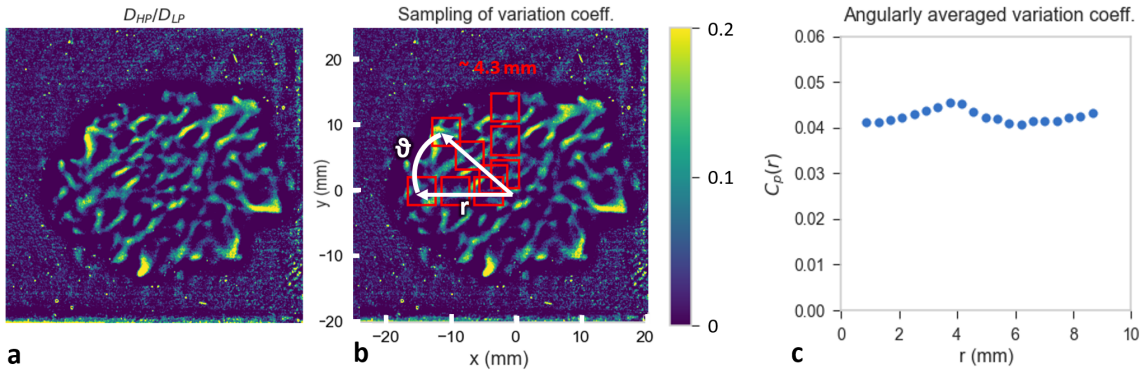

Figure 3.18: Quantification of the modulation strength in the measured proton beam profiles. **a)** Starting point for the analysis is the data representation D_{HP}/D_{LP}. **b)** The variation coefficient $C_p(r,\theta) = |\overline{\frac{D_{HP}}{D_{LP}}}|$ is calculated in dependence on the radius r and angle θ relative to the geometrical center of the beam spot. Red squares of 50 px width ($\sim$ 4.3 mm) are used to sample the variation coefficient. **c)** The angle-averaged variation coefficient $C_p(r)$ is found to be nearly constant for the analyzed data set. Thus, the average variation coefficient $\overline{C_p(r)}$ is obtained by averaging over all radii and is considered as a quantitative measure for the modulation strength on each RCF.

a) Inline prepulse study with 2 µm Ti-foils (see Sec. 3.1):
The highest variation coefficient can be observed in case of no prepulse application, i.e. highest laser contrast. Prepulses applied to the front surface resulted in a reduction of the variation coefficient.

b) Net-like modulations from 2 µm Ti-foils published by Metzkes-Ng et al. (see Sec. 2.3):
The modulation strength is increased in the proton beam profiles measured in 2011, after the laser contrast had been altered. The modulation strength for 2 µm Ti is higher for laser energy of 2.1 J compared to 2.7 J. Thus, the quantitative analysis supports the observation by Metzkes-Ng. et al.[1] of an energy-dependent optimal thickness for a maximal degree of expression of the net-like modulations (see Fig. 2.2).

c) Measurement with ablative prepulses published by Sommer et al.:
P. Sommer et al.[90] performed experiments at Draco 150 TW with ablative prepulses applied to the target front side of 2 µm Ti targets. The ablative prepulses were applied to clean the front surface from contaminants in order to study the source location of the modulated proton beam. No effect of the ablative prepulses on proton beam modulations was found compared to a reference shot. This statement is supported by the quantitative analysis as for both cases the modulation strength is similar.

d) FV targets: Inherent compared to PM cleaned laser contrast (see Fig. 3.6, Sec. 3.2.1):
Analysis of representative shots with FV targets with (250 nm) and without (220 nm) plasma

mirror cleaned laser contrast confirmed stronger beam modulation in the plasma mirror cleaned case. The chosen FV thicknesses are comparable and are both in a regime of negligible light transmitted through the target (cp. Fig. 3.7).

e) FV thickness scan (see Sec. 3.2.2):
The analysis for the FV thickness scan at optimized laser contrast and spectral phase shows a strong thickness dependence. Thicknesses between 120 and 300 nm feature a local maximum of the variation coefficient at around 40 MeV and exhibit higher values for $E_p > 40$ MeV (cp. Fig. 3.19f) compared to the other target thicknesses. Thereby, this thickness range differentiates from the measurements below a thickness of 100 nm in the transparency regime, which show smaller values of $\overline{C_p(r)}$. This behavior coincides with the increased visibility of net-like structures for targets above the onset of transparency observed in the empirical analysis (Sec. 3.2.1).

g, h) Al-coated compared to pure FV targets (see Sec. 3.2.4):
Al coating at the front led to smaller values of $\overline{C_p(r)}$ compared with the uncoated FV targets but similar cut-off energies (see Fig. 3.13) for 120 and 300 nm thickness. Al coating at the rear resulted in strongly reduced cut-off energies. Regarding the modulation strength, Al coating at the rear yielded an unclear picture. For a target thickness of 120 nm $\overline{C_p(r)}$ is similarly low as for Al coating at the front side, but for a thickness of 300 nm $\overline{C_p(r)}$ it is similarly high as for the uncoated FV target.

In conclusion, this analysis turns out to be useful to quantify the prominence of modulated structures and to support the subjective impression of trends. The analysis provides a useful measure of the degree of redistribution in the proton beam compared to an ideally smooth beam spot and reflects the prominence and visibility of structures by eye. On the contrary, this method does not take into account qualitative differences in the modulated structures. Accordingly, comparability of the variation coefficient is limited to beam profiles with structures of similar kind.
Scott et al.[51] suggested that the analysis of the proton beam modulations may provide information on the time history of the magnetic field distribution if the latter is responsible for the proton beam deflection during rear sheath acceleration. In a TNSA picture, this would mean that protons of higher final energy experience magnetic fields at earlier times compared to protons accelerated later and thus to lower final energies. However, one needs to consider that the dose distribution on an RCF layer contains the strong Bragg-peak dose contribution from protons stopped at the corresponding depth but also the plateau dose from the transmitted, higher energetic spectral part. Thus, sophisticated deconvolution methods would be required for the measurements presented here to assign $\overline{C_p(r)}$ to a specific spectral bandwidth.

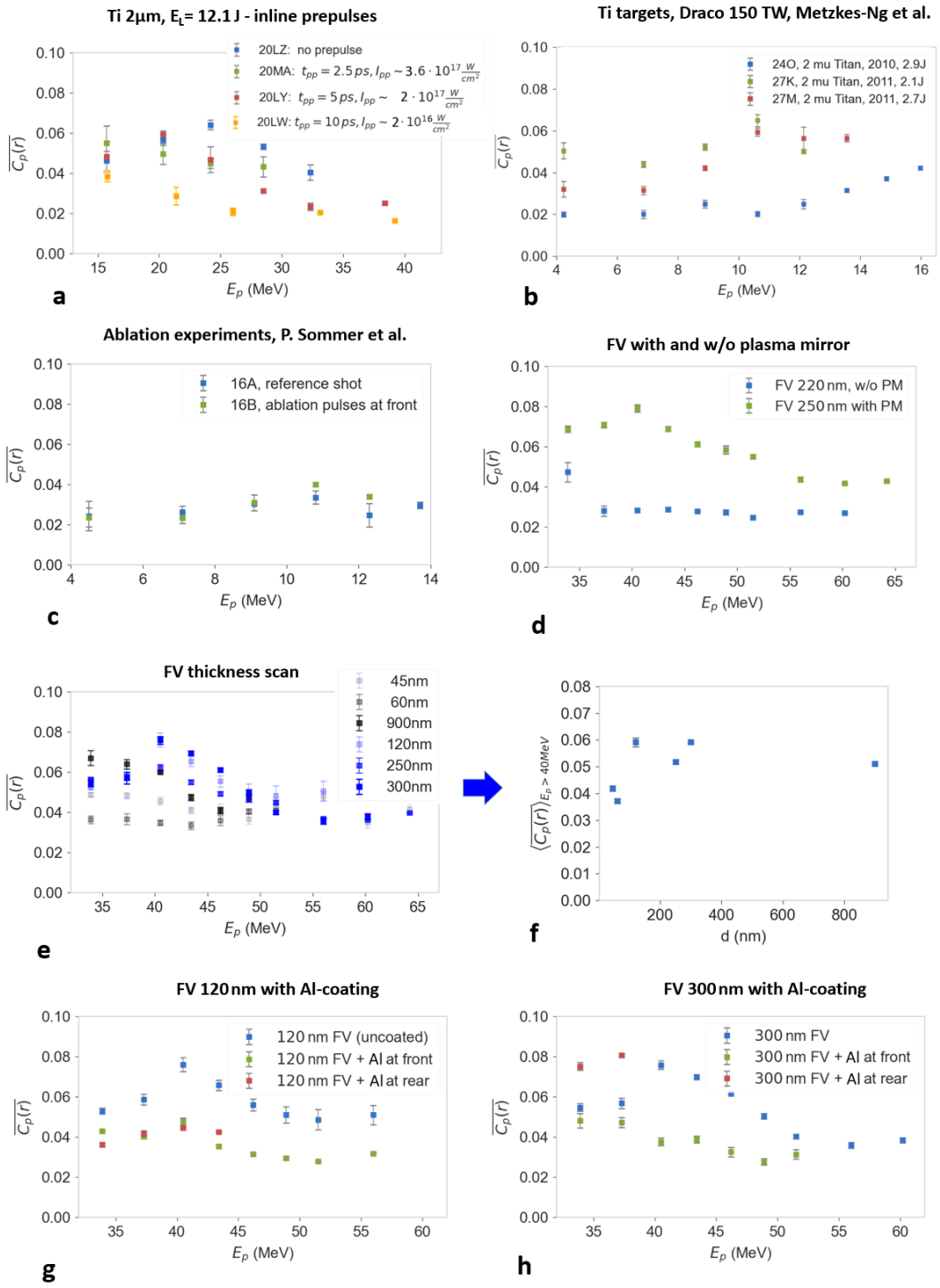

Figure 3.19: Average variation coefficient $\overline{C_p(r)}$ in the transverse dose distribution measured with stacks of RCF for different laser-target configurations. **a)** 2 µm Ti and different inline prepulse settings. **b)** Data from first observation of net-like modulations at Draco[1]. **c)** Data from ablation experiments at Draco with 2 µm Ti targets[90]. **d)** FV with inherent (220 nm) and PM cleaned laser contrast (250 nm). **e)** FV thickness scan at PM cleaned laser contrast. Blue colored data points highlight the increased variation coefficient in the proton energy range between 40 and 50 MeV for thicknesses between 120 and 300 nm. **f)** Variation coefficient averaged for 40 MeV $\leq E_p \leq$ 50 MeV for the FV thickness scan data. **g)** FV targets coated with a 30 nm Al layer at the front or rear side compared to uncoated FV targets for a thickness of 120 nm. **h)** same for 300 nm thickness.

3.4 Conclusions

The presented proton acceleration experiments reveal that modulated structures in proton beam profiles are generated frequently in several different laser-target configurations. In the tested experimental settings, ideally smooth proton beam profiles were found to be a rarity, which became particularly obvious, when the measured proton beam profiles were high pass filtered. Additionally, the subjective trends with regard to the prominence of the modulated structures have been supported by a quantitative analysis of the modulation strength.
The net-like proton beam profiles measured by Metzkes-Ng et al. at the Draco 150 TW laser with few micrometer Ti foils could not be reproduced in the experiments at the Draco PW laser by controlled prepulse application. At PM cleaned laser contrast without prepulse application, streak-like structures along the direction parallel to the horizontal laser polarisation plane were observed. Contrary to our initial hypothesis, an indication was found that prepulse application onto the target front side on the picosecond time scale prior to the main pulse arrival leads to a reduced prominence of modulated structures. This suggests that a high laser contrast or weak front preplasma formation is favorable for the generation of prominent modulated structures.
The same indication was found in the experiments with FV foil targets of few hundred nm thickness, for which prominent net-like structures were observed only at PM cleaned contrast. On one experiment day, the PM virtually played the role of a switch to turn on the generation of net-like structures (see Fig. 3.6). The scan of different FV thicknesses at PM cleaned laser contrast revealed prominent net-like modulations for thicknesses above the onset of target transparency ($\gtrsim$ 100 nm). Yet, modulated structures of arbitrary type were seen in the whole thickness range from 30 to 900 nm. They are seen on all RCF layers distributed over the whole beam spot area. Remarkably, no angular shift of the modulated structures compared to the smooth beam contribution was observed in contrast to the measurements from Powell et al. at oblique laser incidence[60]. Instead, the modulated structures appeared within the beam spot area in the target normal direction, the central beam direction in TNSA.
We consider it rather unlikely that protons accelerated from the target front side, e.g., via RPA-TNSA hybrid schemes, constitute the proton source of the modulated beam structures. In RPA-TNSA schemes, spatio-temporal coupling conditions define the efficiency of the acceleration. However, during the FV thickness scan we observed modulated structures always in the whole spectrum, i.e., up to the last irradiated RCF layer. From this we deduce, that the proton source originates from the target rear side and is rear sheath accelerated.
For a fixed target thickness of 240 nm, a reduction of the laser pulse energy showed the expected reduction in cut -off energies, while net-like structures remained visible in the experimental scan range. This observation demonstrates the robustness of the underlying beam modulation mechanism under variation of the laser intensity.
Another important observation is the stability in the generation of modulated structures for

fixed laser settings. We receive the impression of similarities between beam modulations from shots, when the laser-target conditions are fixed. With *similarity* we here refer to common modulation wavenumbers being located in same regions within the proton beam spot. Yet, we *do not* observe identical proton beam profiles from shot to shot in the sense that identical features are generated at the exact same positions. The latter is rather to be expected in imprinting scenarios, in which spatial information from the laser is imprinted into the laser-accelerated electron distribution[61, 80, 81], and subsequently into the accelerated proton beam. The all-optical imprinting mechanism[58] is most likely to be excluded, as in our experiments we deal with laterally non-restricted foil targets and since we observed the proton beam modulations in a wide thickness range above the onset of target transparency. In our experiments, the laser pulse has been well-focused, i.e., without presence of multiple hot spots, to a FWHM focal spot size of around 3 µm, in which the majority of laser energy is contained (see Fig. 3.20). In the exterior parts of the intensity profile of the UHI laser focal spot we do observe spatial structure in a field of view of around $25 \times 25\ \mu m^2$. Yet, the intensity in these exterior part is largely reduced compared to the FWHM region. From PIC simulations we know that the high-intensity FWHM region of the UHI focal spot dominates the electron transport dynamics through the target. Due to the Fourier limit however, the laser far field in the focal spot plane cannot carry sub-micron-scale spatial information.

Therefore, we consider it more likely that plasma instabilities are the root cause of the pro-

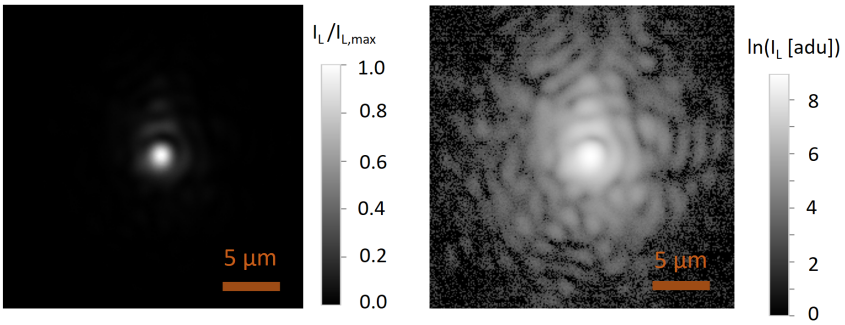

Figure 3.20: Focal intensity distribution in adu on a linear scale **(a)** and logarithmic scale **(b)**.

ton beam modulations. With this regard, electron filamentation via the 2-plasmon decay of surface oscillations in combination with RTI generation at the target front side[42, 47] or via two-stream instabilities seeded at the target rear side[51, 52] come into question. The following discussion aims at finding indications which of these mechanisms is most suitable for our experimental evidences.

First, the case of a Weibel-type instability seeded in a rear side preplasma at the critical density position will be discussed for the case of the recent FV foil experiments at Draco in analogy

to Göde et al.[52]. The growth rate of the Weibel-like instability Γ_W is given by

$$\Gamma_W = \frac{\left(\frac{v_r}{c}\right)\omega_r}{\sqrt{\gamma_r}} \tag{3.1}$$

with v_r, γ_r and ω_r being the mean velocity, relativistic Lorentz factor and plasma frequency of the return current. The growth rate is assumed to be maximized, when the density of the return current equals the critical density n_c. Then, the plasma frequency of the return current equals the laser frequency ω_0 and

$$\Gamma_W = \frac{\sqrt{\frac{n_c}{n_r}}\omega_0}{\sqrt{\gamma_r}}. \tag{3.2}$$

By identifying the Weibel filamentation wavelength λ_W with the Larmor radius of the electrons, the first is given by

$$\lambda_W = \frac{2\pi c\sqrt{\gamma_r}}{\omega_r} \tag{3.3}$$

Adopting a ponderomotive scaling $\gamma_r = \frac{a_0}{\sqrt{2}}$ according to Wilks et al.[11], with $a_0 = 45$ for our laser parameters we find $\lambda_W = 4.8\,\mu\text{m}$. This modulation scale length now needs to be compared with our measurements. To give a specific example, we refer to the previously discussed measurement with the FV target of 120 nm thickness and exploit the RCF layer corresponding to $E_p = 46\,\text{MeV}$. Fig. 3.21 illustrates the geometric situation assumed in the following.

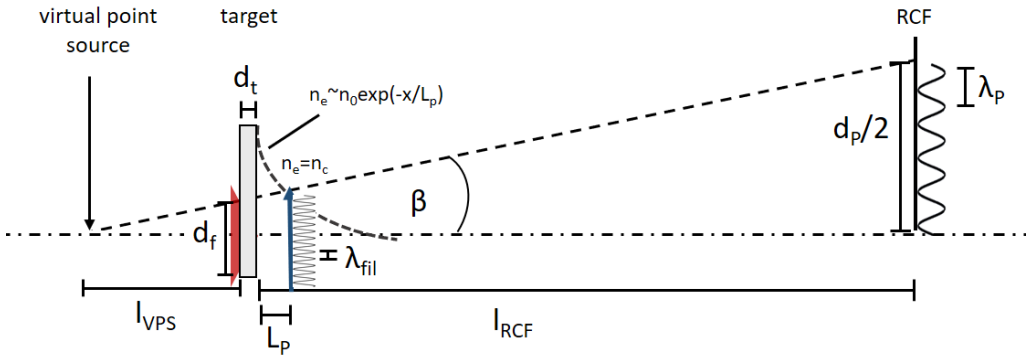

Figure 3.21: Assumed geometric situation for the scenario of a Weibel-type instability seeded at the critical density in a rear side plasma scale length.

First, we aim to define the distance l_{VPS} of a virtual point source from the front side of the target applying the intercept theorem with the constraints of matching the divergence angle β the FWHM focal spot diameter $d_f = 3\,\mu\text{m}$ at the target front side and the proton beam spot diameter $d_p = 28\,\text{mm}$ at the RCF location at a distance $l_{RCF} = 56\,\text{mm}$ from the target.
With

$$2\tan\beta = \frac{d_f}{l_{VPS}} = \frac{d_p}{l_{RCF} + l_{VPS} + d_{target}} \approx \frac{d_p}{l_{RCF}}, \tag{3.4}$$

we find $l_{VPS} = 6\,\mu m$. The dominant proton beam modulation scale length at the RCF position for our measurement is $\lambda_p = 2.6\,mm$ (cp. Fig. 3.16). Strong filamentary, electromagnetic fields can be generated, when a rear side preplasma scale length is larger than $L_p = c\Gamma_W^{-1} = 0.13\lambda_0\sqrt{a_0}$, which for our laser intensity yields $L_p = 0.74\,\mu m$. Thus, we would expect a filamentation scale length λ_f at the rear side critical density position of

$$\frac{\lambda_f}{l_{VPS} + d_{target} + L_P} = \frac{\lambda_f}{(6 + 0.12 + 0.74)\,\mu m} = \frac{\lambda_p}{l_{RCF}} = \frac{2.6\,mm}{56\,mm} \longrightarrow \lambda_f \approx (0.32 \pm 0.05)\,\mu m. \quad (3.5)$$

The uncertainty is obtained by applying Gaussian error propagation assuming $\Delta d_f = \pm 0.5\,\mu m$, $\Delta a_0 = \pm 10$, $\Delta d_p = \pm 2\,mm$ and $\Delta l_{RCF} = \pm 2\,mm$ so that $\Delta l_{vps} = \pm 1.1\,\mu m$ and assuming $\Delta\lambda_p = \pm 0.1\,mm$.
The estimate deduced from our experiments suggests a filamentary modulation wavelength of around 320 nm, which is much smaller than the theoretical Weibel-type filamentation wavelength of 4.8 µm. It should be noted that a simplified picture has been exploited in the calculations made here, while the PIC simulations from Branco reveal that the B-field distribution and evolution is by far more complex. Yet, our estimate of λ_f is a factor of 15 smaller than predicted by the Weibel theory. Thus, we interpret this discrepancy as a strong indication that there is no agreement of this mechanism with our measurements.

Now, we turn the focus to a RT-like instablity generation at the target front side seeded by surface rippling via a 2-plasmon decay of electron surface oscillations. Our estimate of the filamentary modulation wavelength is considerably smaller than the laser wavelength of 800 nm. This observation is not in accordance with a plasmonic enhancement of surface rippling with modulation periods close to the laser wavelength as proposed by Sgattoni et al.[47]. However, on the basis of the PIC simulations from Branco (see Sec. 2.3.4), our findings are compatible with a RT-like instability seeded at the target front surface via a two plasmon decay of plasma oscillations. Fig. 3.22a shows an example of the spatial distribution of the magnetic field in the PIC simulations from Branco. As discussed in the theory part, the generation of the filamented magnetic field at the rear side is preceded by electron filaments, which propagate from the front side towards the rear. The electron source divergence was found to be up to $\sim 35°$ (half-angle) inside the target, which is much larger than the divergence of the TNSA accelerated protons behind the target. The wavelength of the magnetic fields in the simulations by Branco were found to be dictated by the plasma period at the front propagated to the rear. For our target conditions we find an estimate of $\lambda_f \geq 70\,nm$ for the plasma period at the front. While this is smaller than the 320 nm estimate above, it is consistent with it, since we have to consider filament merging.
To sum up, also the simulations from Branco attribute the proton beam modulation itself to the presence of a filamented magnetic field structure at the rear side. However, the simulations suggest that the primary instability is seeded at the target front side. Our estimate of the modulation wave length at the target site deduced from our experiments is compatible

with this assumption.

With magnetic field structures being present at the target rear side in this sceneario (or in case of the two-stream instability), we would need to understand the proton beam modulation in a radiographical picture, in which the accelerated protons probe the magnetic field distribution at the target rear side. In this picture, small deviations in the proton beam modulations obtained from fixed experimental settings could be explained by small variations in the experimental laser-target conditions. Thus, this would explain why we did not observe the occurrence of identical, reproducible beam modulations as one might expect in imprinting schemes.

Now, we need to consider how a modulated magnetic field distribution in vicinity to the target rear side translates into proton beam modulations. Fig. 3.22 shows an example of the spatial distribution and temporal evolution of the magnetic field in the PIC simulations. One can see that higher wave numbers of the filamented structure are located closer to the target bulk due to the effect of filament merging behind the target (see Fig. 3.22a). Consequently, the exact location of the source during the acceleration process determines which spatial wavenumbers of the magnetic field distribution are probed and may be translated into the proton beam profile. If bulk acceleration is possible, e.g., in case of the FV (plastic) targets, one may assume that protons are available at the location of highest wavenumbers in the filamented magnetic field structure directly behind the target rear side. Likewise, one would expect lower spatial wavenumbers in beam profiles from targets for which bulk proton acceleration is not possible, e.g. in case of metal foils. In this case one can assume that protons originating from the contaminant layer are pre-expanded and thus experience lower spatial wavenumbers of the electromagnetic field structure during their acceleration. It was indeed observed that the proton beam modulations obtained in the measurements with Ti foils at Draco PW (see Fig. 3.4) featured lower spatial wavenumbers than those obtained from FV foils.

In addition, the B-field evolves quickly in time (see Fig. 3.22b) with regard to its distribution and field amplitude. As protons propagate through the spatial distribution and transient evolution of the B-field during the acceleration process, the mechanism of proton beam modulations is highly complex. So far, the complex radiography process of the proton propagation through a transient magnetic field evolution has not been considered sufficiently, whereas this process itself could explain many of the beam modulation features. With this in mind, follow-up simulations are needed to study the mechanism how such complex magnetic field structures translate into transverse beam modulation during the proton acceleration.

On the experimental side, the remaining questions are a strong endorsement of the HIBEF efforts of HZDR to combine LPA experiments with X-ray probing techniques at the EUXFEL. In particular, SAXS promises to provide insight on the solid density evolution, e.g., to measure plasma instabilities and to characterize their growth rate. With regard to the remaining questions raised here, SAXS would enable to discriminate between different mechanisms dis-

cussed in literature causing proton beam modulations. Moreover, SAXS in combination with proton beam profile measurements with a Lanex screen would allow to correlate features in the SAXS signals with different types of proton beam profiles. With this regard, Machine Learning could be beneficial to automate the classification of different beam profile types and to find correlations between the SAXS data and beam profile measurements.
Finally, the results from this work suggest the implementation of a plasma mirror setup at HIBEF as the proton beam modulations presented in this work, which are potentially linked to plasma instability seeding and electron filamentation, have been observed at PM cleaned laser contrast conditions.

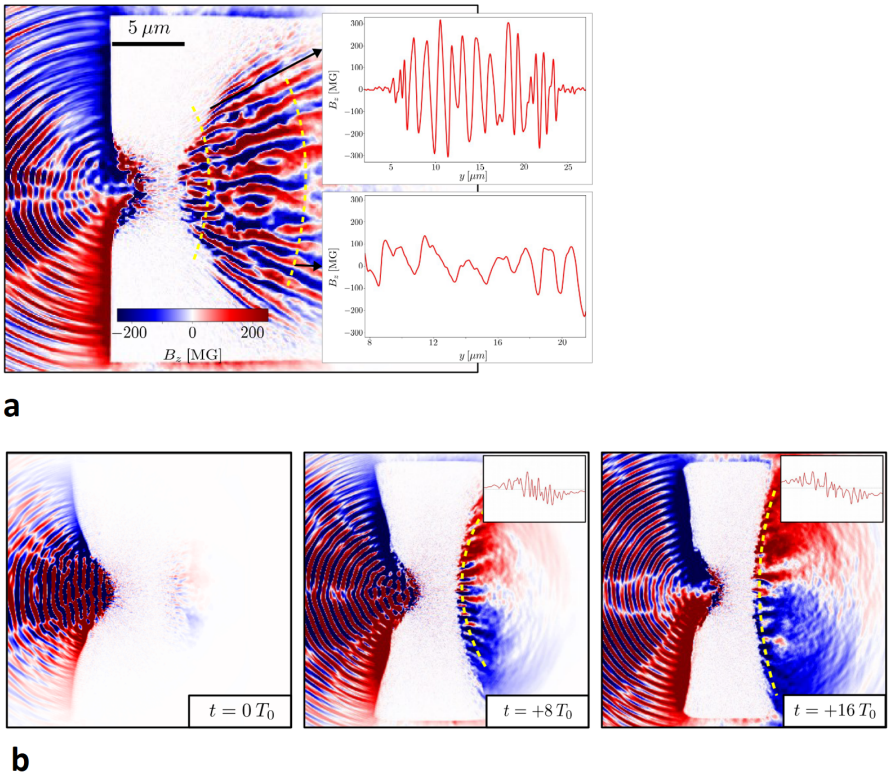

Figure 3.22: a) Spatial B-field structure from simulations from Branco for a hydrogen jet target, $a_0 = 30$, no front side preplasma, rear preplasma scale length $\sim 2.5\,\mu m$, $t \sim 40\,fs$. The two profile lines of the B-field at different longitudinal positions at the rear show the effect of filament merging. The B-field structure shows higher wavenumbers and higher amplitudes closer to the target rear. **b)** B-field distribution at different times ($T_0 = 2.67\,fs$) show the transient evolution of the latter during the time scale of proton acceleration. Figures adopted from Branco[53].

4 Using XFELs in Pump-Probe Experiments of Laser-Solid Interaction

Gaining full control over laser-driven proton acceleration requires in-depth understanding of all relevant processes during the interaction of an UHI laser with a solid. This includes non-linear, transient processes located at the target front side, such as laser absorption, hole boring or the generation of surface plasma waves as well as the transport of the fast, laser-accelerated electrons through the plasma bulk and the generation of plasma instabilities. In the specific example presented in the previous chapter, the identification and characterization of specific plasma instability types are of particular relevance to understand the generation of modulated proton beam profiles and to validate (or falsify) their present interpretation in the literature. This question falls within the general need of the scientific community in laser-plasma research to probe solid density plasma dynamics directly in experiment.
Experimental probe methods in this context require on the one hand to penetrate through solid density plasma, which is not possible with standard optical probes without resorting to higher harmonics. On the other hand, one needs to resolve the transient plasma evolution on the femtosecond scale with a spatial resolution between several hundred down to few nanometers, given by the typical scales of electron density modulations[91]. In light of this challenge, X-ray free-electron lasers (XFELs) promise to be ideally suited as a probe for solid density plasma, as state-of-the-art XFEL facilities such as European XFEL, LCLS or SACLA provide X-ray beams of ultrashort pulse duration down to sub-hundred femtoseconds and of up to 25 keV photon energy allowing to penetrate through solid density plasma. Furthermore, their ultrahigh brilliance[1] and low emittance enable strong scattering signals and thus allow for high signal-to-noise ratios.

The experiments presented in this thesis exploit Small-angle X-ray scattering (SAXS) experiments with XFEL pulses. This technique allows to access the spatial resolution on the sub-micrometer down to few nanometer scale relevant for the above mentioned laser-solid interaction processes. Thus, SAXS fills a gap between the resolutions of sub-nanometers for wide-angle X-ray Scattering and hundreds of nanometers for phase-contrast imaging[16]. Scattering experiments in the context of laser-solid interaction are performed in a pump-

[1] E.g., the European XFEL provides XFEL pulses of $5 \cdot 10^{33}$ photons/s/mm^2/mrad^2/0.1 % bandwidth peak brilliance[92].

probe scheme, where the UHI laser (pump laser) drives the plasma, which is probed by the XFEL pulse at a chosen delay.
As the information on the plasma is contained in the scattering signal, it is necessary to understand its composition. In first order approximation, the SAXS signal off a laser plasma is dominated by Thomson Scattering scattering on free electrons. The approximation of treating electrons as free particles is particularly appropriate when dealing with photons in the x-ray energy regime as long as the photon energy is not in resonance with atomic transitions between deeply bound states[93]. In this case, the SAXS signal in Born approximation is proportional to the absolute square of the Fourier transform of the projected electron plasma density along the XFEL probe direction. Thus, this probe technique is sensitive to the contrast in electron density in the transverse directions relative to the XFEL probe direction.
Measurable electron density modulations may be generated by the laser during its interaction with the solid, or they can be present inherently by use of prestructured targets. The first case represents one of the driving motivations of HZDR to establish SAXS as a standard probe technique at the European XFEL to study transient plasma processes during laser-solid interaction. By means of quantitative calculations, realistic estimations of XFEL probe beam parameters and synthesized data, Kluge et al.[91] demonstrated that various plasma processes, such as surface and electron beam instabilities or shock formation translate into distinct, measurable features on an X-ray detector, when the plasma is probed in orthogonal probe geometry (see Fig. 4.1).
It is apparent that in the case of relativistic laser-solid interaction the plasma dynamics are highly complex as a variety of transient plasma processes may take place simultaneously. Yet, for conducting first systematic SAXS experiments on laser-driven plasma one needs to guarantee a stable generation of a specific process to be studied. Thus, establishing SAXS as a standard probe technique in this context requires proof-of-principle experiments with – to the extent possible – isolated, scientific questions of less complexity.
With this regard, prestructured grating targets have proven to be particularly useful for two main reasons. First, they provide a known, reliable reference signal for every pump-probe experiment due to their inherent electron density contrast in projection along the X-ray probe direction. This is a critical aspect for systematic studies of laser-plasma interaction given the limited beam time at XFEL facilities and the limited shot rates in laser-proton acceleration experiments with solid foil targets. Second, the specific design of prestructured targets is easily adaptable to different science cases to be studied [2]. This has been demonstrated in a first SAXS experiment in the context of UHI laser solid interaction performed in 2016 by Kluge et al.[14] at the Matter in Extreme Conditions instrument at the Linac Coherent Light Source. Here, SAXS was demonstrated to be capable to measure the plasma surface expansion of Silicon grating targets with nanostructures positioned at the front side in a regime of moderate pump laser intensity of around 10^{17} W/cm^2. The experiment was performed in a transmission probe geometry, i.e., both UHI and XFEL probe beam irradiated the grated front surface with a relative angle of 90° at an UHI laser incident angle of 45°. The experimental procedure consists of two *preshots* and a *main shot* at variable, relative delays in order to probe the target in the unpumped (initial) state and upon UHI pump, respectively. With according forward modeling of the projected target electron density, this method then allows to follow the plasma evolution in the Fourier domain, i.e., by changes of the SAXS signal in the main shot compared to the signals in the preshots.
The analytical model of a grating period in real space used by Kluge et al.[14] comprises a

[2] This will be elaborated in detail in the Outlook section of this thesis.

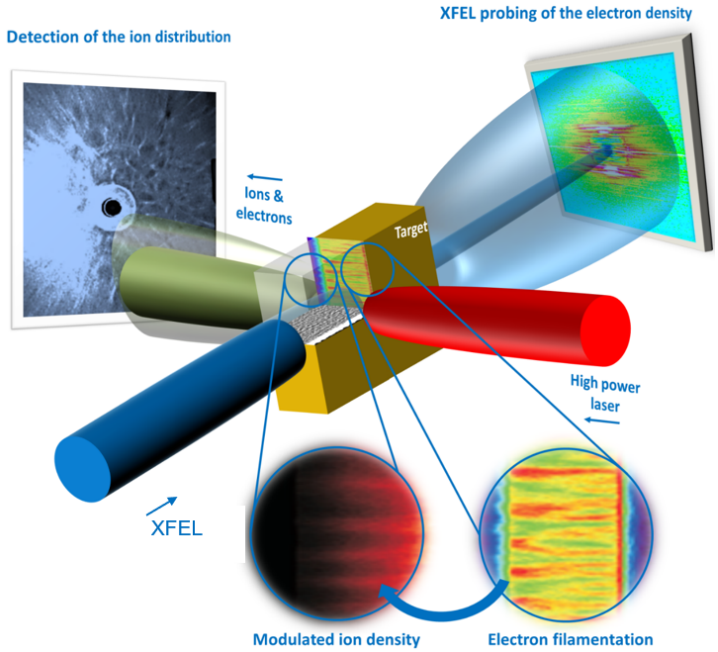

Figure 4.1: Vision of probing laser-induced electron density structures in solid density plasma upon the interaction with an UHI laser pulse. In the scattering geometry proposed here, the XFEL probes the target in orthogonal direction to the drive laser. Electron density modulations, e.g., caused by electron filamentation instabilities result in distinct features in the SAXS signal. Such an experimental scheme promises to measure non-linear plasma processes in solid density plasma directly and may help to develop an understanding of their role in the generation of modulated proton beam profiles. Figure adapted from T. Kluge.

grating period, g, ridge width, b, grating amplitude, h, and a grating edge described by an error function ($\sim erf\left(\frac{x}{\sqrt{2}\sigma}\right)$) parameterized by the smoothness parameter σ. As discussed in [14], the resulting SAXS signal is given by

$$I(q) = \frac{1}{q^2} \cdot \underbrace{I_X r_e^2 \left(2N\frac{N_b}{b}\right)^2}_{A} \cdot \underbrace{\left(\frac{\sin(Nqg/2)}{N\sin qg/2}\right)^2}_{G(q)} \cdot \underbrace{\sin\left(\frac{qb}{2}\right)^2}_{S(q)} \cdot \underbrace{e^{-q^2\sigma^2}}_{E(q)}, \tag{4.1}$$

with I_X, N, N_b being the XFEL intensity, number of illuminated ridges and total number of electrons in an illuminated ridge, respectively. The Debye-Waller term, $E(q)$, is of particular relevance for the method as it defines the exponential signal decay envelope and is sensitive to changes on the nanometer scale of the edge smoothness parameter, σ, when the grating edges smear out due to thermal pressure upon laser pump. A measure for an effective plasma scale length during the grating expansion can then be estimated by the measure $\Delta\sigma = \sigma_{main} - \sigma_{pre}$, where σ_{pre} and σ_{main} refer to the edge smoothness in the preshot and in the main shot, which can be obtained by fitting of the SAXS signals (see Fig. 4.2). As shown by Kluge et al.[14], according analysis at variable probe delay can then be used to extract the plasma surface expansion velocity.

In this thesis, measurements from a follow-up SAXS experiment performed at MEC in 2018 will be presented (see Chapter 5). In this experiment, SAXS was performed for the first time

in the relativistic laser intensity regime at a peak laser intensity of around 10^{19} W/cm^2, which allowed to probe the ultrafast plasma dynamics typically reached in laser-proton acceleration experiments.
The XFEL probing capabilities were further extended by performing resonant SAXS. For this purpose, the XFEL energy was actively tuned to resonant K_α transition energies of a hot copper plasma in order to exploit the sensitivity of the scattering signal on resonant absorption processes, which are linked to the plasma ionization and implicitly to the plasma temperature. Motivated by the high elemental charge state specificity, resonant SAXS or resonant coherent X-ray diffraction (RCXD) at bound-bound transition energies has been proposed for opacity measurements in the context of laser-solid interaction by Kluge et al.[94]. In a scenario of laser-induced ionic structures due to electron filamentation, resonant SAXS in combination with phase retrieval algorithms promises to reconstruct the ion distribution with high spatial, temporal and charge state resolution.
Moreover, opacity measurements promise to benchmark different ionization models implemented in PIC codes, e.g., the Thomas-Fermi (TF) ionization model or Direct Impact (DI) ionization model. To this day, PIC codes do not contain self-consistent models for ionization and recombination required to model the transient, non-thermal plasma evolution during HI laser-solid interaction from first principles. Realistic treatment of the latter requires simulations that solve the atomic transition rate equations on-the-fly, which exceeds existing computational resources. Yet, there are attempts to implement simplified atomic transition rate models in PIC codes, e.g. within the FLYonPIC development[95] for PIConGPU.
In the resonant SAXS experiments, nanostructured Si-Cu compound grating targets[3] and *Cu-only* targets have been employed (see Fig. 4.7). In the target production, a Cu-layer has been added onto the the silicon grating targets by ion sputtering. The manufacturing process results in a buried and rear grating interface, while the rear grating represents a smoothed version of the buried grating. The Cu-only targets have been produced by removing the Si-layer by etching. The analytical description of the Si-Cu or Cu-only grating targets then requires *two sets* of grating parameters, i.e., for the buried (σ_1, b_1, h_1) and rear side grating (σ_2, b_2, h_2). The increased space of fitting parameters impedes to unambiguously distinguish between the expansion of the buried and rear side grating.

Yet, the resonant SAXS patterns contain an additional quality, namely their asymmetry. In this work, a semi-analytical model for the asymmetries in resonant SAXS patterns has been applied in the analysis of the recent experiments at MEC in order to make use of the additional information on the plasma expansion and plasma opacity contained in the asymmetric SAXS signals. In the following sections a short introduction to the principles of resonant SAXS will be given, relevant considerations for the signal composition for resonant scattering on hot plasma and the derivation of the semi-analytical model of asymmetries in resonant SAXS patterns will be presented.

[3] This target system has been employed already in the precedent experiment in 2016.

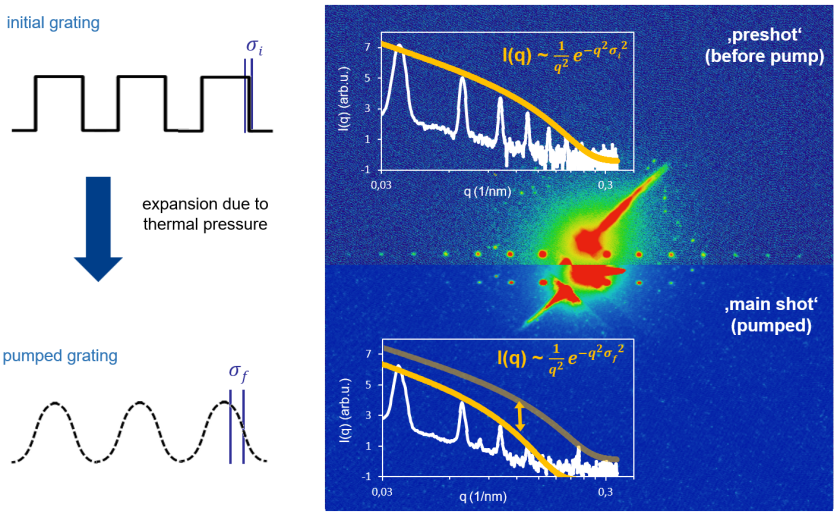

Figure 4.2: Exemplary SAXS signals from a nanostructured Si grating target probed in a transmission probe geometry. Upon HI laser pump, the grating expands due to thermal pressure, which leads to an increase of the smoothness parameter σ in the mainshot compared to the preshot. According to equ. 4.1 this translates into a faster decay of the scattering signal with q or into a reduced number of measurable scattering peaks.

4.1 Resonant Small-angle X-ray scattering

In this section, an overview on the principles of resonant small-angle X-ray scattering and relevant considerations for the signal composition for resonant scattering on a laser-driven plasma will be given based on the text-book introduction by Als-Nielsen[93] and the theoretical framework of Kluge et al.[94].
The fundamental scattering process to be considered, is Thomson scattering, which refers to the elastic scattering of a photon on free electrons. In a classical picture, the electric field of an incoming photon can be described by a plane wave, which forces a free electron to oscillate and to emit a spherical wave in analogy to a small antenna. In a scatter experiment, the quantity of interest is the number of photons per second which are scattered onto a detector covering a solid angle Ω relative to the incident flux. The relevant measure for this is the differential cross-section $\frac{d\sigma}{d\Omega}$ given by

$$\frac{d\sigma}{d\Omega} = r_0^2 \cdot P \qquad \text{with} \qquad r_0 = \frac{e^2}{4\pi\varepsilon_0 m_e c^2}. \tag{4.2}$$

Here, r_0 is the Thomson scattering length of a single electron (or classical electron radius) and P is a factor, which considers the photon's polarization and the relative direction of the scattering plane. This work deals with an XFEL photon source, with the polarization being in the horizontal plane and a scattering experiment in the orthogonal plane, so that $P = 1$. For very small scattering angles as for SAXS, the latter holds true for scattering in the horizontal plane as well. In an illustrative picture, the cross section, σ, can be understood as an effective target area assigned to an object, which is a measure for the probability for a process of interest, e.g., a scattering event. As $r_0 \sim 10^{-5}$ Å, the scattering cross-section for a single electron is very small. Strong scattering signals require on the one hand samples containing

many electrons and on the other hand photon sources of high brilliance.
The next scattering unit to be considered is an atom that contains Z electrons described by an electron density $\rho(\vec{r})$. Different volume elements of the electron density contribute to the scattering with $-r_0\rho(\vec{r})d\vec{r}$ and a relative phase of $e^{i\vec{q}\vec{r}}$, where $\vec{q} = \vec{k} - \vec{k'}$ is the scattering vector and $\vec{k}$ and $\vec{k'}$ are the wave vectors of the incident and scattered wave, respectively. This leads to the definition of the atomic scattering length

$$-r_0 f_0(\vec{q}) = -r_0 \int \rho(\vec{r}) e^{i\vec{q}\vec{r}} d\vec{r}. \tag{4.3}$$

As one can see in equ. 4.3, the atomic form factor, f_0, is proportional to the Fourier Transform of the electron density distribution and is dependent on the scattering vector. In case of small scattering angles ($\ll 1°$) realized in SAXS experiments, all Z electrons of an atom contribute coherently to the scattering signal, so that $f_0 \approx Z$.

In reality, the classical picture has to be extended by quantum mechanical considerations as electrons are bound to the atom and have discrete binding energies. Effectively, not all electrons may be treated as free charged oscillators and absorption processes need to be considered. Accordingly, the atomic form factor needs to be corrected by the dispersion correction terms f' and f'' and becomes the complex-valued quantity

$$f(\vec{q}, E_0) = f_0(\vec{q}) + f'(E_0) + if''(E_0). \tag{4.4}$$

The imaginary part of the dispersion correction, f'', is linked to the total absorption cross section $\sigma_a(\omega)$ by

$$f''(\omega) = -\left(\frac{\omega}{4\pi r_0 c}\right)\sigma_a(\omega), \tag{4.5}$$

and becomes accessible by measurements of $\sigma_a(\omega)$. The real part of the dispersion correction, f', can then be obtained by the Kramer-Kronig relation[4]

$$f'(\omega) = \frac{2}{\pi} \fint_0^{+\infty} \frac{\omega' f''(\omega')}{\omega'^2 - \omega^2} d\omega'. \tag{4.6}$$

It is important to note that, unlike f_0, the dispersion corrections are dependent on the incident photon energy E_0. The contribution of the dispersion corrections to the atomic form factor is strongly increased at X-ray energies that match with electronic transition energies of the sample. For solids at ambient temperature, the most relevant absorption process is the photoelectric effect (bound-free transition). Its cross-section decreases with E^{-3} and has discrete jumps at element-specific absorption edges (see Fig. 4.3). Typically used XFEL photon energies in the multi-keV range correspond to the binding energies of electrons bound in the inner shells (K, L, M).
In a laser-driven, hot plasma, ions can be highly ionized or excited resulting in vacancies in the L shell. When an incident X-ray photon matches with the transition energy between the K and L shell (K_α transition), the cross-section for resonant scattering is significantly increased. In the XFEL pump-probe experiment presented in this thesis, this circumstance has been made use of by actively tuning the XFEL probe energy to resonant energies of these bound-bound transitions. The underlying virtual process of resonant scattering comprises an electron ini-

[4] The symbol $\fint$ refers to the Cauchy Principal Value integral.

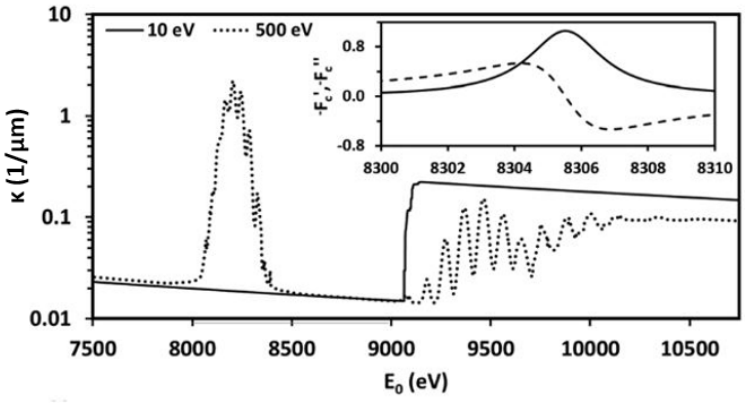

Figure 4.3: Opacity of a copper plasma obtained with FLYCHK simulations in dependence of the XFEL probe energy in a heated and non-heated scenario. The hot plasma of 500 eV shows significantly increased opacity at K_α transition energies below the K-edge. The inset shows the behaviour of the dispersion correction terms for the configuration $1s^2 2s^2$. The figure is adopted from[94].

tially bound in the K-shell which is excited to the L shell followed by relaxation back to the initial state and photon emission with the same energy as the incident photon. Thus, resonant scattering is an elastic scattering process. The K_α transition energy depends on the element's atomic number Z, the number of bound electrons $\Gamma = Z - Q$, where Q is the ion charge, and the specific electronic configuration. For higher ion charge states (or smaller Γ), the transition energy increases as shielding of the nuclear potential is reduced.
This work exploits the sensitivity of the resonant SAXS signal to a variation of the dispersion correction terms in a plasma upon HI laser pump. As described above, the dispersion corrections are closely related to the total absorption cross-section including bound-bound (bb), bound-free (bf) and free-free (ff) transitions. Another commonly used quantity related to the total absorption cross-section is the plasma opacity

$$\kappa = \left(\frac{\rho_m N_A}{A} \right) \sigma_a \tag{4.7}$$

with ρ_m being the element's mass density, A its atomic mass number and N_A Avogadro's number. When a plasma is in a thermodynamic equilibrium (TE) or local thermodynamic equilibrium (LTE) state, the ionization level population distribution is determined by the thermodynamic variables plasma density and temperature and by the Boltzmann and Saha equations[96]. Accordingly, in these cases, the opacity at a specific ionic K_α transition energy is dependent on the plasma temperature.
Fig. 4.4a shows a 2D plot of the opacity (or f''_{Cu}) of a copper plasma in dependence on the XFEL probe energy and plasma bulk temperature. To generate this plot, FLYCHK runs for different LTE temperatures have been performed. Fig. 4.4b shows the opacity as a function of the XFEL probe energy for fixed temperatures. The opacity as a function of the XFEL energy shows local maxima at the K_α transition energies of different ionization states. The latter are termed with the names of ground state ions corresponding to the number of electrons bound in the inner shells only. In a comparably cold plasma of 300 eV, most ions are in a F-like ionization state and the opacity is highest at the corresponding K_α transition energy. When the plasma

is heated further, the abundance of higher ionization states and the opacity at their according (higher) transition energies increase. In the resonant SAXS experiment presented in this thesis, an on-resonant XFEL probe energy of 8165 eV was chosen, which corresponds to the N-like transition energy in copper. Increased opacity at this transition energy requires the copper plasma to be heated to at least 300 eV.

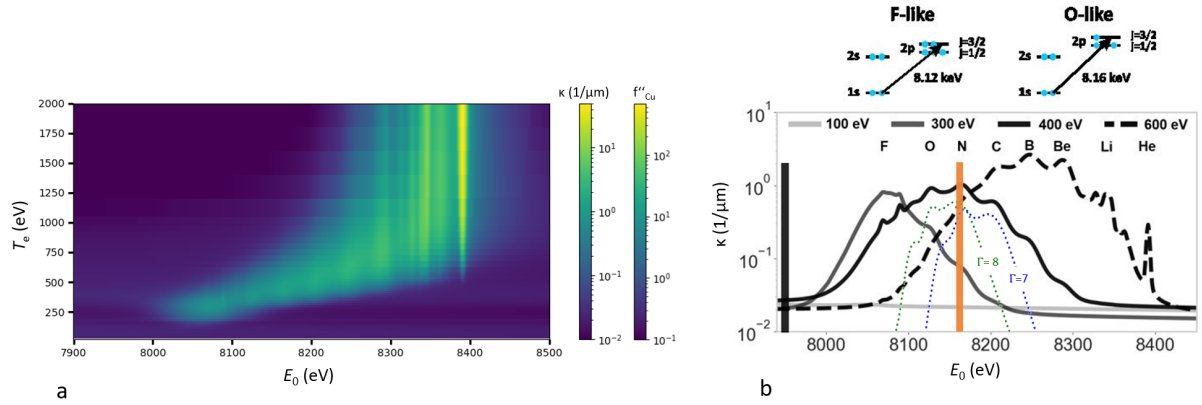

Figure 4.4: a) Opacity κ and imaginary part f''_{Cu} of the ionic form factor of a Cu plasma in dependence on X-ray probe energy and plasma bulk temperature. The colorbar for f''_{Cu} was obtained with equ. 4.5 for a fixed X-ray photon energy of 8200 eV. Thus, the (small) energy dependence in the shown energy interval is neglected here. For given probe energy, the opacity (or f''_{Cu}) is increased when the Cu plasma exceeds a minimal plasma temperature, so that an ionization state corresponding to the X-ray energy is present in the plasma. For XFEL probe energies corresponding to lower ionization states, e.g. F-, O- or N-like, the opacity drops again for higher plasma temperatures, when the abundance of higher ionization states dominate in the ionization state population distribution. **b)** 1D-lineouts of the opacity for fixed plasma temperatures. The black and orange bars highlight the off- (7940 eV) and on-resonant (8165 eV) XFEL probe energies used in the resonant SAXS experiments presented in Section 5.5. The dotted lines indicate the relative contributions of ions in the ionization stage Γ to the opacity. On the top, a schematic of resonant K_α absorption for ions in the F-like (9 bound electrons) and O-like (8 bound electrons) states is given.

However, it is important to note that different ionization states in combination with their electronic configurations may result in similar K_α transition energies. As illustrated with the dotted lines in Fig. 4.4b, at the on-resonant X-ray probe energy of 8165 eV, the contributions of Cu-ions in the ionization states with $\Gamma = 7$ and $\Gamma = 8$ to the opacity are similar for a given LTE plasma temperature of 400 eV. This is due to the fact that the $1s \rightarrow 2p_{1/2}$ (or $2p_{3/2}$) transition energies for ions in the ground state configuration with $\Gamma = 7$ are similar to the excited configuration of an ion with $\Gamma = 8$, as for shielding of the nuclear potential mostly the electrons bound in the K and L shell are relevant.
Moreover, the XFEL probe beam has a finite energy bandwidth. For the XFEL beam available at the HED instrument at the European XFEL the energy bandwidth is in the order of 20 eV ($\Delta E/E \sim 0.2\%$[92]). Consequently, the opacity as a function of the XFEL probe energy results from a whole set of electronic configurations and cannot be attributed to *one* singular transition channel of a particular ionization state. As demonstrated in [94], the parameter that describes the specificity of the resonant SAXS contribution best are configurations γ counting the number of electrons in the lowest shells only.

In order to derive the complete resonant SAXS signal, scattering on free electrons and electrons bound in ions needs to be taken into account. This requires consideration of the ionic form factor for all ions and electronic configurations present in the plasma. As described by Kluge et al.[94], the resonant SAXS intensity distribution, $I(\vec{q})$, from a single-element plasma, can be formalized by

$$I(\vec{q}) = I_0 \frac{r_0^2}{A} \left| \mathcal{FT} \left[\int \left(n_e^{free}(\vec{r}) + \sum_{\Gamma=1}^{Z} n_{i,\Gamma}(\vec{r}) F_\Gamma + \sum_{\gamma=1}^{N-1} n_{i,\gamma}(\vec{r}) F_\gamma^{res}(\vec{r})(E_0, T_e) \right) dz \right] \right|^2 d\Omega, \tag{4.8}$$

with I_0 and A being the incident XFEL intensity and irradiated area. Here, the first sum covers the Thomson term F_Γ for ions with Γ bound electrons and of density $n_{i,\Gamma}$. The Thomson term can be approximated by $F_\Gamma \approx \Gamma$ in SAXS geometry. The second sum treats the dispersion correction terms $F_\gamma^{res} = F_\gamma'(E_0, T_e) + iF_\gamma''(E_0, T_e)$ for resonant scattering on ions in the different configurations γ. In order to obtain F_γ^{res}, one needs to take the weighted sum of all contributing electronic configurations averaged over the XFEL energy bandwidth.
From equ. 4.8 it becomes obvious that predictions of the signal for resonant scattering on a transient, hot plasma are highly complex. Predictions require knowledge of the free electron and ion density distribution for all ion species and ionic configurations and their respective form factors for every point in time and space. Practically, this complete picture is not accessible neither experimentally nor in numeric simulations, which makes approximations unavoidable. The central approximation made in this thesis, is to assign each target material layer *one* constant, average density and set of form factors, f_0, f', f'', (cp. Fig. 4.7). Assuming a constant density per target material layer implies proportional distributions of the free electron and ion density within the layer. With these approximations, the resonant SAXS signal is given by

$$I(\vec{q}) = I_0 \frac{r_0^2}{A} \left| \mathcal{FT} \left[\int \left(\sum_{i \in (Si,Cu,\ldots)} n_i(\vec{r}) \cdot \left(f_i^0 + f_i'(E_0, T_e) + i \cdot f_i''(E_0, T_e) \right) \right) dz \right] \right|^2 d\Omega. \tag{4.9}$$

Here, the sum needs to include the main target material layers (Si and Cu) but also surface contamination layers, such as oxide or hydrocarbon-compound layers.

Fig. 4.5 shows the form factors for Si and Cu for the on- and off-resonant XFEL probe energies used in the experiments in dependence on the plasma bulk temperature. Most relevant for the resonant SAXS experiment is the increase of the opacity for on-resonant XFEL probe energy in the heated copper plasma by more than 2 orders of magnitude. This is reflected by an increase of the imaginary part (f_{Cu}'') of the ionic form factor. As for Si no bb-transition channels are present at this probe energy, the ratio of the imaginary part of the form factors of Cu and Si differs strongly as a function of the plasma temperature (see Fig 4.5c.) The mathematical illustration in Fig. 4.6 shows that asymmetric signals can be generated, if in projection along X-ray probe direction the real and imaginary part of the complex-valued scattering length areal density are differently distributed. This will be discussed in detail in the following section.

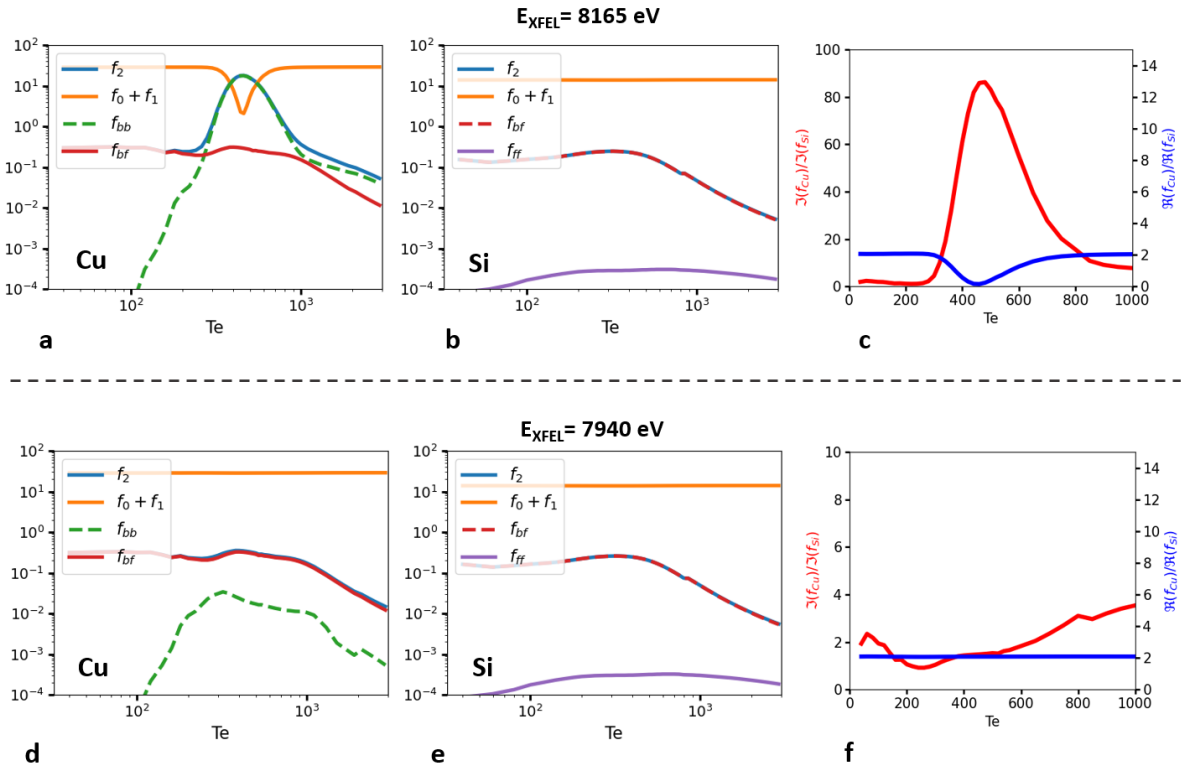

Figure 4.5: Dispersion corrections as a function of the plasma bulk temperature T_e for XFEL probe photon energies of 8165 eV (**a-c**) and 7940 eV (**d-f**) for Si and Cu and the ratios of the real parts ($f_0 + f'$) and imaginary parts (f'') of the form factors f_{Si} and f_{Cu}. Additionally shown are the contributions of f_{bb}, f_{bf} and f_{ff} to the dispersion correction f'', which were derived from the opacity for bound-bound, bound-free and free-free transitions. Of major relevance is the increase of f'' for Cu by several orders of magnitude for the on-resonant XFEL probe energy of 8165 eV, when the Cu plasma is heated to more than around 300 eV and the N-like K_α transition channel opens up. As can be seen in **c)**, this strong increase in opacity changes the ratio of the imaginary parts of the form factor of Cu and Si almost by a factor of 100. This effect does not occur for the off-resonant XFEL probe energy of 7940 eV (**f**). The data was obtained from FLYCHK runs for different LTE temperatures.

4.2 Asymmetric scattering on nanostructured compound grating targets

The complex-valued nature of the ionic form factor is of central importance for this work, as it allows to break Friedel's law or in other words it allows for non-point-symmetric scattering patterns, in the following referred to as *asymmetric* scattering patterns. Friedel's law is well-known from x-ray crystallography. It says, diffraction patterns from a crystal are always centrosymmetric, i.e. $I(\vec{q}) = I(-\vec{q})$, if the structure factor is real-valued, regardless of whether the crystal itself has a symmetry center. This is related to the so-called phase problem in x-ray diffraction. In experiments the measurable intensity on a detector is given by the modulus squared of the amplitude of the scattered wave, so that the phase information is lost. In a simplified example, this implies one cannot reconstruct the absolute orientation of a system

of two scattering objects with each real-valued scattering length from a diffraction experiment. Yet, close to elemental absorption edges, the dispersion corrections need to be added and lead to a complex-valued scattering length. Then, Friedel's law breaks down, if the scattering system in real space is not centrosymmetric. Asymmetries in scattering patterns as a consequence of the resonant scattering terms are exploited to circumvent the phase problem, e.g., by systematically analyzing differences of pairs of reflections (Bijvoet pairs[97]) in order to determine the absolute chirality of molecules. Another application can be found in Multi-Wavelength Anomalous Diffraction (MAD)[98, 99] to determine the structure of macromolecules.
Indeed, asymmetric SAXS patterns have been measured in the two recent XFEL pump-probe experiment at MEC[14, 17]. There is strong interest to understand the generation of asymmetric scattering signals and to make use of additional information contained in measurements of the latter. A semi-analytical model for asymmetry generation with SAXS for the generic case of structured multilayer targets and the specific case of 2-layer grating targets was developed by T. Kluge. In this section relevant considerations for the numerical asymmetry model for the 2-layer nanostructured Si-Cu gratings (see Fig. 4.7) employed in the recent SAXS experiments at MEC will be presented. The following text reuses material from the Supplementary of the related publication in Phys. Rev. Research[17], while the text is streamlined to the specific experimental cases discussed in this thesis.

4.2.1 Mathematical illustration

Firstly, the generation of asymmetric SAXS signals will be illustrated by means of a simple mathematical example (see Fig. 4.6) and the relevant Fourier Transform laws. The Fourier transform of any real valued function has an even real part and an odd imaginary part. The Fourier transform of any imaginary valued function has an odd real part and an even imaginary part. Consequently, the Fourier transform absolute square (i.e. scattering pattern) is symmetric only if the function to be transformed is either purely real valued, purely imaginary valued, or the real and imaginary part are either both even or both odd, or the real part is proportional to the imaginary part. If only Thomson scattering (i.e. scattering on quasi-free electrons) is considered, the scattering length density is purely real valued. For absorption only it is purely imaginary (e.g. in the case bound-bound transitions in highly stripped ions where the electrons have been removed from the scattering volume such as for example in the case of single clusters in a strong laser field).
As described in the previous section, for resonant SAXS on a HI laser-driven plasma, the scattering length density becomes complex-valued, when scattering on *free and bound* electrons needs to be considered. In case of the Si-Cu target configuration, symmetry breaking occurs, when the distributions of the layers are different either spatially or in density, so that in projection along the XFEL-beam path the different materials add up differently at different positions across the XFEL irradiated volume. Two specific origins of differences in the distributions of layered targets with the rear layer replicating the surface of the previous will be discussed in the following: a shift of layers with respect to each other due to a tilt of the target; and random target imperfections either due to small manufacturing jitter or due to surface contaminants. In fact, we attribute the latter two to be the leading origin of the asymmetries in the experiments presented in this thesis.

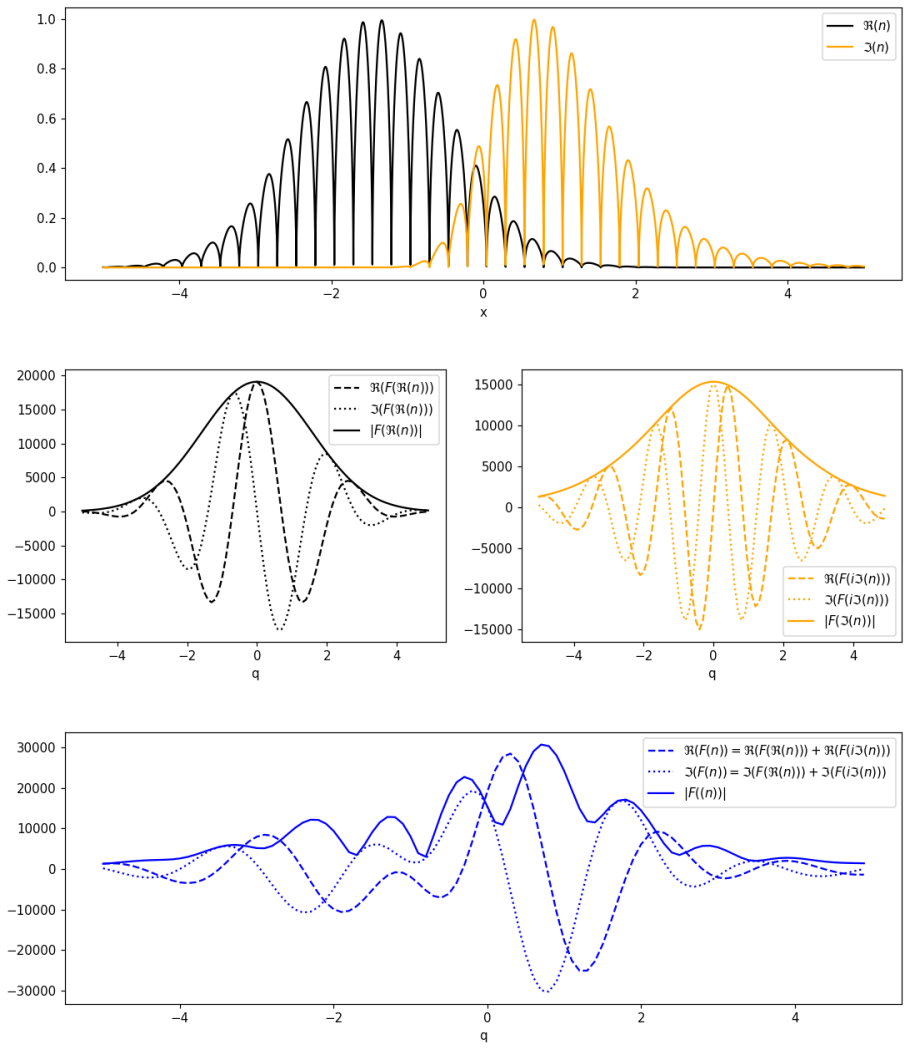

Figure 4.6: Schematics of the generation of asymmetry: An arbitrary complex valued function (top) is Fourier transformed. While the FT absolute value is symmetric when only considering the real part (middle, left) or the imaginary part (middle, right), when considering the complex function it is asymmetric when the real and imaginary parts differ from each other.

4.2.2 Derivation of the asymmetry model for structured multilayer targets

We start with a multilayer target with N layers with arbitrary interface shapes $z_i\left(\vec{r}\right)$ where i is the index of the layer interface and $\vec{r} = (x, y)$ is a vector within the plane perpendicular to the X-ray direction z. By this definition, layer i is located between $z_{i-1}\left(\vec{r}\right)$ and $z_i\left(\vec{r}\right)$, the front surface of the target is described by $z_0\left(\vec{r}\right)$.
The conventional definition of the replication factor of interface i with respect to the interface j is defined by the ratio of the power spectrum distributions (PSD) of layer i and j, and reads

$$R_{i,j}\left(\vec{q}\right) = \frac{\left|\tilde{z}_i\left(\vec{q}\right)\right|^2}{\left|\tilde{z}_j\left(\vec{q}\right)\right|^2}. \tag{4.10}$$

In the following, we use the tilde symbol to indicate the Fourier transform, $\tilde{z}\left(\vec{q}\right) = \mathcal{F}_{\vec{r}}\left[z\left(\vec{r}\right)\right]\left(\vec{q}\right)$. The vector $\vec{q} = \left(q_x, q_y\right)$ denotes the scattering vector $\vec{q} = \vec{k'} - \vec{k_0}$ where $\vec{k_0}$ is the wave vector of the incoming X-ray beam and $\vec{k'}$ is that of the scattered beam, assuming the scattering angle is small and the scattering process is sufficiently elastic, so that $q_z \approx 0$. The replication factor as defined above is 1 for perfect replication, i.e. $\tilde{z}_i\left(\vec{q}\right) = \tilde{z}_j\left(\vec{q}\right)$ and can take on values between 0 (for frequencies contained only in $\tilde{z}_j$) and infinity (for frequencies contained only in $\tilde{z}_i$). For simplicity, in the following we use the term *replication factor* for the complex-valued ratio

$$\chi_{i,j}\left(\vec{q}\right) = \frac{\tilde{z}_i\left(\vec{q}\right)}{\tilde{z}_j\left(\vec{q}\right)}. \tag{4.11}$$

Its difference between two adjacent interfaces reads

$$\Delta\chi_{i,j}\left(\vec{q}\right) \equiv \chi_{i,j}\left(\vec{q}\right) - \chi_{i-1,j}\left(\vec{q}\right). \tag{4.12}$$

With the definition $\Delta\chi_i \equiv \Delta\chi_{i,1}$ the difference between the Fourier components of the interface i and $i-1$ is simply given by $\tilde{z}_i - \tilde{z}_{i-1} = \Delta\chi_i\left(\vec{q}\right)\tilde{z}_1\left(\vec{q}\right)$. The intensity $I\left(\vec{q}\right)$ of the scattered light in Born approximation is proportional to the Fourier transform of the absolute square of the scattering length areal density $\zeta\left(\vec{r}\right)$ multiplied with the complex-valued X-ray field amplitude $a_0\left(\vec{r}\right)$, $I\left(\vec{q}\right) \propto \left|\mathcal{F}\left[a_0\left(\vec{r}\right)\zeta\left(\vec{r}\right)\right]\left(\vec{q}\right)\right|^2$. For a multilayer target with N layers, the scattering length areal density is given by

$$\zeta\left(\vec{r}\right) = \sum_{i=1}^{N} \rho_i\left(\vec{r}\right)\left[z_i\left(\vec{r}\right) - z_{i-1}\left(\vec{r}\right)\right] \tag{4.13}$$

and

$$\rho_i\left(\vec{r}\right) = n_i\left(\vec{r}\right)\left(f_i^0 + f_i' + if_i''\right). \tag{4.14}$$

Here, n_i denotes the atom or ion density in layer i and f_i^0, f_i', f_i'' are the ion form factor and optical corrections. In small angle approximation it is $f_i^0 \approx Z$, where Z is the atomic number of the atom or ion. Note, that this equation is valid without loss of generality, as layers consisting of mixtures of atoms or ions with different atomic numbers and/or different optical corrections could be easily represented by the respective average values. In the following, we assume that the materials of each layer are homogeneous, i.e. the density is not a function of $\vec{r}$, i.e. $\rho_i\left(\vec{r}\right) = \rho_i$. Then, the Fourier transform and its conjugate of equ. (4.13) read

$$\tilde{\zeta}\left(\vec{q}\right) = \sum_{i=1}^{N} \rho_i\left[\tilde{z}_i\left(\vec{q}\right) - \tilde{z}_{i-1}\left(\vec{q}\right)\right] \tag{4.15}$$

$$\tilde{\zeta}\left(-\vec{q}\right) = \sum_{i=1}^{N} \rho_i\left[\tilde{z}_i^*\left(\vec{q}\right) - \tilde{z}_{i-1}^*\left(\vec{q}\right)\right]. \tag{4.16}$$

The latter can be easily seen from splitting z_i into their even and odd components $z_{i,e}$, $z_{i,o}$, respectively: The Fourier transform of an even real-valued function is again even and real-

valued, while that of an odd real-valued function is odd and imaginary, hence

$$\begin{aligned} \tilde{z}_i\left(\vec{q}\right) &= \tilde{z}_{i,e}\left(\vec{q}\right) + \tilde{z}_{i,o}\left(\vec{q}\right) \\ &= \Re\left[\tilde{z}_{i,e}\left(\vec{q}\right)\right] + i\Im\left[\tilde{z}_{i,o}\left(\vec{q}\right)\right] \\ \tilde{z}_i\left(-\vec{q}\right) &= \Re\left[\tilde{z}_{i,e}\left(\vec{q}\right)\right] - i\Im\left[\tilde{z}_{i,o}\left(\vec{q}\right)\right] = \tilde{z}_i^*\left(\vec{q}\right) \end{aligned} \tag{4.17}$$

We will now analyze how the normalized asymmetry

$$\eta\left(\vec{q}\right) = \frac{I\left(\vec{q}\right) - I\left(-\vec{q}\right)}{I\left(\vec{q}\right) + I\left(-\vec{q}\right)} \tag{4.18}$$

is related to the replication factors. The above definitions allow us to give this equation explicitly in terms of the interface layer geometries in a very compact expression if $a_0 = const.$:

$$\eta\left(\vec{q}\right) = \frac{\left|\sum_{i=1}^{N} \rho_i \Delta\chi_i\left(\vec{q}\right)\right|^2 - \left|\sum_{i=1}^{N} \rho_i \Delta\chi_i^*\left(\vec{q}\right)\right|^2}{\left|\sum_{i=1}^{N} \rho_i \Delta\chi_i\left(\vec{q}\right)\right|^2 + \left|\sum_{i=1}^{N} \rho_i \Delta\chi_i^*\left(\vec{q}\right)\right|^2} \tag{4.19}$$

which can be rewritten as

$$\eta\left(\vec{q}\right) = \frac{\sum_{i=1}^{N}\sum_{j=1,j\neq i}^{N} a_{ij}\Im\left(\Delta\chi_i^*\left(\vec{q}\right)\,\Delta\chi_j\left(\vec{q}\right)\right)}{\sum_{i=1}^{N}\sum_{j=1}^{N} b_{ij}\Re\left(\Delta\chi_i^*\left(\vec{q}\right)\,\Delta\chi_j\left(\vec{q}\right)\right)}. \tag{4.20}$$

This equation connects the asymmetry with the replication factors and the material-dependent factors

$$a_{ij} = 2\left(f_j^0 + f_j'\right) f_i'' n_i n_j \tag{4.21}$$

$$b_{ij} = \left[\left(f_j^0 + f_j'\right)\left(f_i^0 + f_i'\right) + f_i'' f_j''\right] n_i n_j. \tag{4.22}$$

Most notably, this relation is independent of the specific geometry of the layer interfaces, i.e. all $\tilde{z}_i$ have completely cancelled out. Only the relative differences, described by the replication factors, are important.
Secondly, the replication factors need to be complex-valued, i.e. the imaginary part must not vanish completely for all replication factors in order for the asymmetry to be non-zero. A simple consequence is that the asymmetry is constant when the target explodes symmetrically. When it expands longitudinally, the densities reduce, but since a_{ij} and b_{ij} are both quadratic in n, this change cancels out. When the interfaces expand laterally, i.e. $\vec{r}' = \vec{c}_i\vec{r}$ (here the prime denotes the positions after expansion), then the replication is only real-valued, $\Delta\chi_i\left(\vec{q}\right) = \tilde{z}_i\left(q/c_i\right)/c_i\tilde{z}_i\left(q\right)$ and therefore even.
Thirdly, at least two layers are needed to generate an asymmetry, since otherwise the sum over $j \neq i$ in the numerator is not defined. This is physically reasonable since the asymmetry is generated by a difference between the spatial areal distribution of the real part of the form factors and optical corrections on the one hand, and the imaginary part of the optical corrections on the other hand. Since by definition there is no difference between them in one layer (up to constant factor), at least two layers are needed. Additionally, in order to generate asymmetry, for each pair of two layers it is required that the ratios between the real and imaginary parts are different, i.e. $a_{ij} \neq a_{ji}$.

4.2.3 Asymmetry model for the 2-layer Si-Cu-compound grating targets

In the following, the case $N = 2$ relevant for the Si-Cu target (see Fig. 4.7) will be considered. During the target production, a copper layer has been sputtered onto a Silicon grating target. As a result of this, the rear Cu layer represents a smoothed replication of the buried Si grating structure. The respective height functions of the flat front and grated buried and rear interface structures will be described by $z_0(\vec{r})$, $z_1(\vec{r})$ and $z_2(\vec{r})$, respectively.

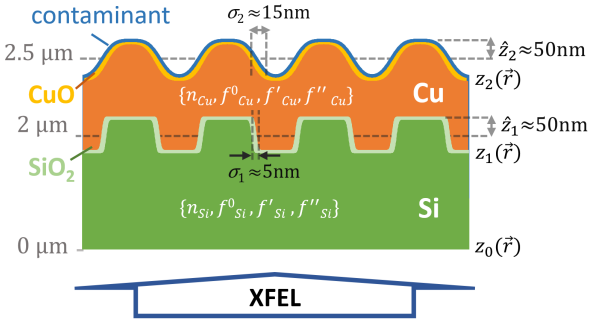

Figure 4.7: Target scheme of the Si-Cu compound grating target.

Using eqn. 4.20 one finds the following expression for a 2-layer target:

$$\eta\left(\vec{q}\right) = \frac{c_1 \Im\left(\chi\left(\vec{q}\right)\right)}{\Re\left[c_2\left|\chi\left(\vec{q}\right)\right|^2 + c_3\chi\left(\vec{q}\right) + c_4\right]} \tag{4.23}$$

with the material constants

$$\begin{aligned} c_1 &= 2n_1n_2\left[(f_1^0+f_1')f_2'' - (f_2^0+f_2')f_1''\right] \\ c_2 &= n_2^2\left[\left(f_2^0+f_2'\right)^2 + \left(f_2''\right)^2\right] \\ c_3 &= 2n_1n_2\left[\left(f_1^0+f_1'\right)\left(f_2^0+f_2'\right) + f_1''f_2''\right] - c_2 \\ c_4 &= n_1^2\left[\left(f_1^0+f_1'\right)^2 + f_1''\right] - 2\left(c_2+c_3\right) \end{aligned}$$

and $\chi\left(\vec{q}\right) \equiv \chi_{2,1}\left(\vec{q}\right)$. For reference, the constants in case of the Si-Cu target are $c_1 = 3.78{\cdot}10^{-3}$, $c_2 = 14.9$, $c_3 = -6.0$, $c_4 = -13.6$. Eqn. 4.23 connects the experimentally accessible asymmetry function $\eta\left(\vec{q}\right)$ with the material properties (form factors and optical corrections in the factors c_i) and the replication factor of the rear surface with respect to the buried interface $\chi\left(\vec{q}\right)$. While χ in equ. 4.23 is generic, one can find a simple expression for the latter for the Si-Cu target geometry. As the rear Cu interface structure is a smooth replication of the buried interface structure, it can be described in real space by a convolution with a Gaussian function. In Fourier space, this convolution translates to

$$\tilde{z}_2(\vec{q}) = \chi_0(\vec{q}) \cdot \tilde{z}_1(\vec{q}) = \hat{\chi} e^{-\sigma^2 q^2} \cdot \tilde{z}_1(\vec{q}). \tag{4.24}$$

Here, $\sigma^2 = \sigma_2^2 - \sigma_1^2$ describes the relative smoothing of the rear grating structure. Real-valued σ correspond to smoothing at the rear, while imaginary values correspond to sharpening. The variable $\hat{\chi}$ is a proportionality factor describing the relative grating height.

In the following, three identified causes for symmetry breaking will be presented:

1. a shift of the rear grating relative to the buried grating
2. noise by small, random fluctuation of geometrical target parameters
3. surface contamination and oxide layers

Shift and convolution with an even function

One way to break the symmetry is a shift of the rear surface with respect to the buried interface by $\vec{r}_s$ which could originate, e.g., due to a small tilt of the target. A shift in real space is expressed in reciprocal space by a multiplication with

$$\chi_s\left(\vec{q}\right) = e^{-i\vec{r}_s\vec{q}}. \tag{4.25}$$

The replication factor of the rear surface is then given by $\chi\left(\vec{q}\right) = \chi_0\left(\vec{q}\right)\chi_s\left(\vec{q}\right)$ and equ. 4.23 can be further simplified to

$$\eta\left(\vec{q}\right) = \frac{c_1\chi_0\left(\vec{q}\right)\sin\left(\vec{r}_s\vec{q}\right)}{c_2\chi_0^2\left(\vec{q}\right) + c_3\chi_0\left(\vec{q}\right)\cos\left(\vec{r}_s\vec{q}\right) + c_4}. \tag{4.26}$$

This relation can be readily solved for $\chi\left(\vec{q}\right)$, which demonstrates that the asymmetry of the scattering pattern directly maps the replication factor in $\vec{q}$-space.

The asymmetry as a function of the three independent parameters determining the replication factor is plotted in Fig. 4.8. Here and in the following we average the absolute value of $\eta(q)$ over the q-range from $0-0.3\,s^{-1}$, where s is the unit spatial dimension (e.g. 1nm). We can see that the maximum asymmetry occurs when $\sigma = 0$ and $\hat{\chi} \approx 0.7$ to 0.75 and $r_s \approx 1.5$s. The specific optimum parameters are material-dependent, but vary only little. For example, increasing the imaginary part or density of either material by an order of magnitude only changes the optimum geometric parameters by the order of 10%. Furthermore, the optimum value of the replication factor amplitude $\hat{\chi}_{opt} \approx 0.7$ is almost independent of $\vec{r}_s$ and σ.

Obviously, one interesting application of the asymmetries is to experimentally restrict the possible range of the geometric parameters. Additionally, the change of the asymmetry upon an external drive can also be used to restrict the change of the replication factor. For example, if the rear layer surface initially perfectly replicates the buried interface ($\hat{\chi} = 1, \sigma = 0$) then from Fig. 4.8 one finds that an increase in the average asymmetry is equivalent to a decrease in χ. If the rear layer does not perfectly replicate the buried interface, e.g. $\hat{\chi} < \hat{\chi}^{opt}$ or $\sigma > \sigma^{opt}$, then this relation is inverted, i.e. an increase of the average asymmetry means an increase in χ.

The Si-Cu targets were designed to investigate the smoothing of a buried layer upon HI laser pump. In conventional transmission SAXS the Fourier components of the buried and rear surface correlations overlap on the detector and cannot be reconstructed separately. The asymmetry offers a way to obtain the smoothing parameter σ which – assuming that upon pumping both target interfaces can only smooth out further – can be used to extract the smoothing of the buried and rear interface separately. Fig. 4.9 shows the average asymmetry in dependence on the buried and rear grating smoothness for different values of $\hat{\chi}$ and

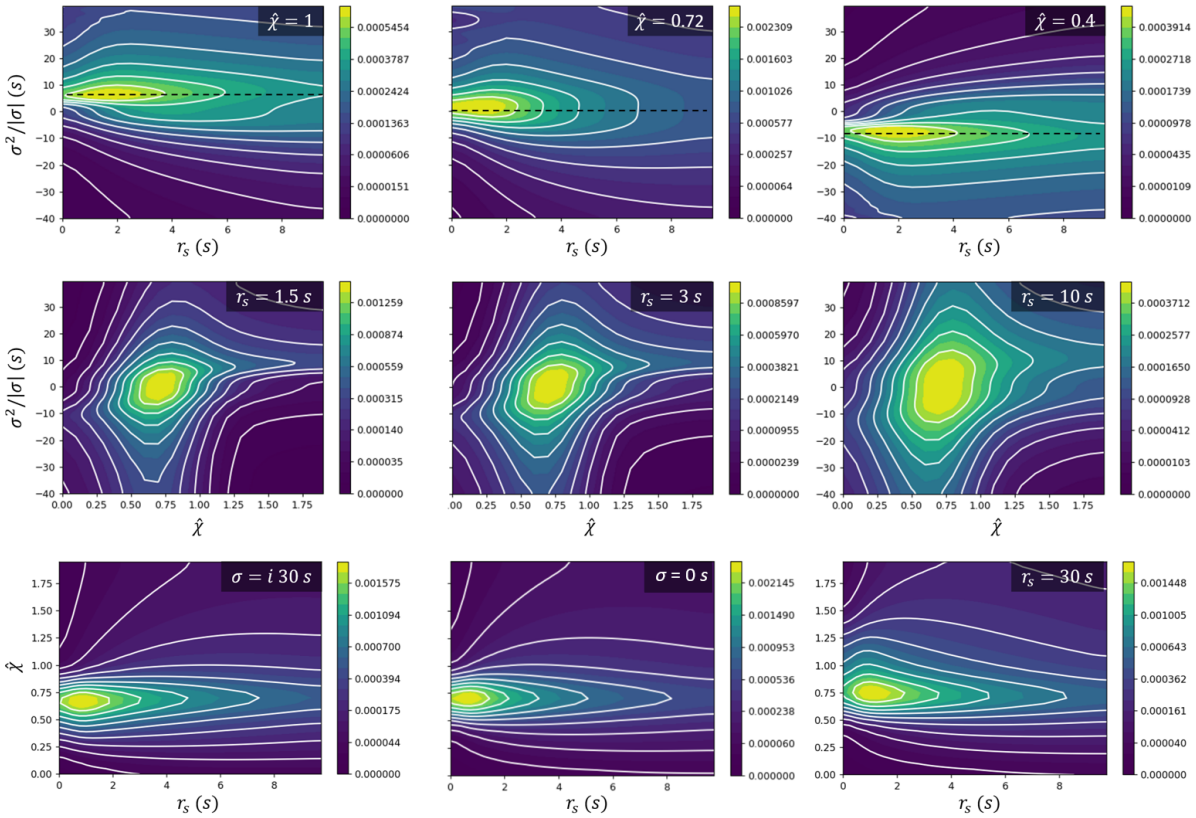

Figure 4.8: Average asymmetry in the q-range from 0 – $0.3s^{-1}$ (as for all figures below) for a two layer Si-Cu compound with flat Si front surface and $\chi\left(\vec{q}\right) = \hat{\chi}\exp\left(-\sigma^2q^2 - i\vec{r}_s\vec{q}\right)$, scanning the three parameters $\sigma, \hat{\chi}$ and r_s.

a fixed shift $r_s = 1.5s$. Depending on the geometric target parameters $(\hat{\chi}, \sigma_1, \sigma_2)$ given in the initial condition, a change in the asymmetry upon laser pump is connected with a dominant smoothing of the buried or rear layer. Clearly, extraction of quantitative information on the expansion by means of asymmetry changes requires modeling of the target geometry as close to reality as possible. This includes consideration of target imperfections as described in the following.

Noise

So far, we have only discussed the case of a perfect target. It is important to realize that small random fluctuations in the target geometry can also lead to a symmetry breaking, i.e., they do not necessarily cancel out and can take the role of the small shift $\vec{r}_s$ from the former subsection. However, simple fluctuations such as introducing a set of different target areas with different $\vec{r}_s^{\,i}$ or different χ_0^i in fact would tend to cancel out due to the linear and odd nature of the numerator of equ. 4.26. What is needed to generate an asymmetry are non-compensated shifts.

In case of the Si-Cu targets, we can assume noise being present in the form of random but small deviations from the perfect grating, i.e., each grating ridge is assigned to a random shift and width. Both are absorbed by random variation of the position of the left/right edge ridge i by $\delta^i_{l/r}$ relative to the edges of the buried layer. In that case and if the edge position variations

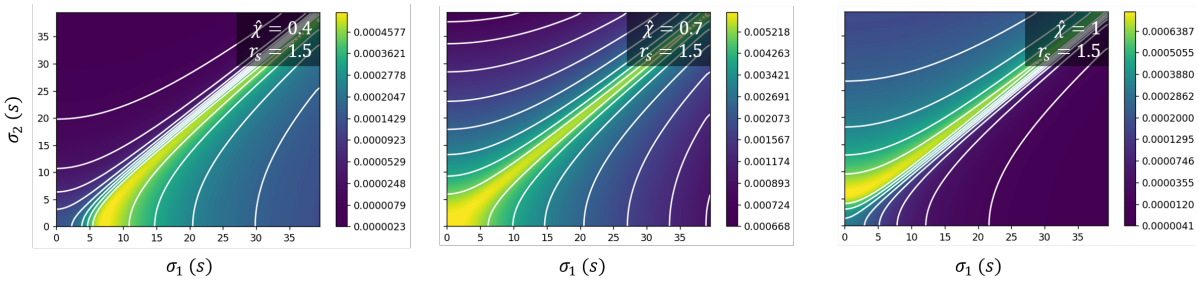

Figure 4.9: Average asymmetry for a two layer Si-Cu compound with flat Si front surface and $\chi(\vec{q}) = \hat{\chi}\exp\left(-\sigma^2 q^2 + i\vec{r}_s\vec{q}\right)$, $\sigma^2 = \sigma_2^2 - \sigma_1^2$ describes the relative smoothing. This exemplifies the specific case where both the buried interface (index 1) and rear surface (index 2) are described by the same function $z(\vec{r})$ and are each subject to a respective smoothing factor $\exp\left(-\sigma_i^2 q^2\right)$ in Fourier space.

all are much smaller than the average grating ridge width w itself, the replication factor can be written as

$$\chi_\delta(\vec{q}) = \chi_0(\vec{q}) \frac{\sum_k \left[\sum_m \left(e^{iq\left(w/2+\delta_{r,2}^{k,m}\right)} - e^{-iq\left(w/2-\delta_{l,2}^{k,m}\right)}\right) e^{i2kqw}\right]}{\sum_k \left[\sum_m \left(e^{iq\left(w/2+\delta_{r,1}^{k,m}\right)} - e^{-iq\left(w/2-\delta_{r,1}^{k,m}\right)}\right) e^{i2kqw}\right]}, \tag{4.27}$$

which at the scattering peak positions $\vec{q}_N = N2\pi/w$ and in small angle approximation simplifies to

$$\chi_\delta(\vec{q}_N) \cong \chi_0(\vec{q}_N) \frac{\sum_k \left[\sum_m \frac{N\pi}{w}\left[(\delta_{l,2}^{k,m})^2 - (\delta_{r,2}^{k,m})^2\right] + i(\delta_{r,2}^{k,m} + \delta_{l,2}^{k,m})\right]}{\sum_k \left[\sum_m \frac{N\pi}{w}\left[(\delta_{l,1}^{k,m})^2 - (\delta_{r,1}^{k,m})^2\right] + i(\delta_{r,1}^{k,m} + \delta_{l,1}^{k,m})\right]}. \tag{4.28}$$

Here, the sum over k goes over all grating ridges, the sum over m denotes the variation of parameters along the individual ridges. We first note that mere scale changes (i.e. $\delta_{r,i}^{k,m} = -\delta_{l,i}^{k,m}$) cancel out the real part of both the nominator and denominator, which means explicitly for real-valued χ_0 that it remains real-valued and no asymmetry is generated. Additionally, as long as local shifts are symmetrically distributed, they cancel out, hence only a net shift of the center of mass averaged over all ridges can generate asymmetry, which then coincides with the former case of equ. (4.25) with r_s substituted by the average net shift of the ridges. This can be seen in Fig. 4.10, where we plot the average asymmetry as a function of the smoothness parameters σ_1 and σ_2 for different numbers n of randomly distributed sub-volumes and $r_s = 0$. Indeed, for $n \gg 1000$ the asymmetry vanishes. Then, the asymmetry is similar to that of a perfect grating. However, this is only true for very large values of n. For smaller n, the random shifts in fact do generate asymmetry, qualitatively and quantitatively similar to the case of a perfect grating but $r_s \neq 0$.

Surface contamination and oxide layers

A structurally different form of asymmetry is generated by another form of noise present in a realistic target: surface contamination in the form of few nanometer thick hydro-carbon compound contamination layers[100] or oxides on the surfaces of the materials forming a

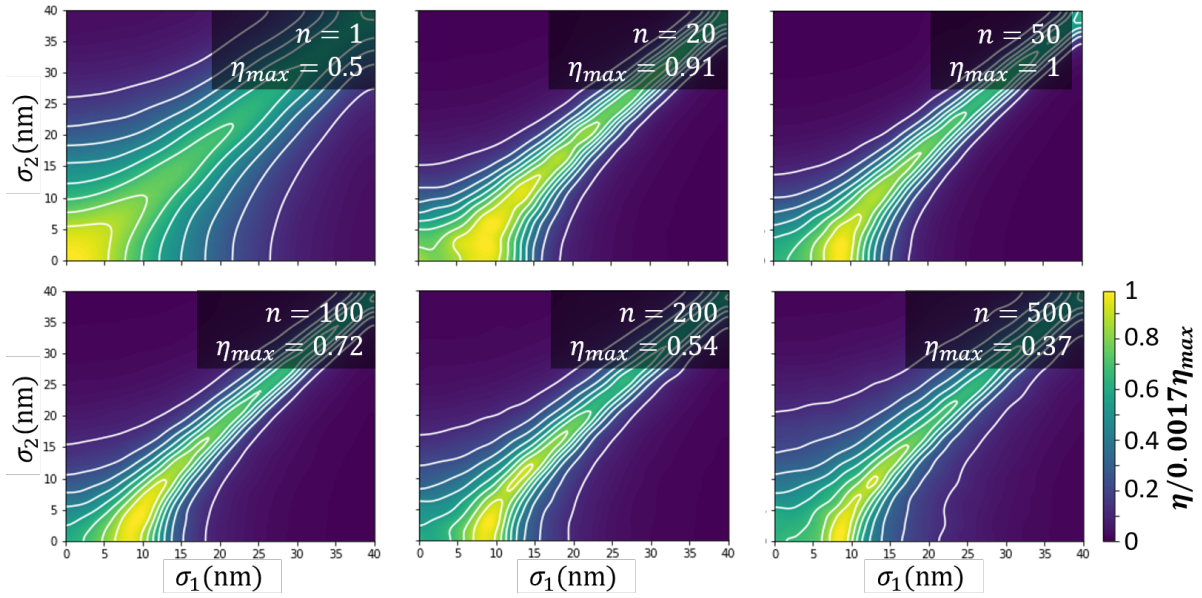

Figure 4.10: Normalized average asymmetry for a SiCu compound target with buried and rear surface gratings, and $\chi\left(\vec{q}\right) = \hat{\chi}\exp\left(-\sigma^2 q^2 - i\vec{r}_s\vec{q}\right)$, $\sigma^2 = \sigma_2^2 - \sigma_1^2$, with $\hat{\chi} = 0.7$. We introduced 10% normal distributed random variations in the ridge edge positions and heights, and varied the number n of sub-volumes with different positions.

thin, yet inhomogeneous layer on the surfaces. Metals tend to oxidize quickly to oxidation layers thicknesses of a few nanometer, Cu e.g. with approximately 5nm [101]. Also silicon can oxidize to a thickness of a few atom layers [102, 103]. Due to their small atomic number Z^x they usually have a very small imaginary part of the optical correction at few keV X-ray photon energy. Having only negligible X-ray absorption in the multi-keV range, we limit ourselves to the case of a contribution of the surface contamination layers to Thomson scattering only and ignore absorption, i.e. $f' = f'' = 0, f_0 = Z^x$. We now want to add the effect that an oxidation layer or surface contamination may have on the asymmetry in equ. 4.23. The surface and oxidation layers need to be considered as additional summands in equ. 4.13. Considering only a single grating ridge of width w, their height functions read

$$z(x) = \frac{1}{2}\left(\mathrm{erf}\left(\frac{x+\frac{w}{2}}{\sqrt{2}\sigma}\right) - \mathrm{erf}\left(\frac{x-\frac{w}{2}}{\sqrt{2}\sigma}\right)\right). \tag{4.29}$$

The oxidation layer projection $\Delta z^x(x)$ is then given, for a layer thickness w^x, approximately by

$$\Delta z^x(x) = \xi\left(\exp\left(-\frac{\left(x+\frac{1}{2}\left(w \pm w^x\right)\right)^2}{2\sigma^2}\right) + \exp\left(-\frac{\left(x-\frac{1}{2}\left(w \pm w^x\right)\right)^2}{2\sigma^2}\right)\right) \tag{4.30}$$

with $\xi \cong 0.4\left(\frac{w^x}{\sigma}\right)^{0.843}$. The plus sign refers to surface contamination while the minus sign refers to oxidation layers. In equ. (4.19), we now have to add the term $\tilde{\zeta}^x = \rho^x \Delta\chi^x(q)$, where

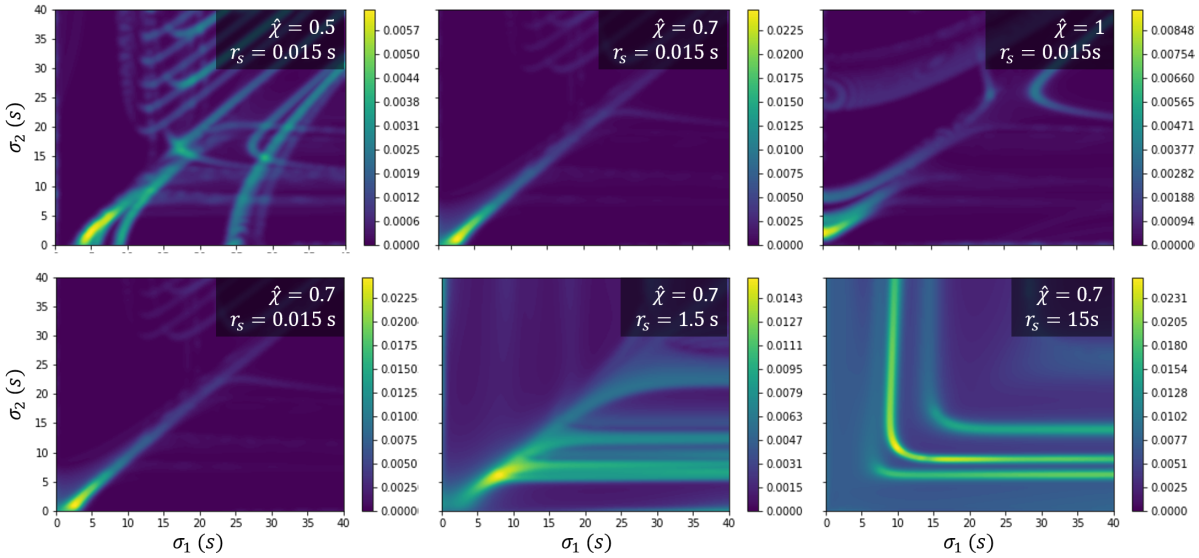

Figure 4.11: Average asymmetry for a SiCu compound target with buried and rear surface gratings, and $\chi\left(\vec{q}\right) = \hat{\chi}\exp\left(-\sigma^2 q^2 - i\vec{r}_s\vec{q}\right)$, $\sigma^2 = \sigma_2^2 - \sigma_1^2$, as before. We added CuO and hydro-carbon-oxide (thickness of both 5% of $\hat{z}_1$) and SiO (thickness 0.3% of $\hat{z}_1$) surface contamination layers to all applicable surfaces. Those values correspond to typical oxidation layer thicknesses ($\approx$ 5nm and 0.3nm for Cu0 and SiO, respectively) and hydro-carbon-oxide layer thickness ($\approx$ 5nm), when a grating height of $\hat{z}_1 = 100$nm is adopted.

$\rho^x = Z^x n_i^x$ and $\Delta\chi^x(q)$ at the scattering peak positions reads

$$\Delta\chi^x(q) = \Delta\tilde{z}^x(q)/\tilde{z}_1\,(q) \cong \frac{\xi\sigma q\cos\left(\frac{1}{2}q\left(w \pm w^x\right)\right)e^{-\frac{1}{2}q^2(\sigma^2-\sigma_1^2)}}{\sin\left(\frac{qw}{2}\right)}.$$

The random fluctuations can then be introduced in the same manner as carried out above in equ. (4.27).
Following the derivation of eqn. 4.23 above, we can give a similar equation for the asymmetry of a two-layer target including surface layers,

$$\eta\left(\vec{q}\right) = \frac{\Im\left(c_0^x + c_1^x\chi\left(\vec{q}\right)\right)}{\Re\left[c_2^x\left|\chi\left(\vec{q}\right)\right|^2 + c_3^x\chi\left(\vec{q}\right) + c_4^x\right]} \tag{4.31}$$

where the constants c_i^x are now not only material specific constants but rather depend also on the surface structure itself via $\Delta\chi^x(q)$

$$\begin{aligned}
c_0^x &= 2n_1 n_x f_1'' Z^x \Delta\chi^x\left(\vec{q}\right)\\
c_1^x &= c_1\\
c_2^x &= c_2\\
c_3^x &= c_3 + n_2 n_x (f_2^0 + f_2')Z^x\Delta\chi^x(q)\\
c_4^x &= c_4 + 2n_1 n_x (f_1^0 + f_1')Z^x\Delta\chi^x(q) + (n_x Z^x)^2\left|\Delta\chi^x(q)\right|^2.
\end{aligned}$$

The asymmetry from this equation is plotted in Fig. 4.11 for different values of $\hat{\chi}$ and r_s for a Si-Cu sandwich target assuming a flat front side, a grating interface and a grating rear surface and no tilt, buried grating height 100s, grating period 300s. We find that especially r_s has a large effect. While for small values of r_s the pattern is similar to that without surface contamination, albeit now much larger in absolute terms, for larger r_s the pattern changes significantly.

Again one can convince oneself that the imaginary part of $\Delta\chi^x$ is vanishing for $r_s = 0$ and symmetric random fluctuations, so that again no asymmetry is expected if no tilt is present. As before, for values $n \lesssim 1000$ we find numerically that the asymmetry is not vanishing. A surprisingly stable asymmetry pattern can be found that depends only little on the random geometric variations but more pronounced on the geometry of the target layers, which therefore is accessible for measurements without the detailed knowledge of the noise and contamination layers. The underlying reason for this effect is that the random variations of positions in fact do introduce an imaginary part of the replication factor - both of the target layers (χ) as well as the surface contamination ($\Delta\chi^x$). Its compensation depends on the illuminated target size and amount of randomness. If both are small, there simply is not enough variation in order to be fully compensated. One can conclude that for very large illuminated target areas and small correlation lengths of the noise the asymmetry generated by the noise disappears and the asymmetry is given by equ. (4.23), (4.26), or (4.31).

For all other cases the variation has to be taken into account numerically. In Fig. 4.12 we show the numerically calculated asymmetry maps for the Si-Cu sandwich target. We vary the copper oxide layer thickness relative to the buried interface grating height, $\Delta\hat{z}^{CuO}/\hat{z}_1$, the width ε of the random gaussian distribution around the mean values of the grating ridge position offset (mean=0) and CuO layer thickness (mean=0.05), and the q-range over which the average of the asymmetry is taken.
Regardless of the specific values of the surface layers or the grating heights, we can find certain general laws:

§1: The asymmetry for a two material compound target with surface contamination, flat front surface and given σ_1, σ_2 is a strong function of the randomness ε and the q-range over which the asymmetry is averaged.

§2: The choice of the 'randomness' ε has great impact on the asymmetry for small values of ε, while it converges to a stable pattern for larger values.

§3: The asymmetry is largely independent on variations of the exact geometry, i.e. the pattern is stable with respect to different mean values of the relative surface layer thicknesses (see panel a) or ridge heights (b).

§4: The asymmetry with surface contamination and small randomness is distributed qualitatively similar to the case without the surface contaminants, but much larger and blurred.

§5: When ε is increased to above 10%, the asymmetry pattern changes qualitatively similar to how it would change without randomness (see. Fig 4.11), but now being blurred due to the randomness.

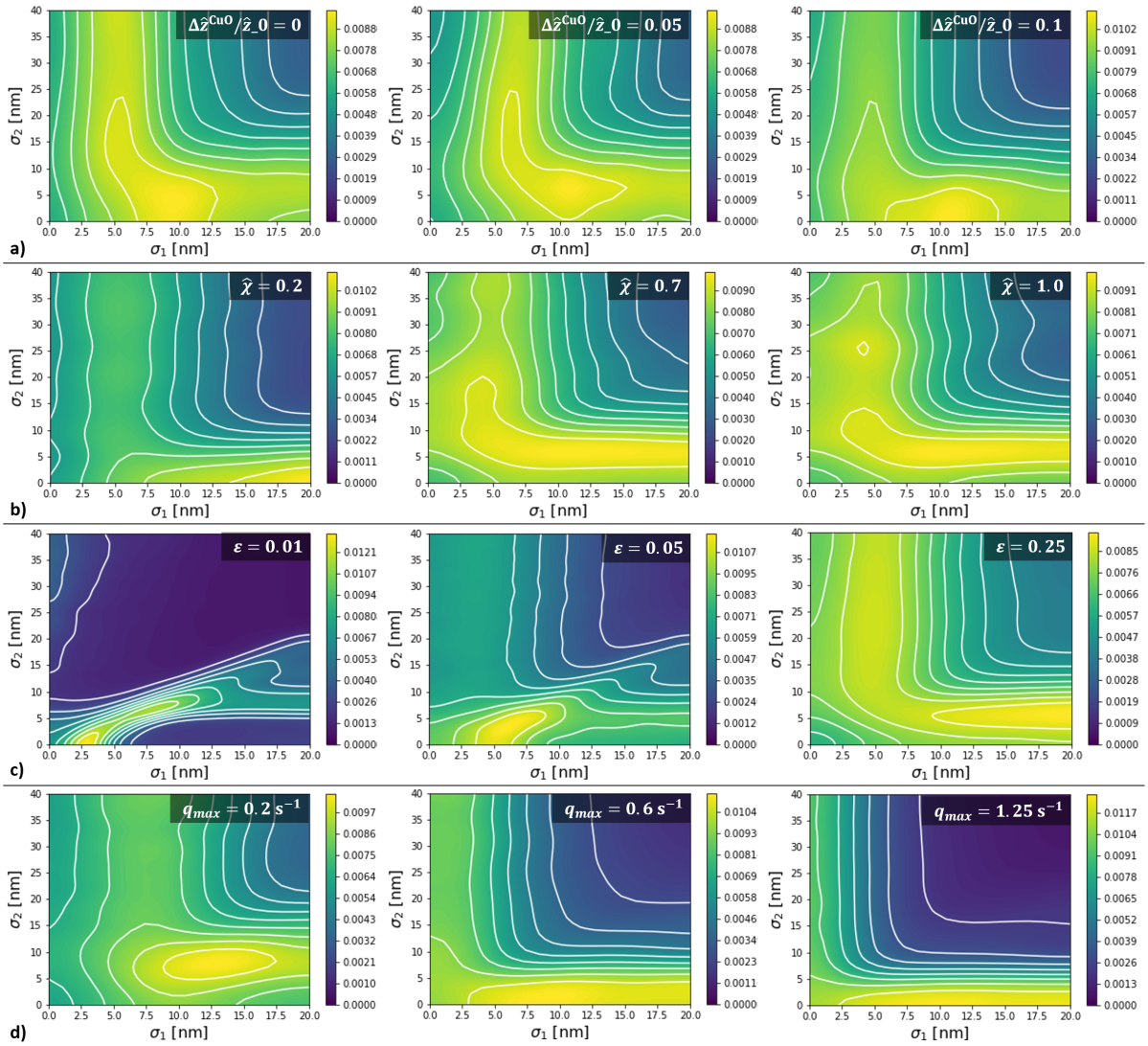

Figure 4.12: Average asymmetry in the q-range from 0 – $0.3s^{-1}$ for a SiCu compound target with buried and rear surface gratings, and $\chi(\vec{q}) = \hat{\chi}\exp(-\sigma^2 q^2)$, $\sigma^2 = \sigma_2^2 - \sigma_1^2$ with $\hat{\chi} = 0.7$. We added CuO, hydro-carbon-oxide (both thickness 5% of $\hat{z}_1$) and SiO (thickness 0.3% of $\hat{z}_1$) surface contamination layers to all applicable surfaces, as before. We introduced $\varepsilon = 10\%$ normal distributed random variations in the ridge edge positions, heights and layer thicknesses with $n = 100$ sub-volumes with different positions. In a) we vary the thickness of the CuO oxidation layer; in b) we vary $\hat{\chi}$; in c) we vary ε; in d) we vary the q-range over which the average is taken.

5 Probing Ultrafast Laser-Plasma Processes Inside Solids with Resonant Small-Angle X-ray Scattering

An XFEL pump-probe experiment for measuring HI laser-solid interaction has been performed at the Matter in Extreme Conditions (MEC) endstation of the Linac Coherent Light Source (LCLS) at the Stanford Linear Accelerator Center (SLAC) in 2018. The experiment was realized in the frame of a big collaborative effort and goes back to theoretical[91, 94] and experimental[14, 104] groundwork of previous years. The results were published in Phys. Rev. Research[17] by the author of this thesis and will be presented in this chapter. Large parts of the following text can therefore be found in the main text and supplementary text of the respective publication.

In the experiments, 1D (multilayer) grating targets with structures positioned at different depths were used, as they yield high signal at high q components that correspond to nm resolution. The pre-structured targets provide a comparably clean initial condition and allow us to follow the changes induced by the rapid plasma dynamics via the pump-probe technique in a transmission probe geometry (see Fig. 5.1). We used the MEC short-pulse HI pump laser[1] to generate solid-density plasma, i.e., induce fast, non-thermal melting and ionization/excitation. The LCLS XFEL beam was used to diagnose the plasma dynamics by means of SAXS. Both the HI drive laser and XFEL probe beam axes are oriented at 45° with respect to the target surface normal, in a plane parallel to the grating ridges.

With SAXS, the complex plasma dynamics becomes accessible via the temporal evolution of the nanoscale Fourier components of the scattering length density $\rho(\vec{r}, t) = n_e + n_i f_i$, where n_e

[1] This is a Titanium:Sapphire-based high power laser system based on chirped pulse amplification.

and n_i are the free electron and ion densities, respectively, and f_i is the ion form factor. The use of structured targets has so far proven to be particularly useful and sensitive to nanoscopic, ultrafast dynamics of the plasma electron density (n_e) employing SAXS[13, 14].
In the present work we also measure in a small-angle transmission geometry, but here we take advantage of a different, so far ignored, property of the scattering patterns, namely their asymmetry. This enables us to measure the ionic contribution to the scattering, which is primarily given by the imaginary part of ρ via X-ray absorption, e.g., due to bound-bound transitions. This means we gain access to the plasma opacity simultaneously to the plasma density distribution via (resonant) SAXS. We demonstrate that the asymmetry arising from differences in the distributions of electrons and ions can be used to reconstruct spatial correlations between 3D structures and their respective material properties. Tuning the narrow-bandwidth XFEL photon energy to a specific atomic bound-bound transition energy allows us to turn on and off the asymmetry effect[94]. Since the opacity depends on the existence of specific ionization states in the plasma, which in turn depend on the plasma temperature, with according modeling this method resembles an ultrafast in-situ temperature measurement, similar for example to X-ray Thomson Scattering (XRTS)[18] but with the potential of high spatial resolution similar to the element-specific imaging technique in [19].

Two specific science cases have been investigated, with each according targetry and experimental settings of the XFEL probe and optical pump laser. In *case 1*, Si-Cu compound grating targets were employed to investigate the heating and expansion of a buried grating structure at a moderate HI laser intensity of $3-7\cdot10^{17}$ W/cm^2. The Si-Cu target consists of a 2 µm thin silicon grating, which was covered with a copper layer of a few hundred nanometers thickness. With this, the buried grating structure at the Si-Cu interface imprints also at the Cu-vacuum interface in the form of a smooth replication. As the buried grating structure is located at 2 µm depth, it is protected from direct laser interaction by the Si membrane. *Case 2* was performed with the goal to measure opacity changes in a highly ionized Cu plasma following the HI laser pump. For this part of the experiment, the MEC optical laser allowed to enter the relativistic intensity regime with a laser intensity of $2-4\cdot10^{19}$ W/cm^2. To generate highly ionized Cu, pure Cu targets have been employed in order to irradiate the Cu-layer directly (details in Sec. 5.1.3).

5.1 Experimental setup and methods

5.1.1 Optical laser

The optical and XFEL configuration for *case 1* is the same as described in [14]. The optical MEC short-pulse laser is a Titanium:Sapphire-based high power laser system based on chirped pulse amplification. It provided ultrashort pulses ($\tau = 80$ fs) at a central wavelength of 800 nm with an energy of 0.4 J or 1 J before the compressor which corresponds to 180 mJ and 460 mJ on target, focused by an off-axis parabolic mirror to a spot size of 30 µm × 16 µm FWHM.

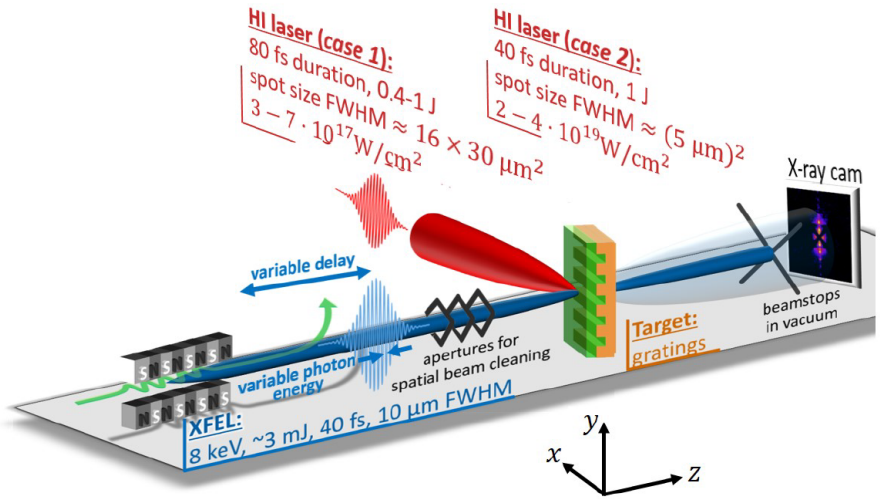

Figure 5.1: Schematic of the Resonant-SAXS experimental setup. A near infrared HI laser pulse is focused onto a silicon-copper compound target with a grating interface (*case 1*) or copper grating (*case 2*). The profile edges are assumed to follow an error function, $z \propto \mathrm{erf}\left[(y - y_0) / \left(\sqrt{2}\sigma_{1/2}\right)\right]$. Both the HI drive laser and XFEL probe beam axes are oriented at 45° with respect to the target surface normal, parallel to the grating ridges and polarized as shown.

For *case 2*, the laser was compressed and focused more tightly ($\tau = 40$ fs, spot size of 5 µm FWHM) resulting in two orders of magnitude higher peak intensity on target of approximately $2 - 4 \cdot 10^{19}$ W/cm^2.

5.1.2 XFEL

We used two ranges of X-ray photon energies: off-resonant, i.e., 7990 eV (*case 1*) and 7940 eV (*case 2*) and resonant, i.e., 8065 eV (*case 1*), 8165 eV (*case 2*). At the off-resonant X-ray energies, the cross sections are dominated by photo-ionization (bf transitions) and Thomson scattering. Photon energies on Cu-K$_\alpha$ bound-bound resonance transitions allowed us to probe the ionization state of Cu via X-ray absorption. The fundamental frequency of the LCLS X-ray beam was focused with compound refractive lenses into the MEC experimental area to spot sizes of 20 µm FWHM (*case 1*) and between 5 and 10 µm FWHM (*case 2*). The 3rd harmonic was only weakly focused and hence its intensity is significantly reduced on target. For *case 2* a high harmonic rejection mirror system was used. While in *case 1* the XFEL pulse intensity had to be attenuated by various Si and Cu absorbers to ensure the scattering signal was within the dynamic range of the PIXIS XF 2048B camera, for *case 2* we positioned small absorber plates (Ti and Mo, transmission of ~10 %) in the low q range ($\lesssim 0.05$ nm^{-1}) in front of the camera to selectively attenuate the first scattering peaks only. This allowed us to use much higher XFEL transmissions for *case 2*, between 20 % (preshots) and 100 % (main shots), than for *case 1* (1 %

and 5 %, respectively). Consequently, we could measure the signal to higher values of q. The delay time error of ± 114 fs between the optical laser and the XFEL is given by the r.m.s. jitter of the XFEL.

To record the SAXS signals, we used a PIXIS XF 2048B X-ray camera. For absolute photon numbers we calibrated it using an Am^{241} and an Fe^{55} source. The system resolution is dictated by the PIXIS point spread function and XFEL beam divergence, which are both between 2 and 3 pixels on the detector.

5.1.3 Target samples

For the experiments we used grated layer targets (see schematic drawing and scanning electron microscopy (SEM) images in Fig. 5.2). For *case 1*, we employed compound gratings of a 2 µm thin silicon support membrane ($n_e = 7 \cdot 10^{23}\,\mathrm{cm}^{-3}$, $f'' = 0.33$) covered with a copper layer ($n_e = 2.4 \cdot 10^{24}\,\mathrm{cm}^{-3}$, $f'' = 0.63$). First, a grating was inscribed into the rear surface of the Si membrane and subsequently covered with copper of a few hundred nanometers thickness. With this the buried grating structure at the Si-Cu interface imprints also at the Cu-vacuum surface yielding a replication factor $\chi\left(\vec{q}\right) = \hat{\chi}\exp\left[-\left(\sigma_2^2 - \sigma_1^2\right)q^2/2\right]$ between the buried and rear side grating structures. Note that the grating interface is protected from direct laser interaction by the Si membrane. For *case 2* this membrane was etched away almost completely, leaving only the copper foil with a sharp grating at the front and its less sharp imprint at the rear.

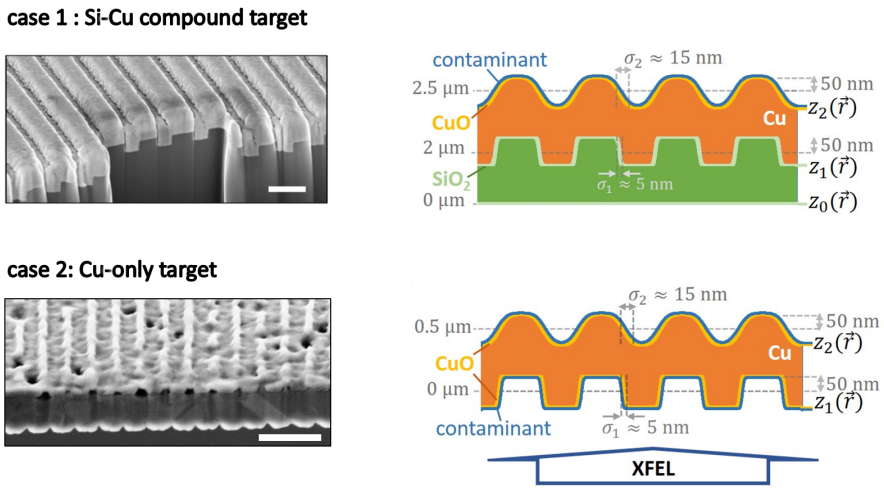

Figure 5.2: left: SEM images of representative target sample cross sections prepared by focused ion beam cuts. **top:** Si-Cu – dark: Si, light gray: Cu; **bottom:** Cu. White bar is 500 nm. **right**: Schematic models of target configurations for case 1 and 2 (not to scale).

5.1.4 Spatial and temporal overlap

The procedure to overlap the optical and XFEL pulses in space and time was the same as described in Ref. [14]. The delay timing between the optical and XFEL pulse was performed with a so-called *time tool*, a spatial encoding timing diagnostic[105, 106]. For this, we used a YAG crystal placed in the interaction plane. The XFEL pulse was used to irradiate the YAG crystal at an angle of 45° in order to create free charge carriers. The resulting changes in the refractive index were then probed by the optical laser in order to correlate the position of the shaded area with the relative time delay. The same concept was used upstream of the XFEL beam to determine the delay time-jitter.
To achieve the spatial overlap of the XFEL and optical laser at the interaction point, we first created a hole into a (sacrificed) target membrane positioned in the interaction plane with a burst of XFEL pulses. Then, we back-illuminated the hole with the optical laser and used the focus diagnostic imaging system to define the interaction point in the interaction plane. Subsequently, we steered the optical beam onto the hole position by movements of the off-axis parabola (OAP). Since these movements changed the optical beam path, the procedure was completed by checking the delay timing with the time-tool once again. The spatial overlap of the XFEL pulse with the grating targets is complementary confirmed before each pump-probe experiment by the non-destructive XFEL-only preshots.

5.1.5 K_α emission spectra

The presence of nitrogen-like ions and therefore the resonant absorption transitions in the Cu targets (*case 2*) and their absence in the case of Si-Cu compound targets (*case 1*) was independently verified by K_α emission spectra. Those were measured by a spectrometer employing a 2x4 cm^2 large HOPG crystal with a mosaicity of $m = 0.8°$ observing the rear side of the target. This instrument provides a spectral resolution better then 10 eV, the data are spatially and temporally integrated. Fig. 5.8 presents the measured spectra for the two different targets highlighting the K_α emission line of N-like Cu at 8165 eV.

5.2 Semi-analytical model for the asymmetry in the resonant SAXS patterns

As described in the theory section 4.2, SAXS patterns are usually symmetric since the scattering length density ρ, is often either purely real-valued, purely imaginary or the real and imaginary parts are proportional to each other. Hence, any asymmetry

$$\eta(\vec{q}) = \frac{I(\vec{q}) - I(-\vec{q})}{I(\vec{q}) + I(-\vec{q})} \tag{5.1}$$

in a SAXS intensity pattern $I(\vec{q})$ can be directly linked to the different distribution of the real and imaginary parts of ρ in the target. In the case of hard X-rays, the real part is primarily due to Thomson scattering, while the imaginary part arises from the opacity, i.e., the combined absorption cross section of atomic resonant bound-bound (bb), bound-free (bf), and free-free (ff) transitions. In the scattering length density the opacity enters via complex valued optical corrections $f_i' + if_i''$ that need to be added to the ion form factor f_i. Based on the considerations discussed in the theory section, we aim to make us of additional information contained in the asymmetry in the measured SAXS signals, with appropriate modeling for the targets used in the experiments.

For the targets used in the experiments, we can express the two structured contours of the front (or buried) and rear surface interface as $z_1(\vec{r}, t)$ and $z_2(\vec{r}, t)$, respectively. Correlations between both contours are expressed in reciprocal space by a replication factor $\chi(\vec{q}, t)$ with

$$\chi(\vec{q}, t) = \frac{\tilde{z}_2(\vec{q}, t)}{\tilde{z}_1(\vec{q}, t)}. \tag{5.2}$$

A replication factor of $|\chi(\vec{q}, t)| = 1$ means perfect replication, i.e. the two interfaces have identical profiles. For complex form factors, it follows from the Fourier Transform symmetry properties that the scattering intensity is asymmetric for two or more materials being asymmetrically distributed with respect to each other. In the theory section we show an intuitive example and demonstrate how to calculate the asymmetries for the target configurations relevant for our experiments. The main finding is that for two-layer targets the replication factor $\chi(\vec{q})$ and the respective optical properties of the target materials can be determined from the asymmetry in the scattering patterns via

$$\eta\left(\vec{q}, t\right) = \frac{c_1 \Im\left(\chi\left(\vec{q}, t\right)\right)}{\Re\left(c_2 + c_3 \chi(\vec{q}, t)^2 + c_4 \chi(\vec{q}, t)\right)} \tag{5.3}$$

if the phase of the XFEL is plane[2]. This equation connects the asymmetry with the structural parameters (entering in χ) and the material properties such as density and opacity (entering in c_i, defined in eqn. 4.23, Sec. 4.2). A simple way to break the symmetry is a shift of the rear layer by $\vec{r}_s$ with respect to the front layer structure, e.g., by a tilt of the target: $\chi\left(\vec{q}\right) = e^{-i\vec{r}_s\vec{q}}$. In experimental conditions, surface inhomogeneities (i.e. random variations in the geometric parameters) and contamination layers of oxides and hydrocarbon compounds are unavoidable and take the role of the small shift. Their influence can be absorbed into the parameters c_i, as demonstrated in the theory section. In fact, for the targets used in the experiments the random variations do not cancel out due to their number not being large enough. This

[2] This assumption is valid for our experiments since as shown in the Appendix (see Fig. 7.6) the preshots of the same target show similar asymmetries. Therefore, the effect of the phase on the asymmetry, if present at all, can be assumed to be constant, i.e. all changes of the asymmetry between pre- and main shots cannot be attributed to the influence of a phase, which will therefore for reasons of simplicity be neglected.

circumstance is identified as the leading cause of the asymmetry in the experiments. The fluctuations are taken into account in the calculations for the expected asymmetries of our targets in figures 5.6 and 5.9f. The existence of hydrogen and oxides on the target surface can be experimentally shown by the data recorded separately by a Thomson parabola spectrometer (see Appendix, Fig. 7.7).

The target grating smoothness σ_1 for the buried grating of the silicon-copper compound targets and the front grating of the pure copper targets were taken from the preshots in [14], as there we used the same grated silicon carrier membrane. There, the grating smoothness parameter was measured to be 5.4 nm. The rear surface smoothness σ_2 of the Cu coating in the present experiment cannot be accessed independently via SAXS, yet we can restrain the possible value of σ_2. It must be larger than σ_1 and based on the analytic analysis of the asymmetry, the results are consistent with a smoothness of $\sigma_2 \approx 10-15$ nm and $\hat{\chi} = 1$, which is also supported by the SEM images (Fig. 5.2). The buried grating height was set to the manufacturing specification of 100 nm. The exact thickness of the contamination layers cannot be determined by asymmetry measurements, but their approximate values can be extracted from literature[100, 101, 103, 107]. We assume imperfect gratings with CuO, hydro-carbon-oxide (both thickness 5% of $\hat{z}_1$) and SiO (thickness 0.3% of $\hat{z}_1$) surface contamination layers at all applicable surfaces. We further assume $\varepsilon = 10\%$ normal distributed random variations in all the geometric parameters: ridge edge positions, ridge heights, layer thicknesses and smoothnesses, as well as contamination thicknesses, with $n = 100$ sub-volumes with different parameters on a target in the XFEL focus. For the modeling and stability analysis for these choices the reader is referred to Fig. 4.12 in theory Section 4.2.

In principle, defects or other geometric effects on the nanometer level that we resolve with SAXS are deviations from our simple grating model and can potentially introduce or change the asymmetry. However, any defects that change the asymmetries of our measurements need to have a correlation on the scale of the grating period, or else their effect on the scattering signal at the peak positions will be greatly reduced as compared to the effect of the surface contamination which are correlated greatly with the grating pattern structure by definition. Secondly, if the defect's effects are changing with delay, and they were what is dominating the signal instead of our model, then at least the qualitative finding of a reduced replication (case 1) and increased opacity (case 2) with increased delay would still remain true, though now the quantitative model would be wrong. However, especially the Si-Cu compound targets of case 1, due to easier and experienced production routines, are almost perfect with virtually no imperfections as visible on the few ten nm level.

5.3 Calculation of the experimental average asymmetry

Experimentally, we extract the asymmetry by comparing the scattering signal in the scattering peaks for each peak pair at $+q$ and $-q$. The absolutes of the individual asymmetry values are

then averaged in order to have a simple quantity to easily compare simulations with theory. Additionally, the uncertainty margins of the averages are much smaller than those of the individual asymmetry values. Fig. 5.3 illustrates the workflow from the raw experimental data to the average asymmetry values $\overline{|\eta|}$.

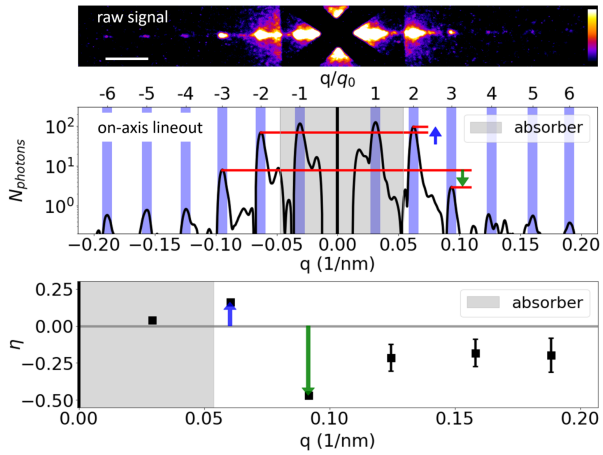

Figure 5.3: Data processing steps from raw signal to asymmetry on the example of a resonantly probed HI laser driven Cu grating foil. **Top**: raw data (log color scale), white bar corresponds to 0.03 nm^{-1}; **middle**: background corrected lineout through center, blue bars indicate the q-ranges for integration of the photon number per peak; **bottom**: asymmetry of peak pairs whose averaged absolute values give the average asymmetry $\overline{|\eta|}$.

The raw data PIXIS signal was summed within a square of 17 pixels width for each individual scattering peak and then corrected by the corresponding mean of two background values determined at both sides of the signal axis. The result was normalized to photon numbers N_{ph} by a conversion factor of 1/164 photons per ADU which was calibrated offline (as in [14]). The total uncertainty is given by the Poisson error $\sqrt{N_{ph}}$, the statistical background error given by the standard deviation σ_{Bg} of 12 background values allocated at both sides of the peaks and the dark field error σ_{DF} (measured analogously). Applying Gaussian error propagation the individual peak error is then given by $\Delta N_{ph} = \sqrt{N_{ph}(q) + (\sigma_{Bg} - \sigma_{DF})^2 + (\sigma_{DF})^2}$. For each pair of scattering peaks situated at $+q_i$ and $-q_i$ with photon numbers $N^+_{ph,i}$ and $N^-_{ph,i}$, respectively, the individual asymmetry was calculated according to $\eta_i = \left(N^+_{ph,i} - N^-_{ph,i}\right) / \left(N^+_{ph,i} + N^-_{ph,i}\right)$. This peak-wise asymmetry was then averaged over all q_i in the q-range that is experimentally accessible, which helps to reduce the error margin in the experiment compared to the single measurement η_i. The error of the average asymmetry is then again derived by applying Gaussian error propagation.

In *case 1* we averaged the asymmetry $\overline{|\eta_0|}$ over the experimentally accessible q-range from 0.05 to 0.15 nm^{-1}, preshot asymmetries are around 0.13 ± 0.5. The accessible q-range between 0.015 and 0.2 nm^{-1} in *case 2* was larger due to the use of absorbers. The average preshot asymmetry here ranged from 0.03 to 0.09 with uncertainties in the range from 0.01 to 0.06.

Experimental results

Before we present our results regarding the opacity measurements, we need to study the role of the replication factor. In fact, we will demonstrate that the asymmetry can be used to obtain nanoscopic information about interface structures lying behind each other. We performed dedicated experiments for both cases: *Case 1* is an experiment measuring the relative expansion dynamics of interface structures placed at different depths. Here, the laser pump intensity was chosen small enough that only structural changes occur, i.e. only the replication factor χ varies while the optical properties of the compound materials remain constant. In *case 2* we focus on changes in the asymmetry caused by variations of the optical properties of the target material (via c_i). Here, a combination of high pump laser intensity and short time scales yield a rapid heating, ionization and consequently an increase of X-ray opacity at resonant bb transitions.

5.4 Buried Layer Heating (Case 1)

For this part of the experiment, Si-Cu compound targets were pumped by the HI laser at an intensity of 3 and 7×10^{17} W/cm^2, respectively. At this comparatively low intensity, the resulting heating is sufficiently low such that collisional ionization of the rear copper layer is suppressed to less than the L-shell, i.e., no K_α transition channels are available. Additionally, a slow effective expansion of either one or both the buried grating interface and rear grated surface (i.e. an increase of σ_1 or σ_2) can be deduced from a decrease of the scattering yield at large values of q with a Debye-Waller-like procedure similar to that presented in [14]. Comparing the SAXS patterns obtained after irradiation by the HI laser pulse (main shots) with those obtained before from the target without HI irradiation (preshots), we observe a reduction of scattering yield at large q-values corresponding very roughly to an effective expansion of up to 10 nm (see Appendix 7.2). However, the expansion values cannot be fitted separately for the interface and rear surface with a reasonable accuracy, since the projected total electron density of the target with two grating surfaces simply has too many free parameters. On the other hand, no significant reduction of the yield was found for the shortest probe delay of 250 fs, indicating that the interface expansion indeed is a slow process over a few ps, as expected for those low laser intensities.

To confirm the expectations of low heating and slow expansion, we performed particle-in-cell (PIC) simulations to model the laser interaction and prompt electron heating in the first 500 fs after the laser hit the target, followed by a hydrodynamic simulation (performed by J. Nikl) to study the plasma response over the few ps range up to the 16 ps delay of the latest probe, see Fig. 5.4. While at the front surface the high electron temperature generates a

strong ablation pressure which leads to shock formation and ablation, heat diffusion towards the target rear leads to a partial temperature equilibration over the whole compound target depth to around 5 eV within the following few hundred femtoseconds. The key finding here is that the PIC simulation predicts a very slow expansion of the buried interface and the rear surface, which is in fact below the simulation's resolution within the first 500 fs.
With the hydrodynamic simulations we then observe the formation of a strong shock over the next 5 ps that propagates toward the target rear with an initial velocity of 169 nm/ps, slowing down to 45.5 nm/ps after 20 ps. This means that the rear side interface remains intact during the whole range of our probing delays and the scattering patterns are changing only due to the thermal expansion of the grating surfaces.

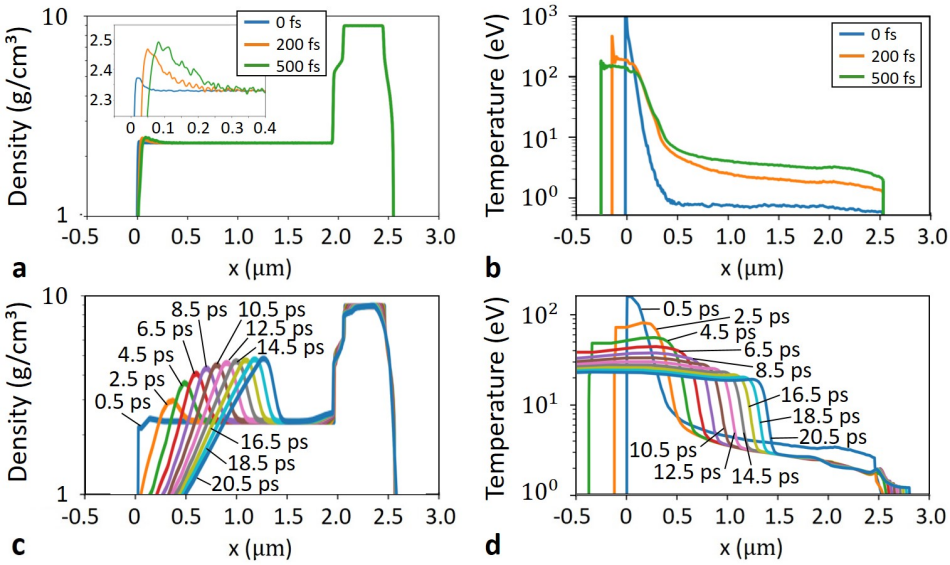

Figure 5.4: Density (a,c) and temperature (b,d) evolution of the Si-Cu compound target for the average optical laser intensity in the XFEL probe area from Particle in cell simulations (a,b) and hydrodynamic simulations (c,d). Laser strength was set to $a_0 \cong 0.3$, which corresponds to the average intensity in the XFEL focal spot for 1 J laser energy. Laser is incident from the left.

We now analyze the asymmetry parameter $\overline{|\eta|}$ in the experimental scattering patterns in order to obtain additional information about the relative expansion of both gratings. We observe a reduction of $\overline{|\eta|}$ compared to its value $\overline{|\eta_0|}$ in the respective XFEL-only preshots, which is dependent on a combination of the delay and energy of the HI laser (see Fig. 5.5). As expected, since no K_α channels are available, the reduction of the asymmetry is not substantially different for the different XFEL photon energies (i.e. on- and off-resonance). The observed reduction of the asymmetry must therefore be predominantly due to a change of the replication factor, i.e. expansion of the grating surfaces.
In Fig. 5.6 the expected asymmetry calculated with our asymmetry model is shown as a function of the two respective smoothness parameters σ_1 and σ_2 of the buried and rear grating

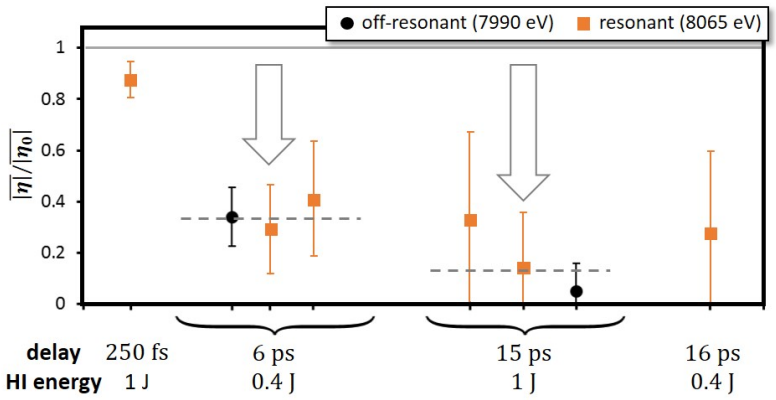

Figure 5.5: *(Case 1)* Average asymmetry $\overline{|\eta|}$ relative to that of the preshots $\overline{|\eta_0|}$ for single shots on Si-Cu compound targets with $3–7 \times 10^{17}$ W/cm^2 laser intensity. Dashed lines are respective weighted averages.

ridges. We adopt a replication factor of

$$\chi(\vec{q}) = \hat{\chi} \exp\left(-\Delta\sigma^2 q^2/2\right) \tag{5.4}$$

with $\Delta\sigma^2 \equiv \sigma_2^2 - \sigma_1^2$, i.e. a smoothing of the rear interface (index 2) with respect to the buried interface (index 1). With the initial conditions for the un-pumped targets, it can be readily seen that a reduction in asymmetry is linked to an expansion of the grating interfaces. Specifically, the experimentally observed reduction by more than a factor of three reduces the range of possible values for σ_1 and σ_2. The result we can therefore infer is that the observed drop in asymmetry is characteristic for the expansion of the buried grating interface (σ_1) by at least 10 nm and cannot be explained by an expansion of the rear grating interface (σ_2) alone.

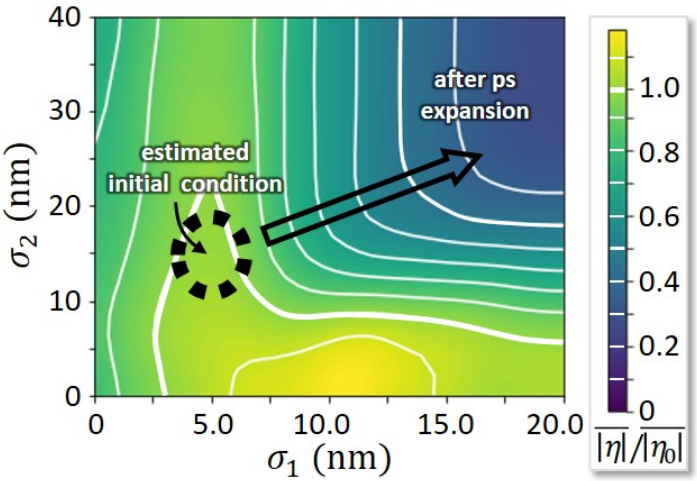

Figure 5.6: Analytical modeling of the asymmetry $\overline{|\eta|}$ averaged over the experimentally accessible q-range and normalized by the corresponding preshot asymmetry $\overline{|\eta_0|}$ for *case 1* (Si-Cu compound targets).

5.5 Ultrafast opacity changes during relativistic laser-solid interaction (Case 2)

The second case considered here focuses on ionization-induced changes in the asymmetry of the SAXS signal as a novel tool to measure the temporal evolution of plasma opacities. For this purpose, we used Cu grating targets irradiated by a higher laser intensity with a peak intensity of $(3 \pm 1) \times 10^{19}$ W/cm^2. With the first grating positioned now directly at the front surface, the dynamics are no longer dominated by heat diffusion and shock propagation through the silicon layer as before in *case 1*, but by the faster direct laser-solid interaction and by electrons being accelerated into the target by the laser.
The relevant timescales for these processes are shorter, on the order of a few hundred femtoseconds only. Consequently, we start to observe a slight decrease of the asymmetry due to the grating expansion for shots with XFEL probe delay already at 200 fs at off-resonant probe energies of 7940 eV, see Fig. 5.7. For resonant scattering, we tuned the energy of the XFEL to an open atomic transition in the plasma, namely to 8165 eV. This is the energy of the K_α transition in nitrogen-like Cu (K1L6 - K2L5)[108]. During the first 300 fs after pump laser irradiation the scattering patterns in those resonant shots exhibited large values of asymmetry compared to the XFEL-only preshots, cf. Fig. 5.7. The increase in asymmetry only disappears at larger delays. At the largest delay of 1 ps the main shots and preshots show a similar asymmetry again. This is to be expected, due to the onset of expansion/destruction of the front side grating and ion-electron recombination (see below). In order to understand these obser-

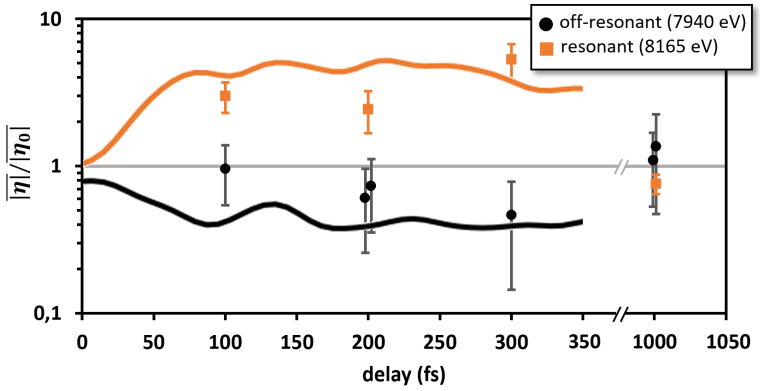

Figure 5.7: *(Case 2)* Average asymmetries $\overline{|\eta|}$ relative to that of the preshots $\overline{|\eta_0|}$ for the shots on Cu targets with $2–4 \times 10^{19}$ W/cm^2 laser intensity. Solid lines show the asymmetry prediction obtained from PIC simulations for off-resonant (black) and resonant (orange) XFEL energy (see Section 5.5). Vertical error bars were obtained by Gaussian error propagation (see Sec. 5.3). The delay time error of ±114 fs is given by the r.m.s. jitter of the XFEL.

vations, we refer to the discussion of *case 1*. We expect the rear side grating to remain virtually unchanged while the front surface should thermally expand during the first few hundred femtoseconds after the laser irradiation, which is also supported by PIC simulations described

later. Consequently, the reduced asymmetry in the main shots at off-resonant XFEL energy during the initial 300 fs after the HI irradiation compared to the preshots can be explained by a reduced replication factor χ.
On the other hand, this means that the increase of asymmetry for the on-resonance shots at the same delays can not be caused by geometric changes and hence must be due to a change of the opacity at the resonant energy. Since the cross section of bf transitions is similar within the range of the two XFEL photon energies used in the experiment (within ±25%), the large increase of asymmetry in the on-resonance shots must be due to an increase of f''_{Cu} due to bb transitions[104], whose existence was measured independently with a temporally and spatially integrating spectrometer (see Fig. 5.8).
We therefore conclude that an NLTE equivalent plasma temperature is above 300 eV, which is the necessary temperature to sufficiently ionize copper to open the N-like K_α resonances at the probe photon energy of 8165 eV[108, 109]. Thus, temperatures above 300 eV lead to increased plasma opacity and contribution of resonant scattering (see Fig. 4.4 & 4.5 in Sec. 4.1). However, the plasma evolution is more complex, i.e. non-thermal and highly transient, as the laser-plasma interaction is pulsed, relativistic and non-linear. A detailed insight of the plasma dynamics requires PIC simulations, which are described in the following sections.

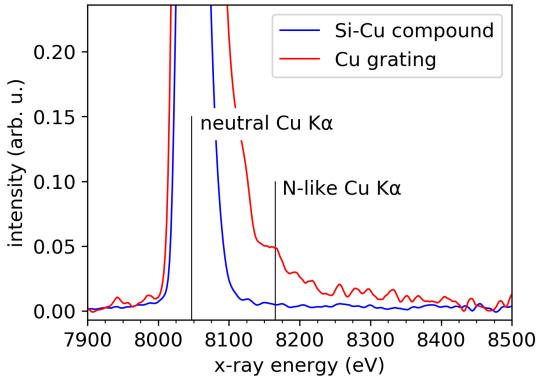

Figure 5.8: The measured K_α emission spectra for two different target compositions. In the case of a pure Cu target, the emission at the probe photon energy of 8165 eV is clearly visible showing the presence of Ni-like ions. In contrary, in the Si-Cu compound target the ionization of Cu is far not that high as most of the laser energy is absorbed in the Si layer.

To determine the statistical significance of the resonant main shots on copper being more asymmetric than the preshots and the off-resonance shots being less asymmetric, we calculated the conditional probability of the respective shots being either all more asymmetric ($p_>$) or all less asymmetric ($p_<$). Here, $p_> = \prod_{j \in M} p^j_>$ is the product of the individual probabilities for each shot j of the set M of resonant or off-resonant shots. For example, the probability for the asymmetry being increased on the resonant main shots is given by $p_>/(p_< + p_>)$. To calculate $p^j_>$ ($p^j_<$) we assume a Gaussian distribution of the probability density distribution for the real value being around the measured value, with the width of the Gaussian given by the respective measurement error bar, and integrate from $\eta/\eta_0 = 1$ to ∞ (from 0 to 1). The result is $p_>/(p_< + p_>) = 0.999999902$ for the resonant shots with delays equal to or less than 300 fs, which is larger than the 5σ level.

Simulation-assisted modeling of the asymmetry

In order to model the plasma evolution, we performed a set of simulations[3] with the PIC code Smilei[110]. SMILEI is a collisional PIC code that treats binary collisions within a cell between charged particles and includes ionization via field ionization and collisional direct impact ionization[111]. It does not include recombination, hence the simulation gets unphysical when the plasma is first heated and then is cooled down too much, which is the case after a few hundred fs.
The simulations are used to obtain the predicted asymmetry evolution (solid lines in Fig. 5.7) by means of the semi-analytical model for the asymmetry. The asymmetry prediction requires a detailed analysis of the plasma expansion, ionization and opacity changes in order to extract the geometrical and optical parameters that enter the model. As the plasma dynamics are strongly dependent on the pump laser intensity, the UHI focal spot is modeled by performing a set of simulations with different a_0. The main results of the simulation studies are displayed in Fig. 5.9.

Opacity evolution

During the first few tens to hundreds of femtoseconds following the HI laser irradiation, electrons are quickly accelerated and heated leading to ionization and expansion of the grating. The simulations for $a_0 = 3.7$ down to $a_0 = 1.2$ show average free electron energies in the range of multiple keV; for the nominal average intensity in the grating area, i.e. $a_0 = 2.1$, it peaks at approximately $\langle E \rangle = 5\,\mathrm{keV}$, cf. Fig. 5.9e. It is important to mention that in thermal equilibrium an electron temperature in this range would suggest that Cu should be ionized up to He- and H-like states. Consequently, we could not expect to observe N-like K_α transitions. However, due to the non-equilibrium transient situation and limited ionization rates, the average ionization can be seen and is known[112] to be much less during the first several hundred femtoseconds. Thus, the ionization distribution contains significant contribution of N-like ionized Cu and the respective K_α resonance in fact *can* be driven by the XFEL.

The asymmetry prediction requires the extraction of the plasma opacity in order to derive the evolution of f' and f''. Since the PIC simulations cannot calculate the plasma opacity self-consistently, we have to resort to approximate the opacities using a separate software SCFLY[109]. Opacities were calculated using SCFLY for a range of NLTE plasma temperatures, as shown in Fig. 4.4 in the theory section 4.1. However, since SCFLY like any available atomic rate solver cannot take into account the multidimensional and kinetic non-thermal and tran-

[3] The spatial and temporal resolution for the simulations was set to $\Delta x = \Delta y = c\Delta t/\sqrt{2} = \lambda_{laser}/800$. The simulation box was set to 1.6 µm in longitudinal (z-), and 200 nm in transverse y-direction. The laser polarization was set in y-direction, the transverse envelope was a planar. We took the finite waist into account by running the simulation several times with different laser intensity and averaging the respective contribution to the asymmetry. All other laser and target geometry parameters were set according to the experimental values shown in Fig. 5.1 & 5.2. We placed 20 ions per cell, copper being preionized to charge state 1+, and added electrons to start with a neutral plasma.

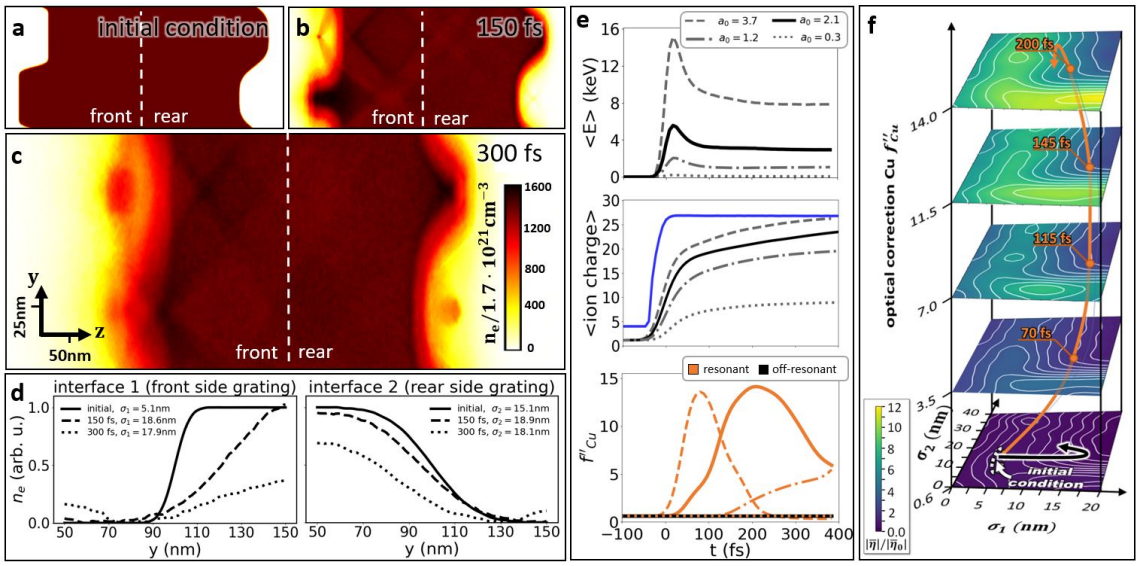

Figure 5.9: Simulation of the Cu target (*case 2*) for the average intensity of $1.3 \cdot 10^{19}$ W/cm^2 ($a_0 = 2.1$). **a-c)** Total electron density distribution at three different times. **d)** Transverse electron density integrated through the front (left) and the rear (right) half of the target normalized with respect to the initial grating amplitude. **e)** Temporal evolution of the average kinetic electron energy (top), average copper ionization charge state (middle), and fitted opacity using SCFLY[109] (bottom) for resonant (orange, 8165 eV) and off-resonant (black, 7940 eV) XFEL energy. For reference the evolution is given also for different laser intensities, corresponding to different parts of the laser focal spot. The blue curve in the middle illustrates the overestimation of the average Cu charge state, if a plasma in temperature equilibrium is assumed. The curve was obtained with SCFLY runs, where the time history for the electron and ion temperature was taken from the PIC simulation for $a_0 = 2.1$. **f)** Calculated asymmetry based on the simulation results for resonant (orange) and off-resonance (black) XFEL energy.

sient nature of the plasma, we cannot simply identify the NLTE temperature used in SCFLY with the average kinetic electron energy in the PIC simulation. This becomes obvious when comparing the average copper charge states from SCFLY and PIC. The blue curve in Fig. 5.9e shows the average Cu charge state obtained with SCFLY runs, where the time history for the electron and ion temperature was taken from the PIC simulation for $a_0 = 2.1$ and a temperature equilibrium is assumed. The black curve represents the average ionization obtained from the Smiley PIC simulations. Comparison of these curves makes it obvious that assuming an equilibrium situation greatly overestimates the ionization.[4] Therefore, in order to approximate the opacities we took the approach to rather match the average charge states for each PIC time step from a series of SCFLY runs with different NLTE temperature. From the matching SCFLY run we then extract the respective opacity and finally obtain the evolution of f''_{Cu} plotted in the bottom graph of Fig. 5.9e for the on- and off-resonant XFEL probe energy.

[4] SCFLY is known to overestimate the low temperature ionization due to the Thomas-Fermi ionization model used there not being applicable in the low temperature limit.

Expansion dynamics

In order to obtain the geometrical model parameters of the front (1) and rear (2) grating structures, the total electron density was integrated along the target depth in the front and rear half. Then, the grating contours were fitted with an error-function according to $z \propto h_{1,2} \cdot \mathrm{erf}\left[(y - y_0) / \left(\sqrt{2}\sigma_{1/2}\right)\right]$ to extract $\sigma_{1,2}$, $b_{1,2}$ and $h_{1,2}$. As can be seen in Fig. 5.9a-d for the case of $a_0 = 2.1$, the grating quickly expands within the first 300 fs. As expected, the front grating being irradiated directly expands more quickly than the rear structure. In first order, the grating expansion is characterized by an increase of the edge smoothnesses and a reduction of the grating amplitudes. Yet, a closer look at the expansion dynamics (see Fig. 5.11) reveals a slight, initial increase of the grating amplitude, which goes along with a reduction of the ridge width. This behaviour can be explained by a predominant expansion of the plasma in the direction along the surface normal of the grating contour.
Additionally, the simulations reveal higher order effects by the generation of plasma jets[14]. These jets are emitted from the bottom parts of the grating edges (well-visible in Fig. 5.9 b) and superpose in the center of the grating pits, which eventually leads to an additional, interleaved grating structure which is not included in the analytic model. Therefore, for each simulated pump laser intensity, the analysis is stopped at those simulation time steps, for which the density amplitude induced by the jets exceeded 20 % of the prestructured grating amplitude included in the model.

Calculation of the expected asymmetry evolution

Finally, we can now calculate the expected change of the asymmetry in the SAXS scattering patterns when the XFEL energy is tuned to the nitrogen-like K_α resonance energy of 8165 eV. For the specific expansion parameters of the grating surfaces we can use equ. 5.3 with the appropriate extension for including surface contaminants to compute the asymmetries for the respective increased absorption. For the specific example of the average laser intensity ($a_0 = 2.1$), the asymmetry evolution is shown in Fig. 5.9f. The predictions for the experimental results take into account the spatial laser intensity profile. We do this in Fig. 5.7 by first computing the asymmetric scattering amplitudes for each of the simulations with different laser intensity (see Fig. 5.10). We then calculate the total scattering signal by summing up those individual contributions, weighted by the areal fraction of the respective laser intensity in the focal spot (see Table 5.1). The resulting semi-analytical predictions for the asymmetries of the scattering pattern can be seen to be in excellent agreement with the experimental results.

Summarizing, the fact that the asymmetry is increased during the first 300 fs after HI laser irradiation is due to a combination of electron acceleration to an average energy well above 300 eV, moderate delayed grating expansion, and comparatively slow ionization not reaching saturation during the brief heating time before the bulk cools down again.

a_0	3.7	2.1	1.17	0.32
weighting factor	0.41	0.24	0.26	0.09

Table 5.1: Relativistic laser amplitudes a_0 and weighting factors used for the final asymmetry prediction. The UHI focal spot was modeled by a 2D-Gaussian distribution with a FWHM of 5 µm and was discretized by the simulated a_0. The individual weights were obtained by calculating the relative number of photons being assigned to the covered areas.

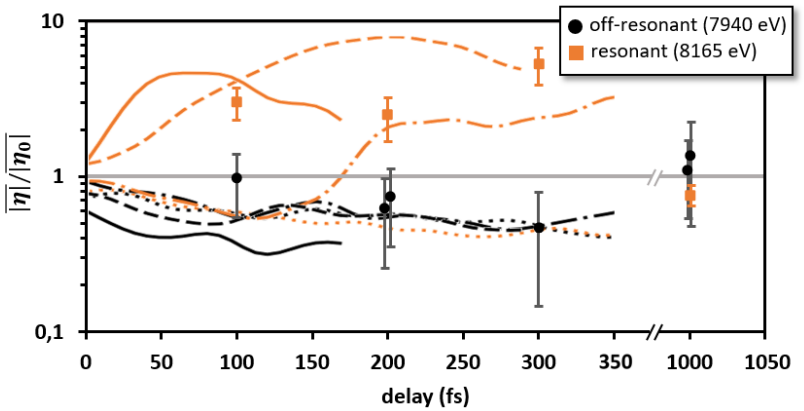

Figure 5.10: Evolution of the asymmetry obtained from the individual PIC simulations for different values of the relativistic laser amplitude a_0 for off-resonant (black) and resonant (orange) XFEL energy. Solid: $a_0 = 3.7$, dashed: $a_0 = 2.1$, dash-dotted: $a_0 = 1.2$, dotted: $a_0 = 0.3$.

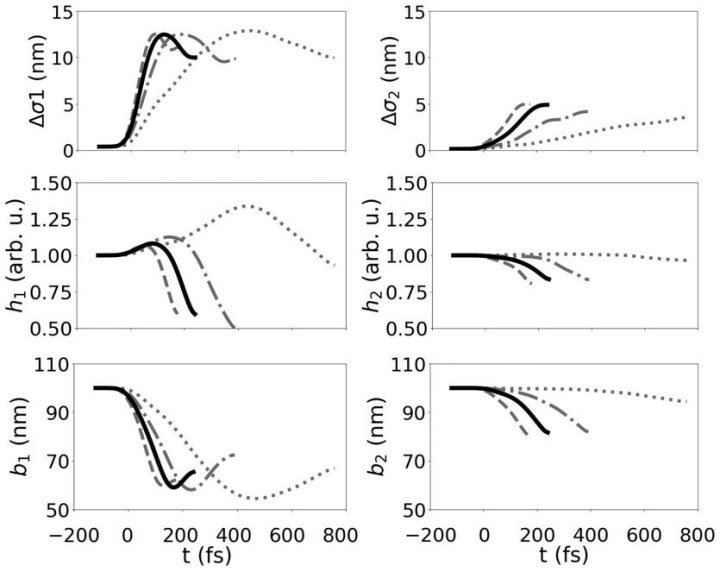

Figure 5.11: Dynamics of the geometrical parameters of the front (index 1) and rear (index 2) grating obtained from the Smilei simulations for different values of the relativistic laser amplitude a_0. Solid: $a_0 = 3.7$, dashed: $a_0 = 2.1$, dash-dotted: $a_0 = 1.2$, dotted: $a_0 = 0.3$. The fit parameters were obtained by fitting the integrated total electron density in the front and rear half of the target with error functions ($z \propto h_{1,2} \cdot \mathrm{erf}\left[(y - y_0) / \left(\sqrt{2}\sigma_{1/2}\right)\right]$). **upper row:** relative smoothing of the edges $\Delta\sigma_{1,2} = \sigma_{1,2}(t) - \sigma_{1,2}(t_0)$, **middle row:** grating amplitudes $h_{1,2}$ and **lower row:** width of the grating ridges $b_{1,2}$.

5.6 Summary and perspectives

Two specific science cases have been investigated in a recent experiment at MEC by performing resonant SAXS in a pump-probe scheme. By means of a semi-analytical model for the asymmetry in the scattering patterns, it was demonstrated in both cases that the asymmetry contains additional information on the expansion and opacity changes in laser-pumped, nanostructured compound grating targets. In the first case, the buried layer expansion in a Si-Cu compound grating target was investigated on the ps time scale upon laser irradiation of around 10^{17} W/cm^2, i.e. close below the relativistic intensity regime. At this intensity level, the asymmetry model is only sensitive to changes of the replication factor via grating expansion as ionization is suppressed and thus no K_α transition channels are available. By comparison of the measured asymmetry with the numerical model, a buried layer expansion of around 10 nm could be revealed. Combined PIC and hydrodynamic simulations revealed that a compression wave is generated at the target front side and travels towards the rear. Yet, during the range of probe delays ($\lesssim$ 16 ps) the compression wave does not reach the buried grating interface meaning the grating expansion is purely driven by thermal pressure. Extraction of the buried layer expansion in the experimental transmission probe geometry is non-trivial as due to projection along the XFEL probe direction, the SAXS signal is linked to the modulus square of the Fourier Transform of the *2D areal density* only. Thus, this insight represents a valuable extension of the SAXS capability compared to the precedent MEC experiment[14], which was dedicated to the front surface expansion only.

In the second case, resonant SAXS has been performed for the first time in the relativistic intensity regime, at around 10^{19} W/cm^2, in an LPA experimental context. Here, the asymmetry is sensitive to the grating expansion and opacity changes in a Cu-grating target at the same time. PIC simulations revealed that expansion and opacity changes take place on the ultrafast time scale of tens of femtoseconds. Experimentally, at on-resonant XFEL probe energy we observed an increase in the asymmetry in the main shots compared to the XFEL only preshots within the first 300 fs after laser pump. We connected this increase of the measured asymmetry in the main shots at on-resonant XFEL probe energy with the existence of N-like transition channels in the heated copper plasma. With this, resonant SAXS shows its capability to probe specific ionization states in a laser-heated plasma. This is a highly relevant capability, which promises to benchmark ionization models in the future.

The detailed analysis of the performed PIC simulations revealed that in the relativistic intensity regime, the plasma passes through highly non-thermal equilibrium states. Specifically, we find that the ionization within the first few hundred femtoseconds is not in a thermodynamic equilibrium with an NLTE plasma temperature corresponding to the average kinetic energy of the electron ensemble found in the PIC simulations. We found that FLYCHK runs imply saturation of the plasma ionization, i.e., no presence of K_α transition channels, if the average kinetic electron energy is identified as the NLTE plasma temperature used as the input for the FLY-

CHK runs. A saturation of the plasma ionization however was found to be in a disagreement with the average ionization found in the PIC simulations and with the presence of the N-like ions confirmed by K_α emission spectroscopy. Instead, we were able to deduce a lower limit NLTE equivalent plasma temperature of 300 eV, which matches the average ionization found in the PIC simulations and is thus compatible with the existence of N-like transition channels. The potential of resonant SAXS to probe specific ionization states and to serve as an in-situ thermometer, could be even further extended in the future with multicolor XFEL probing becoming available: In the presented experiments, only one XFEL probe energy per probe pulse could be used, i.e., only one measure point for the asymmetry and implicitly for the opacity was available. Yet, the opacity as a function of the plasma temperature is not bijective (see Fig. 4.4) so that we could only deduce a lower limit for the plasma temperature. A second measure point would allow to further constrain the temperature estimates. Two-color XFEL probing is currently in the process of being commissioned at EuXFEL and promises to allow for that in near future.

As mentioned above, the plasma in the relativistic intensity regime passes through non-thermodynamic equilibrium states. This circumstance made the extraction of the plasma opacity (or f'') non-trivial in the analysis presented here. In a first order approximation, we have dealt with this challenge by performing FLYCHK runs in the post processing of the PIC simulations by varying the plasma temperatures until the average ionization from FLYCHK is matched with the one from the SMILEY PIC code. This inconvenient and challenging analysis emphasizes the need for treating atomic rate equations simultaneously in the PIC simulations, which endorses the ongoing efforts to implement atomic rate equations within the FLYonGPU project[95] for PIConGPU.

Moreover, the PIC simulations revealed a strong intensity dependence of the expansion and opacity changes on the pump laser intensity, which implies a high relevance to take the intensity profile of the UHI focal spot into account. In particular, the generation of plasma jets leads to deviations from simple grating models and implies intensity-dependent probe delay time constraints for the applicability of our model. With the advance of available laser intensities at XFEL facilities, it may be advisable to work with larger focal spots so that plane wave approximations for the UHI-solid interaction are ensured and averaging effects of the XFEL scattering signal are suppressed.

In the analysis presented in this work we made use of the average symmetry in the SAXS signals in order to reduce the uncertainty margins. With stronger signals and further improved background suppression, e.g., by using a SAXS mirror[113], used to separate the scattering signal from the plasma self-emission, this promises to make use of the full $\vec{q}$-dependency of the asymmetry, in which the spatial information is encoded. The presented experiments represent a contribution to extending the XFEL probing capabilities, and are a stepping stone for the long term goal to study laser-induced structures systematically at XFEL facilities (see Ch. 6).

Potential and challenges of grating targets as an experimental platform

Although the targets used in our experiments represent quite specific cases, grating targets as one representation of periodically prestructured targets are of general relevance for the following reasons: First, prestructured targets have the practical benefit to provide a contrast in the electron density projected along the probe direction, which allowed yielding signal at high values of q and high signal-to-noise ratios. The strong signals in the unpumped preshots provided reliable reference signals, that allowed us to systematically follow slightest changes of the sample following pump laser irradiation in reciprocal space.
Secondly, and even more importantly, prestructured grating targets represent an experiment platform that can be adapted to a wide range of scientific problems: Transverse grating structures can be freely placed along the longitudinal target dimension (along probe direction) allowing to study the plasma response following pump laser irradiation at variable positions in depth. Gratings at the front or rear side can be used to measure the respective plasma scale lengths which are known to strongly affect the acceleration performance in laser-driven ion acceleration. Buried grating structures can be used to study the dynamics of the plasma bulk. In this context, multilayer structures at different positions in depth are ideal for studies of isochoric or shock heating and are thereby of relevance for laser-fusion science. Moreover, gratings can have a wide range of defining parameters such as their spatial extent and orientation, a chirp or their degree of regularity. A variable number of buried gratings with different orientations could be used to increase the depth information. Multifield grating targets with several grating areas of different orientations, for example, would allow measurements at variable distances to the driver. In summary, their versatility makes grating targets an experiment platform and thus one of the central pillars for HIBEF at European XFEL.

Of course, the question arises whether our results regarding the asymmetry model can be also translated to more complex target systems. Given that one finds an analytical representation of the target, the proposed method would equally apply for more complex targets. However, for more complex or unknown target structures, this would become progressively more difficult. Hence, this method cannot unconditionally be translated to arbitrary other complex structures. However, the known target configuration used here, enabled to describe the target, derive the semi-analytical asymmetry model and to understand the relevant physics that leads to the asymmetry in the SAXS patterns.
For more complex 2D structures, one would resort to phase retrieval which would allow to obtain the real and imaginary part almost model-free. Both iterative and neural-network-based phase retrieval techniques are developed in our group for future experiments, but it can be shown that 1D gratings, as presented here, are not well-suited for phase retrieval. Fig. 5.12 shows an illustrative example for a 2D phase retrieval carried out by H. Meissner. The upper row reflects the real and imaginary part of the 2D distribution of an object exemplified by the letters A and B. The upper right plot shows the phase of the complex phase object, whereby the imaginary part has a phase of 90°. The lower row shows the reconstruction of the real

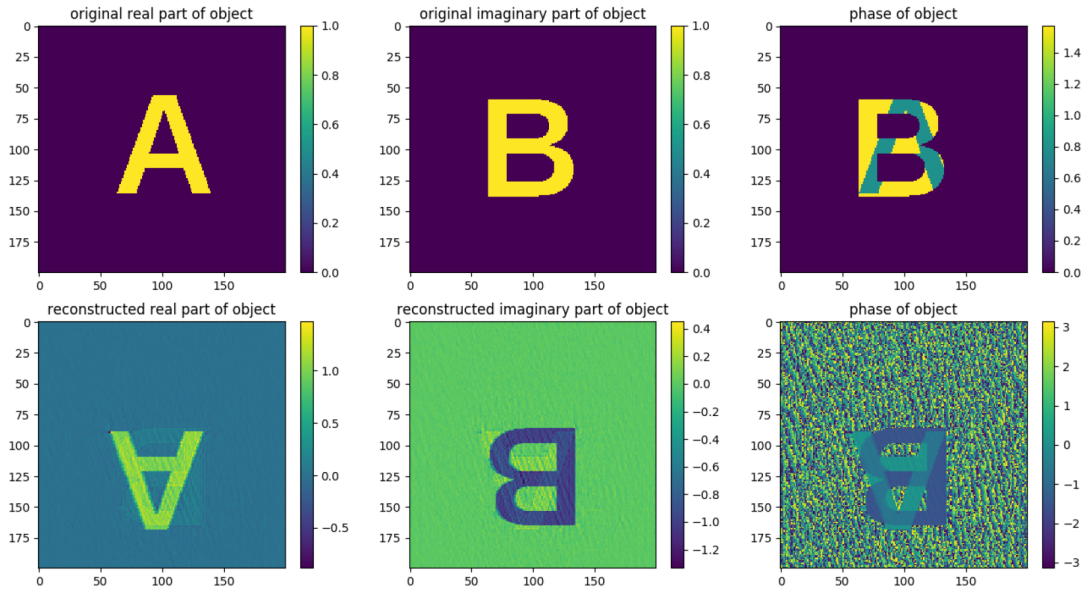

Figure 5.12: Reconstruction of the real and imaginary part from a complex phase object via phase retrieval algorithms (performed by H. Meissner).

and imaginary parts and the phase of the object by phase retrieval from the modulus square of the Fourier transform of the original phase object. This example illustrates that phase retrieval in principal enables the reconstruction of the real and imaginary parts of an original object. Yet, it does not provide the relative phase of the reconstructed real and imaginary parts, here reflected in form of the mirror image. Also one needs to emphasize that idealized cases have been considered in the example shown here. In realistic experimental conditions, e.g., when noise in the data is present or when the distributions of the real and imaginary parts of the object are distributed similarly and are, for example, only slightly shifted relative to each other, the reconstruction by phase retrieval is much more complicated. Thus, forward modeling and making use of an analytic consideration of the asymmetry as carried out in this thesis is nonetheless useful as it provides a physical model, which thus helps in the interpretation of the underlying physics.

6 Outlook

This thesis is a contribution to the long-term goal to study transient, laser-induced plasma effects with resonant SAXS with XFELs in the experimental context of laser-proton acceleration by means of two complementary approaches. The central part of this thesis is a pump-probe experiment performed at MEC with the goal to establish resonant SAXS with XFELs as a ultrafast probing technique for UHI laser-solid interaction with femtosecond and nanometer resolution. In the experiments simplified science cases, namely the buried layer heating and ultrafast opacity changes in laser-driven matter, have been investigated based on the testbed provided by nanostructured (compound) grating targets. Complementary, different laser-target conditions have been tested at the Draco PW laser system at HZDR to investigate their influence on transverse beam profile modulations, and in particular on the generation of net-like modulated beam profiles, systematically. The background for these experiments is related to net-like proton beam modulations measured more than ten years ago in experiments with the Draco 150 TW laser system at HZDR. These modulations were linked to the generation of different plasma instabilities in solid density plasma, in particular the Rayleigh-Taylor instability or a Weibel-like instability, but the clear identification of one of the possible candidates is still today impeded by the lack of experimental methods to probe solid density plasma with the required resolutions of femtoseconds and nanometers. These unclarities are one of the main drivers for the efforts made at HZDR to establish XFEL probing techniques for laser-solid interaction processes.

To our knowledge, the experiment performed at MEC is the first published experiment that has combined an XFEL with a relativistically intense laser beam in order to generate and probe hot dense plasmas. We developed an analytical model that demonstrates the connection between the optical properties of a material and surface replications in the X-ray longitudinal direction with the asymmetry in the scattering pattern. We demonstrated the application of asymmetric scattering on disentangling the expansion of a buried layer from the one at the rear, and to extract the plasma opacity by tuning the XFEL energy to the nitrogen-like bb transition of copper. Using an idealized target model including random surface contamination,

the asymmetry changes upon laser pumping were found to be in quantitative agreement with an expansion of a grating buried 2 µm deep in silicon of at least 10 nm at low laser intensities, and a heating of a thin copper target to at least 300 eV at HI laser intensities exceeding 10^{19} W/cm^2. The experimental results demonstrated the potential of resonant SAXS to measure the spatial distribution of the electron density and plasma opacity as well as correlations both in the XFEL transverse *and* longitudinal direction via the replication factor.
The presented experiments employ simple 1D grated targets in longitudinal transmission geometry. However, this is part of our broader efforts to develop structured targets as a platform for HED XFEL experiments. One could easily envision longitudinal structures and transverse probing, chirped gratings or complex 2D structures buried for example inside the target. The principles of resonant SAXS can also be transferred to Grazing Incidence SAXS[114] or other scattering geometries. This flexibility makes periodically structured targets a versatile experiment platform that can be easily adapted to a wide range of science cases to measure the density and opacity gradients on surfaces or inside the compressed and heated solid.
Additionally, grating targets themselves are an interesting research object. Samples with a grated front surface, for example, have shown complete laser absorption resulting in enhanced ion acceleration and extreme ultra-violet (XUV) generation [115, 116]. Multilayer structures containing few hundred nanometers to micron deep buried structures with nanoscale spatial modulation are ideal for studies of isochoric or shock heating of solid-density plasmas [94, 117], or can serve as an ex-situ model of DT layered fusion capsules or to study the role of surface roughness and modulations by adding a grating layer.

Establishing the experiment platform of structured targets is also a stepping stone towards bringing HI laser relativistic plasma fundamental physics experiments to XFELs, e.g., by probing the nanoscopic evolution of structures self-generated during and after the laser irradiation[91, 104]. With regard to this, the experiments performed at the Draco PW laser system provided a preparatory work for future proposals for beam time at EuXFEL. The presented measurements are an impetus to establish systematic studies with regard to the influence of different laser-target parameters on the generation of net-like proton beam modulations.
A controlled variation of the laser contrast by prepulse application onto the front surface of few micrometer thick Ti targets did not reproduce the net-like proton beam modulations observed by Metzkes-Ng et al[1]. However, we found a frequent generation of net-like modulations in conditions of high laser contrast, i.e., with a plasma mirror setup, in combination with FV foil targets of few hundred nm thickness. A scan of the target thickness revealed that these modulations occur in a wide range of thicknesses, while the most prominent visibility was observed for thicknesses above the onset of target transparency in a thickness range between around 100 and 300 nm. For a fixed target thickness of 240 nm, the net-like modulations were found to be unaffected by changes of the laser pulse energy on target within the range of 4 to 18 J ($20 \leq a_0 \leq 45$). Variation of the atomic number of the target front

side material showed an impact on the frequency of the proton beam modulation. Yet, a lack of knowledge on the proton source size impeded to make conclusions with regard to the wavenumber of the ion density at the source location assumed to be at the target rear. Therefore, we suggest to repeat the experiments with targets containing a periodically pre-structured rear surface in order to deduce the source size from the expected beating pattern in the proton beam profile.

Another major observation is that the modulations showed similarities for shots at comparable laser-target settings. Yet, we did not observe identical beam profiles obtained from different experiments as expected in imprinting scenarios, in which spatial information from the laser is translated into the electron beam and subsequently into the proton beam. From that, we deduce that these beam modulations need to be understood in a radiographical picture, in which the accelerated proton beams probe electromagnetic field structures located at the target rear, which potentially have their origin in instability-induced electron beam filamentation.

As expected, these observations could only provide answers to a limited extent with regard to the underlying mechanism. The remaining uncertainties are a strong endorsement of the HIBEF program to establish resonant SAXS as a probing technique for solid density plasma in the context of LPA experiments. The measurements at Draco at high laser contrast give an indication for a frequent generation of plasma instabilities. This suggests to use SAXS in future experiments in similar experimental conditions in an orthogonal probe geometry. With regard to follow-up experiments at HED at EuXFEL it may be advisable to install a plasma mirror setup to enable high laser contrast conditions and to use micro stripe targets, i.e., foil targets of limited lateral extent allowing for an orthogonal probe geometry. Also, it will be of benefit in combining SAXS with a proton beam profiler to find correlations between the SAXS signal and the generation of transverse proton beam modulations.

7 Appendix

7.1 Spatial frequency filter applied on proton beam profiles

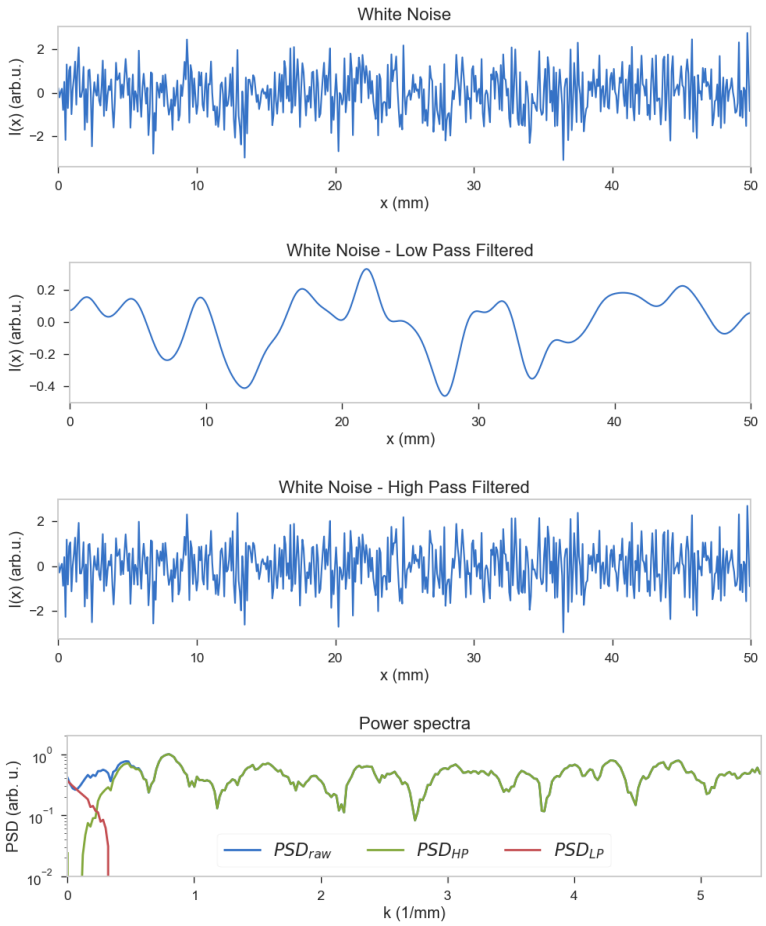

Figure 7.1: a) White noise signal given by 580 values randomly drawn from a standard normal distribution. The sampling rate is chosen to match with the resolution of the analyzed RCFs (580 px/ 50 mm). **b)** Low pass filter (LP) of white noise signal obtained with a Gaussian filter (scipy.ndimage.gaussian_filter1d) with $\sigma = 10$ ($\widehat{=}$ 1)mm chosen as the standard deviation for the Gaussian kernel. **c)** High pass filter (HP) of white noise signal by subtracting the low pass filtered signal from the raw signal. **d)** Power spectral density (PSD) for all three cases. For the chosen Kernel size, PSD_{HP} differs from PSD_{raw} only for wavenumbers $k \lesssim 0.5\,\mathrm{mm}^{-1}$. Transferred to the RCF analysis of the proton beam profiles, this means all modulated structures with wavelengths smaller than 2 mm remain preserved after high pass filtering.

7.2 Integrated SAXS data

Case 1

The integrated peak heights for *case 1* (Si-Cu compound grating targets) are shown in Fig. 7.2 and 7.3 for 0.4J and 1J, respectively.

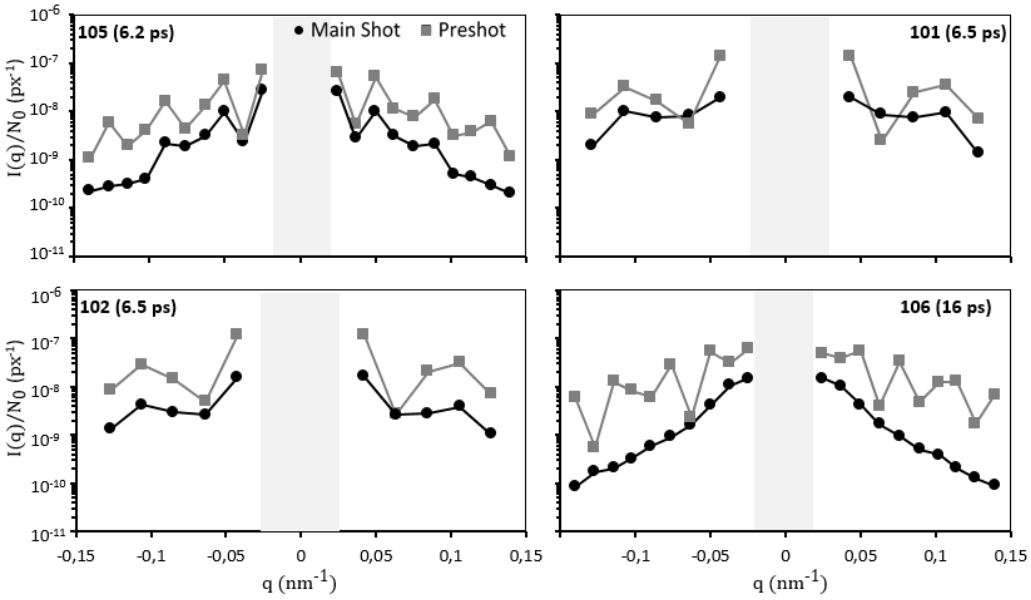

Figure 7.2: Scattering amplitudes in the first maxima, normalized to the number of photons in the respective XFEL pulse. Gray: asymmetric XFEL-only preshot, black: symmetric main shot with HI laser + XFEL. Lines are guides to the eye. The gray shaded area could not be evaluated due to over-saturation. Note, that due to the semi-logarithmic scale asymmetries visually appear much smaller than they are. Significant reduction of intensity at large q, connected to an increase of the grating smoothness, can be seen only after 16 ps.

Case 2

Fig. 7.4 and Fig. 7.5 show exemplary shots from the data set of shots on Cu targets at resonant and off-resonant XFEL energy. Consecutive preshots on the same Cu grating exhibit similar average asymmetry values in the scattering patterns. Hence the ratio of the average asymmetry in the scattering signal of two consecutive preshots is close to 1 as can be seen in Fig 7.6. This means that differences in the asymmetry between preshots and main shots cannot be explained by a random variation of the XFEL phase, as this would influence also the preshots. The fact that the changes of the asymmetry we describe in the main text for the main shots are not occurring in the preshots hence means that those changes are in fact correlated to the HI laser generated plasma dynamics rather than the XFEL-only properties.

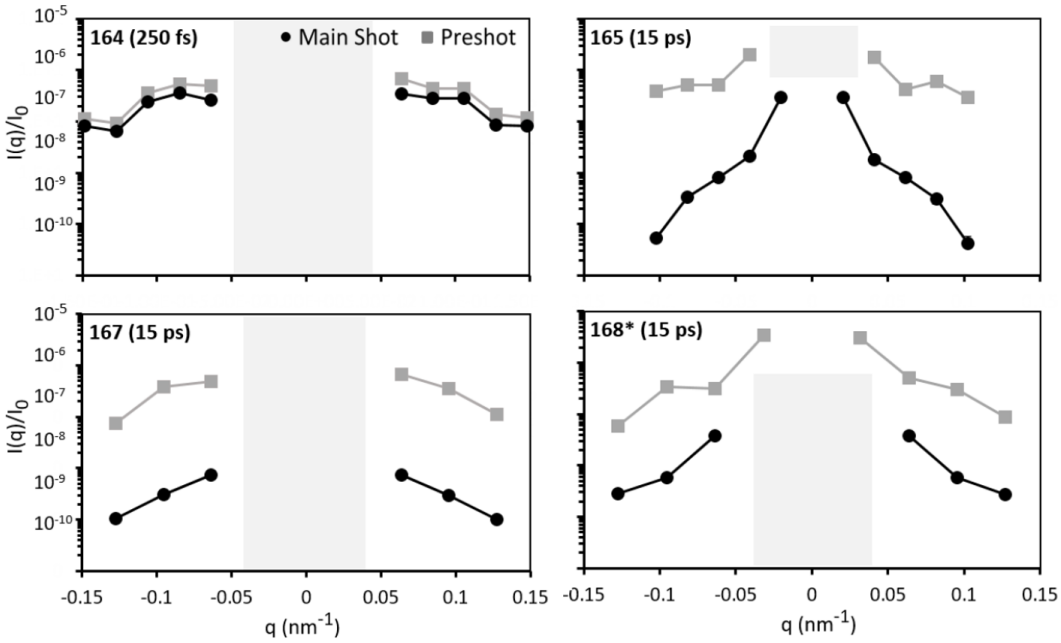

Figure 7.3: Scattering amplitudes in the first maxima, normalized to the number of photons in the respective XFEL pulse. Gray: asymmetric XFEL-only preshot, black: symmetric main shot with HI laser + XFEL. Lines are guides to the eye. The gray shaded area could not be evaluated due to over-saturation. Note, that due to the semi-logarithmic scale asymmetries visually appear much smaller than they are. (*shot 168: $E_0 = 8000$eV).

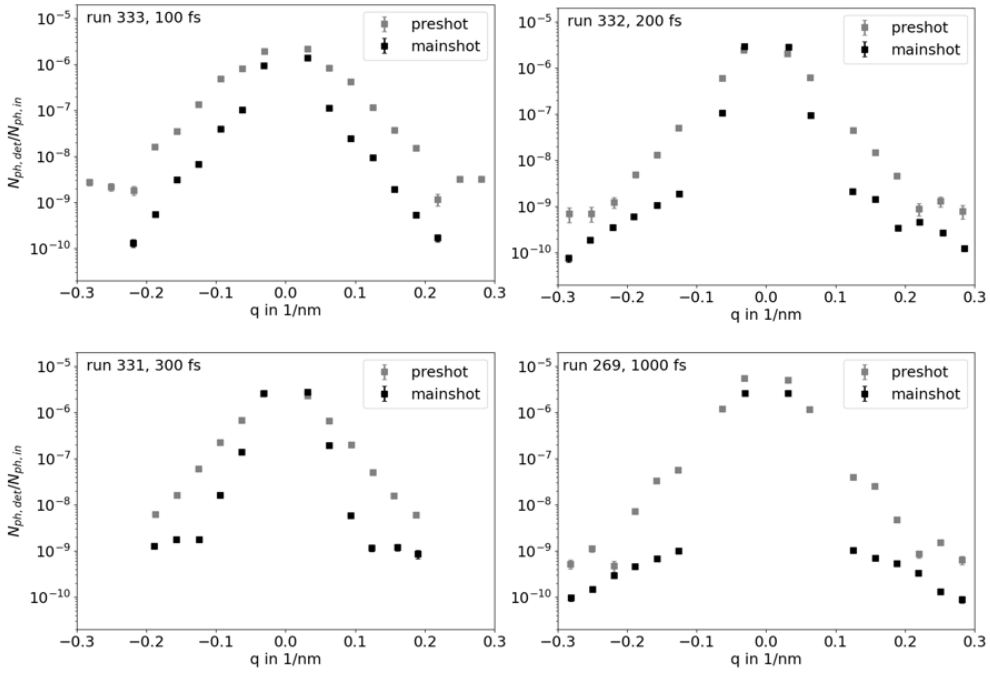

Figure 7.4: Scattering amplitude in the first maxima, normalized to the number of photons in the respective XFEL pulse for exemplary shots on Cu targets at resonant XFEL energy.

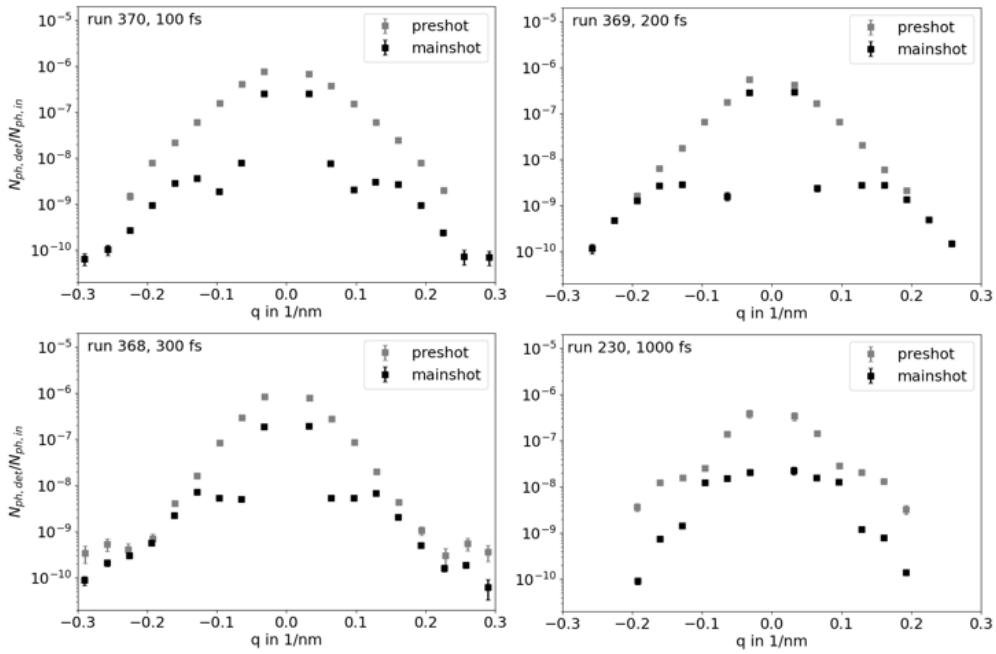

Figure 7.5: Scattering amplitude in the first maxima, normalized to the number of photons in the respective XFEL pulse for exemplary shots on Cu targets at off-resonant XFEL energy.

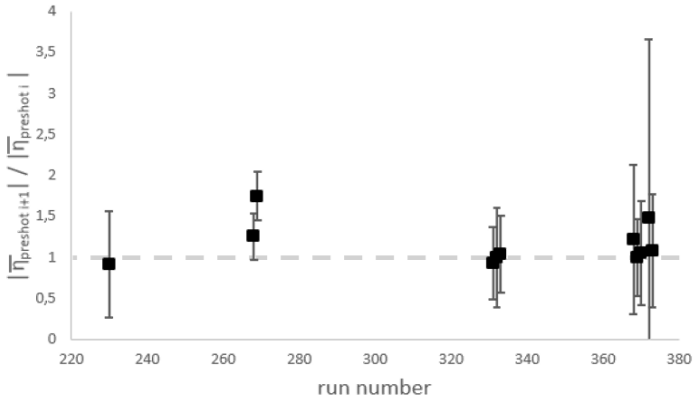

Figure 7.6: Ratio of the average preshot asymmetry between two consecutive preshots on the same Cu grating target indicated by their respective run number. The scattering signals of preshots on Cu grating targets yield similar average asymmetries and hence ratios close to 1.

7.3 Existence of surface oxides

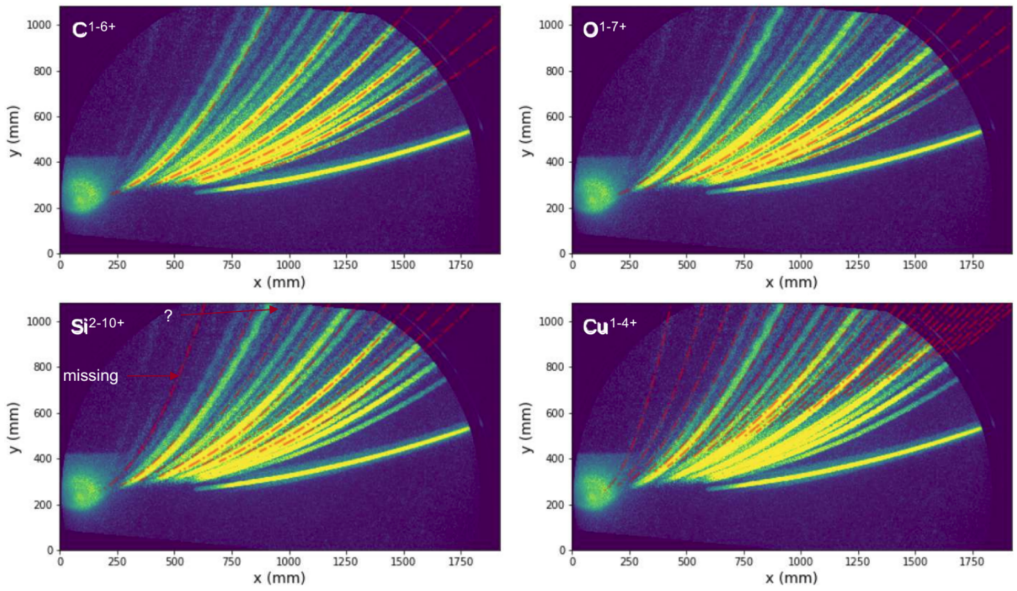

Figure 7.7: Thomson parabola traces and the respective trace identification as C and O ions. Also, there are traces of H^+, $Cu^{1+} - Cu^{4+}$ as well as remnants of the Si carrier membrane visible.

The existence of oxides on the target surface can be shown by the data recorded by a Thomson parabola. It was positioned behind the target along the target normal direction with a distance of approximately 93cm between the ion source and the 500 μm-pinhole.

The ion traces for an example shot on a copper grating target are shown in Fig. 7.7. They show the existence of oxygen ions in the accelerated ion cloud, which means that oxygen was present on the surfaces either in a layer of hydro-carbon compounds and/or CuO on the surface. There is also a strong hydrogen line visible, giving evidence to a hydro-carbon compound layer on the target surface.

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Hiermit versichere ich, dass ich die vorliegende Arbeit ohne unzulässige Hilfe Dritter und ohne Benutzung anderer als der angegebenen Hilfsmittel angefertigt habe; die aus fremden Quellen direkt oder indirekt übernommenen Gedanken sind als solche kenntlich gemacht. Die Arbeit wurde bisher weder im Inland noch im Ausland in gleicher oder ähnlicher Form einer anderen Prüfungsbehörde vorgelegt. Die Promotionsordnung, Technische Universität Dresden, Bereich Mathematik und Naturwissenschaften vom 23.02.2011 mit letzten Änderungen vom 23.05.2018 erkenne ich an.
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